

Ophthalmology

World of Revision

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BASIC ANATOMY OF THE EYE

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General Features

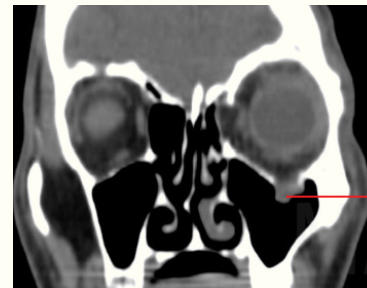
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Shape : Aspherical (Oblate spheroid).

Volume : 6 mL.

Location : Orbit → 4 walls.

- Weakest wall : Floor.
- Thinnest wall : medial wall.
- m/c orbital fracture : Blow out fracture.



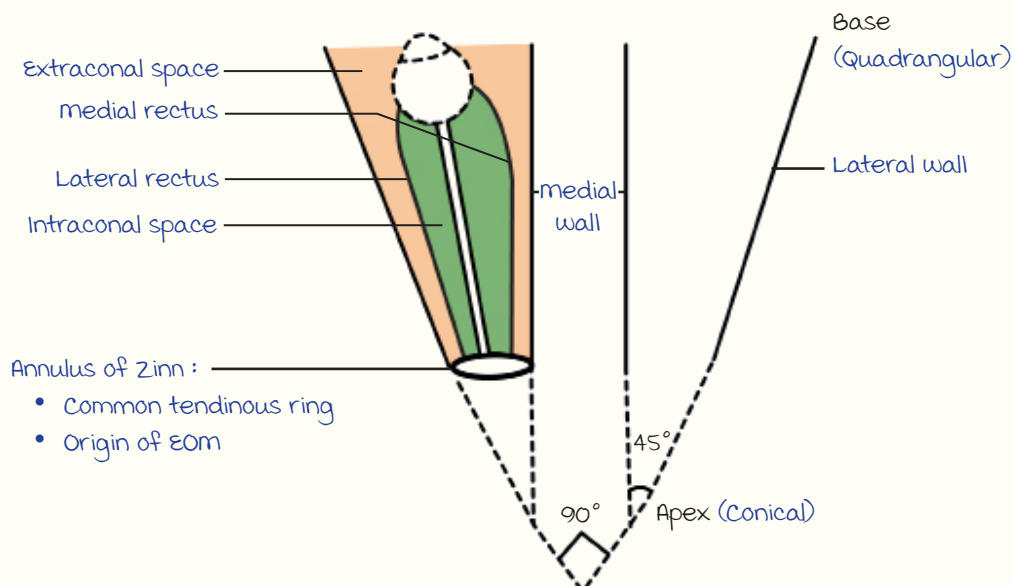
Blow out fracture

Note : Tear drop sign Infraorbital nerve anaesthesia Double diplopia (In superior & inferior eye movements)

Routes of Local Anaesthesia & Drug Administration

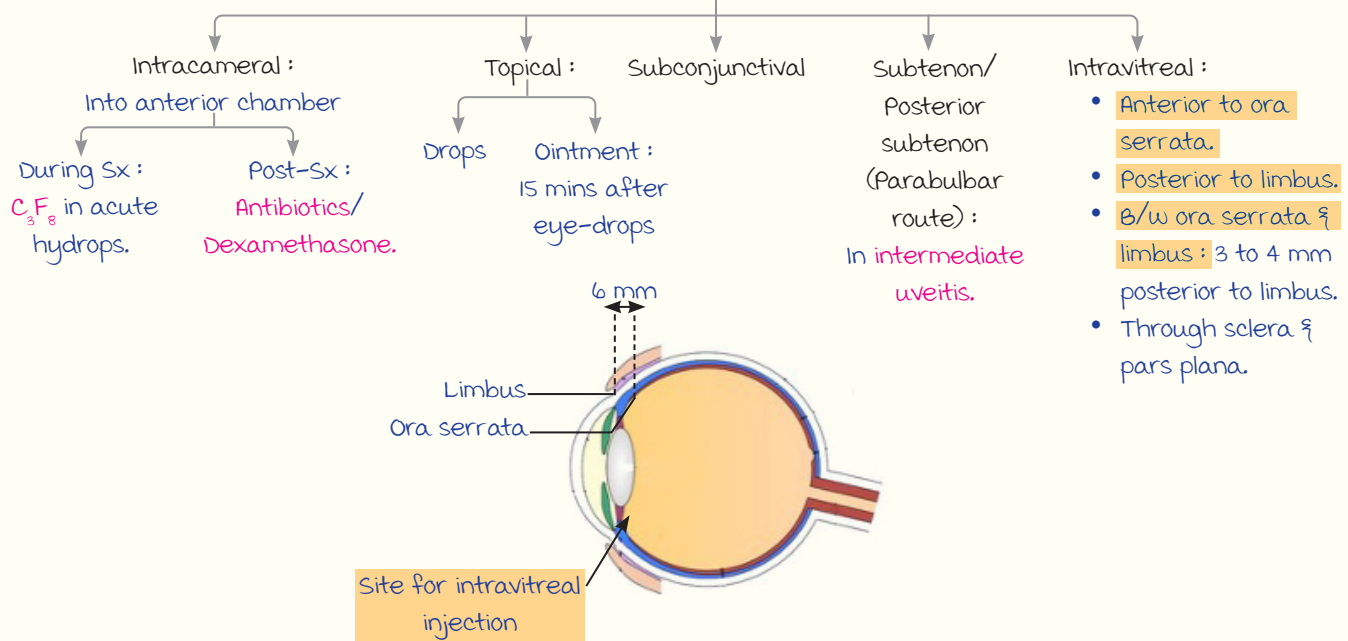
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Routes of Local Anaesthesia :



- Retrobulbar ■ : Intraconal space (Retrobulbar hemorrhage)
- Peribulbar ■ : Extraconal space → 24 G needle (↓ complications).

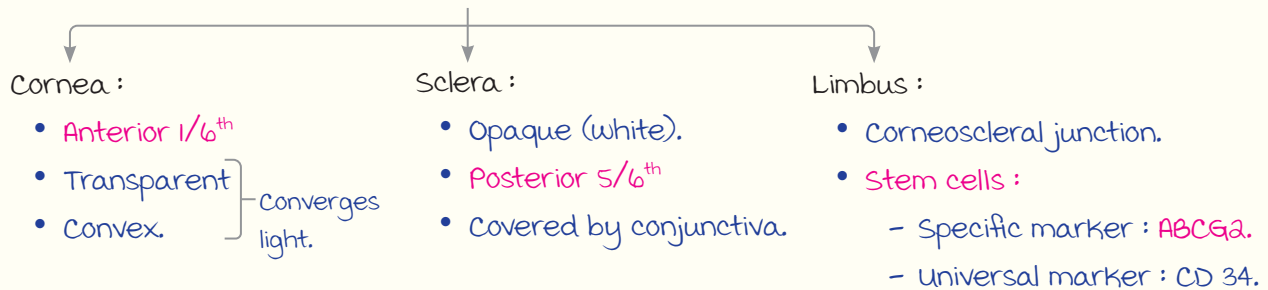
----- Active space ----- **Routes of Drug Administration :**



Layers of the Eyeball

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Outermost :

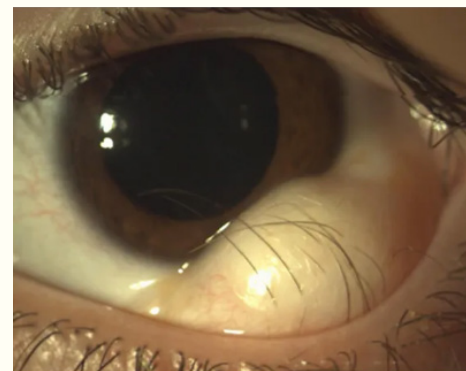
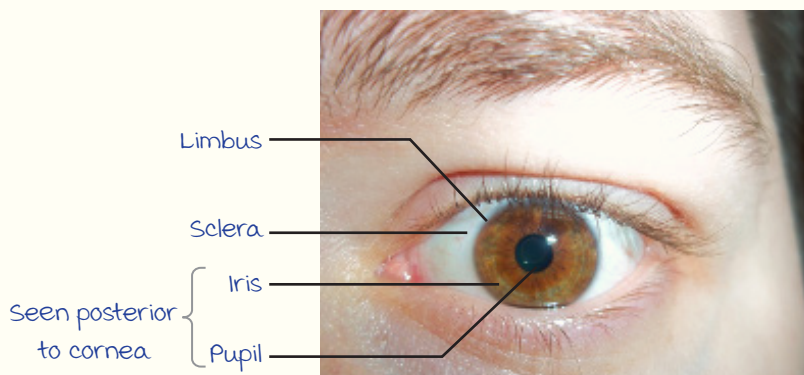


Applied aspects :

1. Limbal dermoid :

- Benign tumour.
- **Choristomatous** tissue ⊕ (Hair follicles, teeth, etc.).

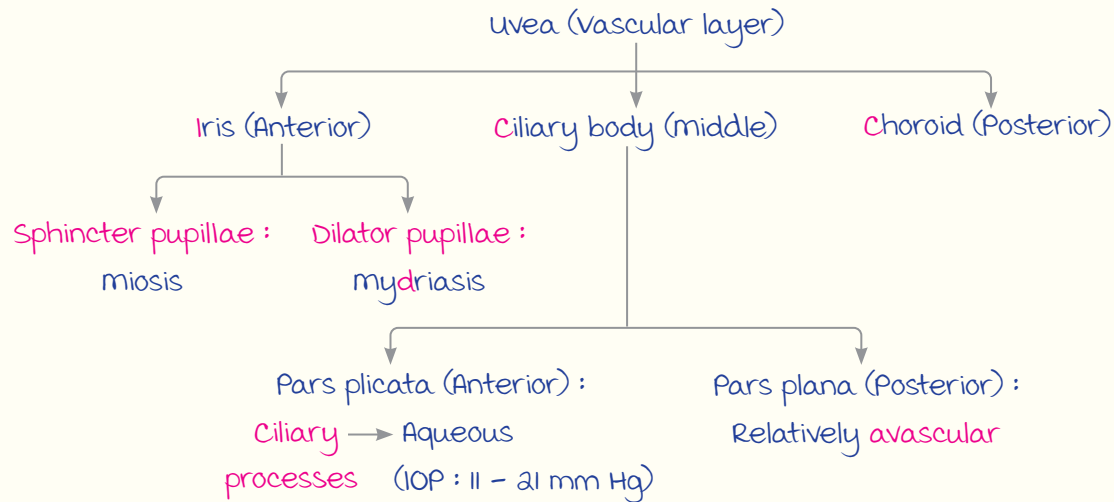
2. Pterygium : Conjunctiva grows over cornea.



Limbal dermoid

middle 👁 :

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👁 Sphincter Pupillae	👁 Dilator Pupillae
Parasympathetic : III CN ↓ Inferior branch to inferior oblique muscle ↓ Short ciliary nerve	Sympathetic : Hypothalamus ↓ 1st order neuron Cilio-spinal (Budge) center (C8 - T1) in Spinal cord Horner's syndrome ↓ 2nd order neuron (Pre-ganglionic) Superior cervical ganglion ↓ 3rd order neuron (Ascends along ICA to cavernous sinus) Long ciliary nerve Pancoast Tumor

Inner :

Retina : Neuro-sensory layer.

Horner's Syndrome 👁

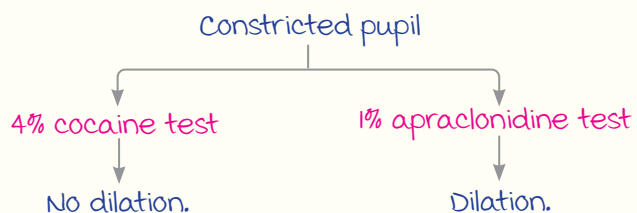
00:19:45

Oculosympathetic palsy.

Features :mnemonic : **HI MAPLE**.

- **H**eterochromia iridis : Hypochromic → Congenital only.
- **I**nferior eyelid elevation.
- **M**iosis : Dilator pupillae paralysis.
- **A**nhidrosis (I/L).
- **P**tosis : muller's muscle palsy.
- **L**oss of cilio-spinal reflex.
- **E**nophthalmos.

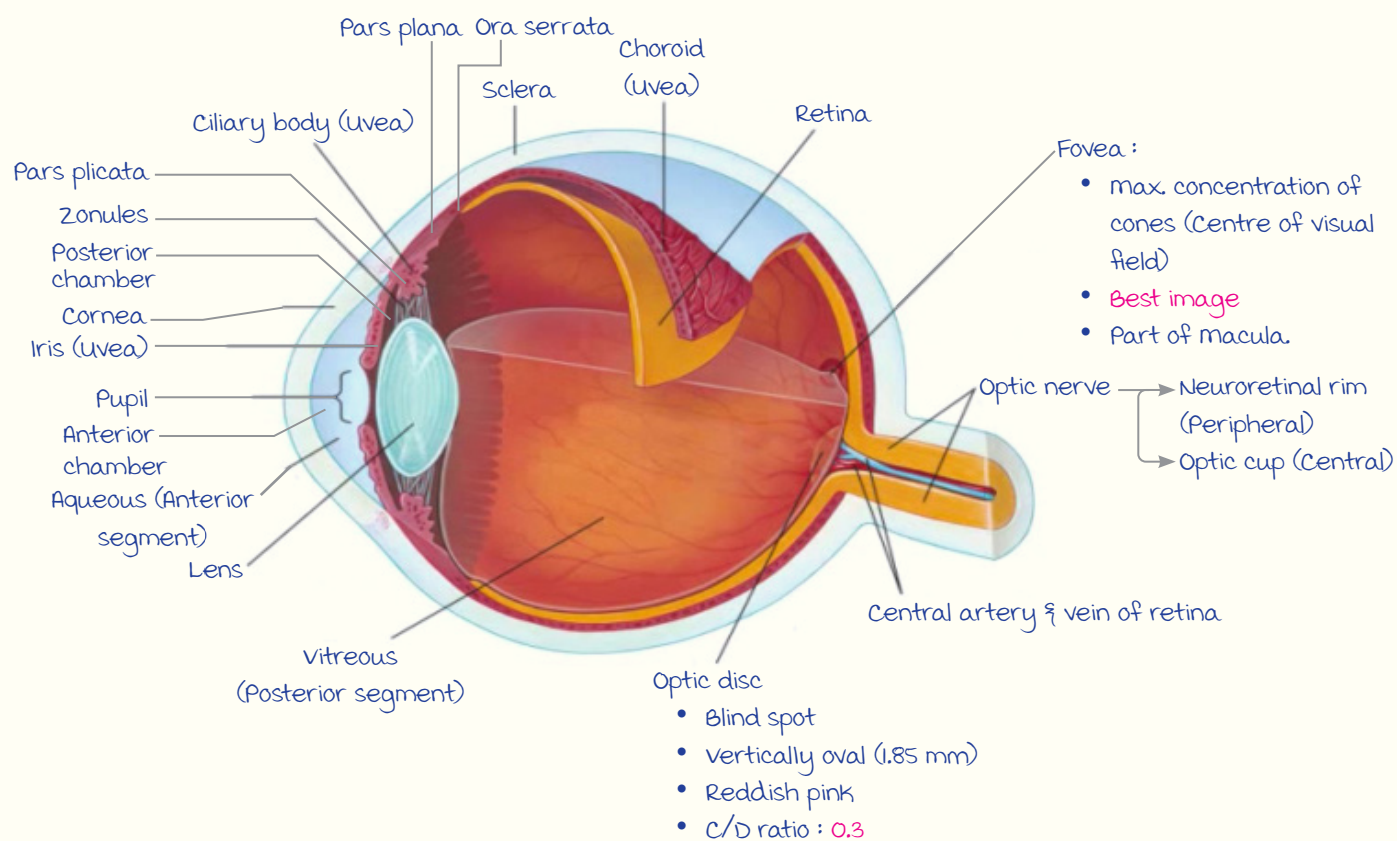
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Diagnosis :

Localization :

Cross - Section of Eyeball

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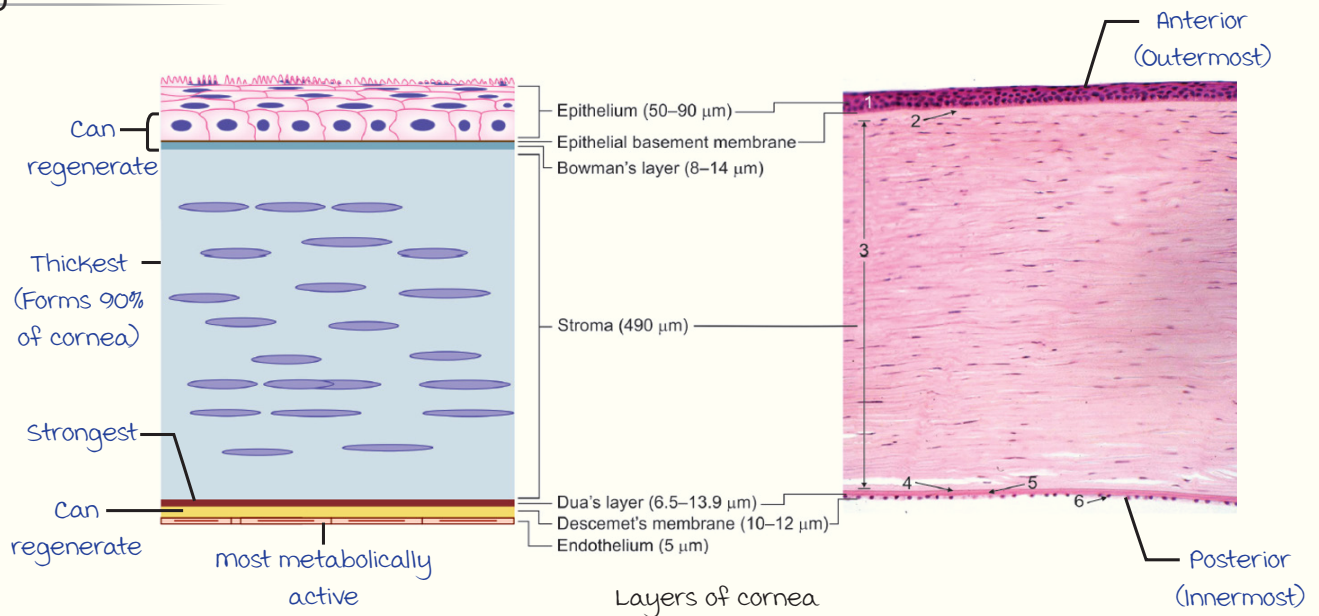
CORNEA AND SCLERA

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Anatomy of Cornea

00:00:55

Layers of Cornea :



1. Epithelium :

- Non-keratinized stratified squamous epithelium.
- Layers :
 - Superficial : Squamous (microvilli $\oplus$ $\rightarrow$ Adhesion to innermost/mucin layer of tear film).
 - Basal : Columnar (mitosis $\oplus$ $\rightarrow$ **Regeneration**).

2. Bowman's layer :

- False basement membrane (**PAS -ve**).
- Acellular
(No regeneration $\rightarrow$ Heals by scar formation $\rightarrow$ **Corneal opacity** $\rightarrow$ LOV).
- Corneal opacities  :

	Pupillary margin	Iris details	Depth of involvement
Nebular (Faintest)	Visible	Visible	Bowman's & superficial stroma
macular	Visible	Not seen	$\leq 1/2$ stroma
Leucoma (most opaque)	Not seen	Not seen	$\geq 1/2$ stroma

Nebular opacity : **maximum discomfort** (Induces irregular astigmatism).

Leucoma : **maximum visual loss**.

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3. Stroma : made of collagen (m/c Type I) & GAGs (m/c keratan sulphate : Corneal transparency)
4. Dua's layer : Strongest corneal layer.
5. Descemet's membrane (Dm) :
- Secreted by underlying endothelium.
 - Regeneration (+).
 - Shwalbe's line : Peripheral termination of Dm.
6. Endothelium :
- metabolism (+).
 - maintains corneal transparency.

No. of endothelial cells (Cells/mm ²)	Significance
2400-3000	Normal
<2400	Corneal compensation (Polymegathism & polymorphism)
<500 (Critical point)	Corneal decompensation → Edema → Hazy cornea → LOV

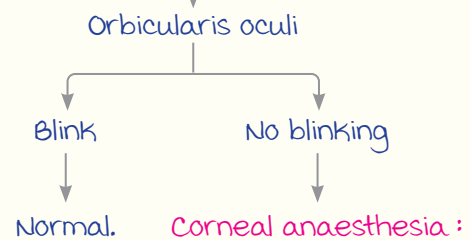
Counted using specular microscopy

Nerve Supply (Sensory) :

TON : Trigeminal → Ophthalmic → Nasociliary.

Test for corneal sensations :

Cotton wisp → Touch cornea → CN V (Afferent) → Brain → CN VII (Efferent)



HDL :

- Herpes.
- Diabetes.
- Leprosy.

General Properties of Cornea :

Corneal shape : Convex (Converging) } Curvature ∝ Power

Corneal power : +43D to +44D

Corneal colour : Transparent (Appears black).

Special Investigations for Cornea

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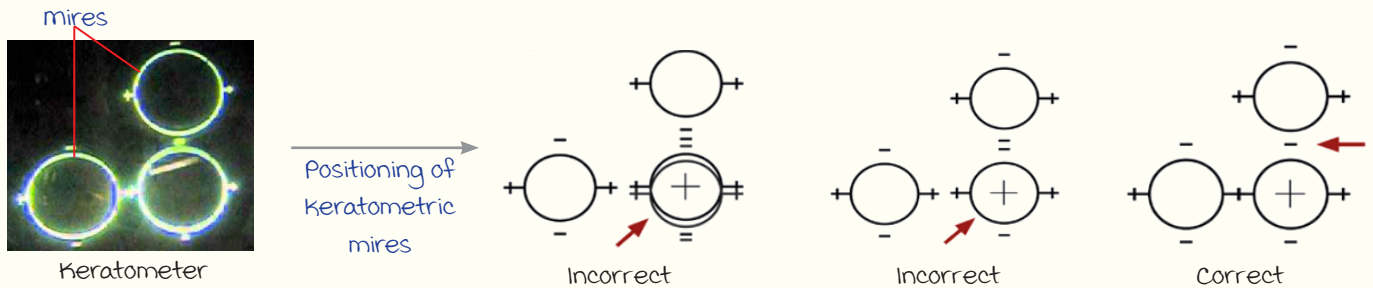
1. Pachymetry :

- measures corneal thickness (CT).
- Normal CT ~ 540 microns/0.5 - 0.6 mm.
- IOP measurement :
 - $IOP \propto CT$.
 - Post-LASIK → Thin cornea → False low IOP.

2. Keratometry :

measures corneal curvature :

- Astigmatism : Vertical curvature $\neq$ Horizontal curvature.
- Keratoconus.
- Calculation of IOL power.



3. Topography :

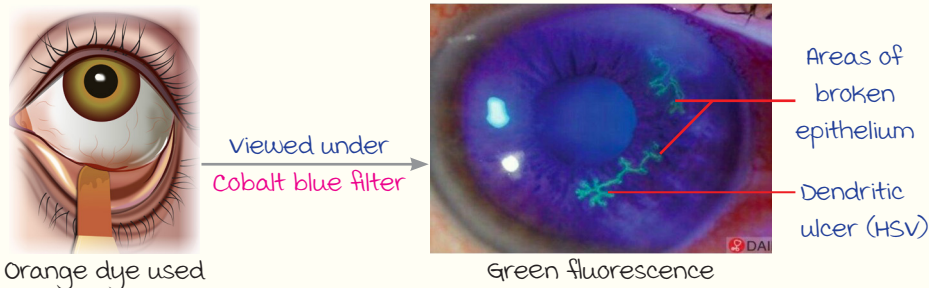
Examination of corneal surface using **placido disc**.



4. Staining :

a. Fluorescein dye :

Stains areas of **broken epithelium** (ulcer base).

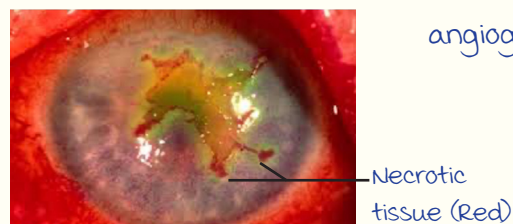


• Other uses :

- Goldmann's applanation tonometry.
- Seidel's test (To locate site of perforation).
- Tear film break up test (To detect dry eyes) : > 3 sec.
- Fundus fluorescein angiography (FFA).

b. Rose Bengal dye :

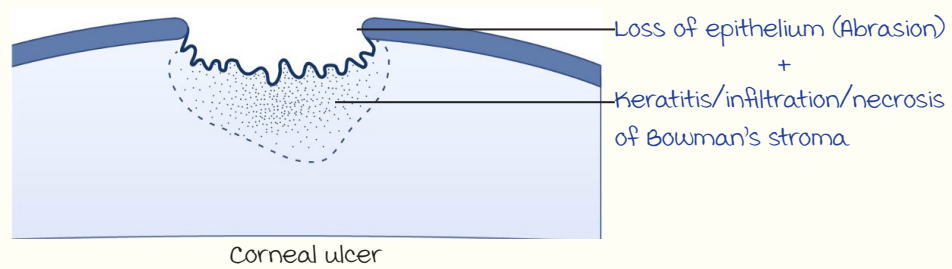
Stains areas of **necrotic tissue** (ulcer margins)



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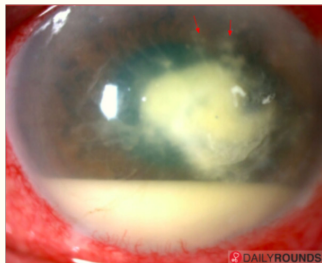
Corneal Ulcer

00:25:22

**Bacterial Corneal Ulcer :**

Pneumococcus/
S. pneumoniae :

- m/c cause (India).
- **ulcus serpens** : Snake shaped ulcer.
- **Hypopyon** : mobile & sterile collection of pus in anterior chamber.



Hypopyon

Nocardia :

- Seen in traumatic cases.
- Wreath/pin head pattern ulcer.



Wreath/Pin head pattern ulcer

Pseudomonas :

m/c cause in **contact lens users**.

Note :

Organisms that can penetrate intact cornea :

Mnemonic : **High Level CNS**

- **Haemophilus aegyptius**
 - **Listeria**
 - **Corynebacterium diphtheriae**
 - **Neisseria gonorrhoeae**
 - **Neisseria meningitidis**
 - **Shigella**
- } m/c, most severe

management 👁 :

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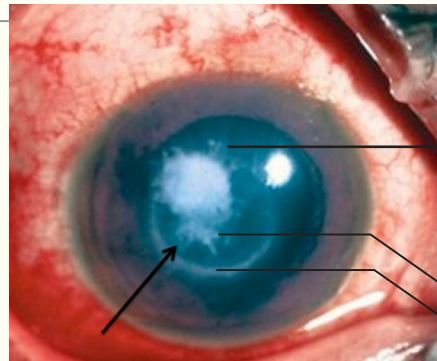
mnemonic : **SS**

1. **S**crabe base of ulcer using blunt edge of 15 no. blade.
2. Gram **s**taining & culture.
3. **S**tart fortified antibiotics.
4. **S**upportive therapy : **A**tropine → Cycloplegia relieves pain.
5. Avoid **s**teroids.

Fungal Corneal ulcer :

Etiology : ←

- m/c cause : **Aspergillus** (Filamentous, septate)
- H/o : Trauma with **vegetative material**



Fungal corneal ulcer

→ Signs :

>>

Symptoms :

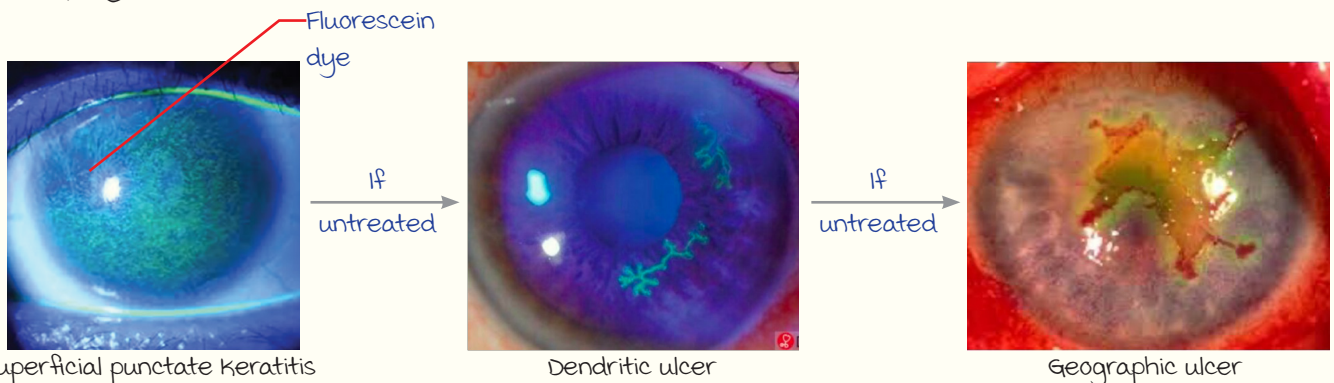
- Dry ulcer
- Satellite lesion
- Hypopyon :
 - **Immobile**
 - **unsterile**
- Feathery margin
- Wessley immune ring

Treatment :

- DOC : **5% Natamycin** eye drops
- Rest of the management is similar to bacterial corneal ulcers
- Amphotericin B (Candida)

Herpetic Keratitis :

Disease progression :



Rx :

- Acyclovir (3%)/Ganciclovir **eye ointment**.
- Steroids : **C/I**.

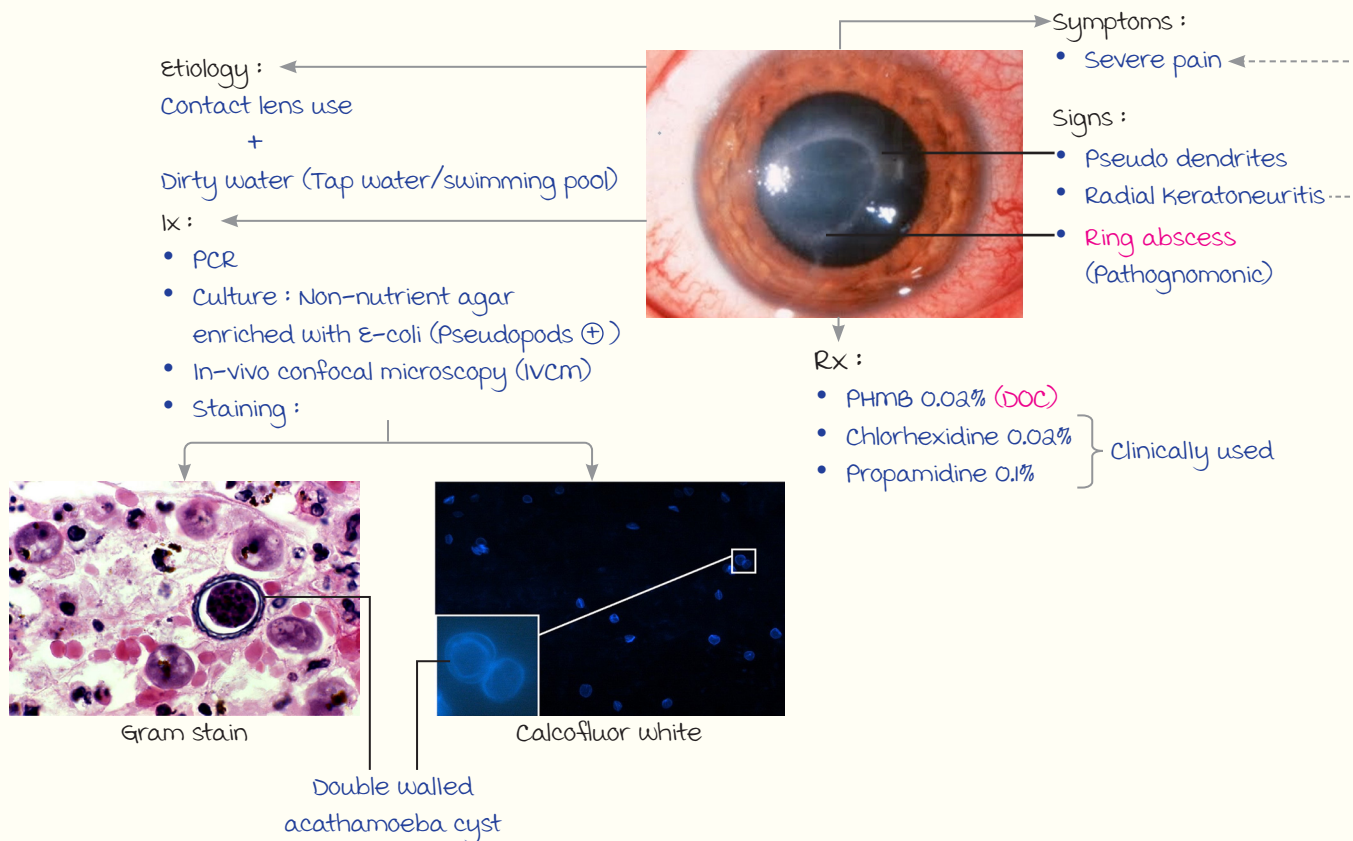
----- Active space ----- Other lesions d/t herpes :

- Necrotising stromal Keratitis
 - Disciform Keratitis/endothelitis
 - metaherpetic keratitis/neurotrophic ulcer : CN V atrophy/palsy → Corneal anaesthesia.
- Intact epithelium → Rx : 3% Acyclovir eye ointment + Topical steroids.

Note :

1. Neuroparalytic ulcer (Not d/t herpes) : CN VII palsy → Loss of blinking (Lagophthalmos) → Exposure keratitis → ulcer.
2. Lagophthalmos seen in Leprosy & Bell's palsy.

Acanthamoeba Keratitis :



Corneal Dystrophies

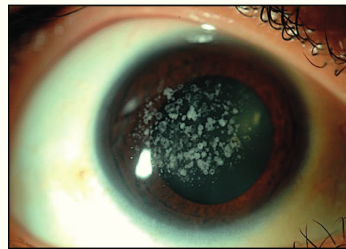
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- B/L non-inflammatory opacifying disorder.
- Autosomal dominant.

Types 👁 :

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Dystrophy	Site	Features
Cogan's epithelial basement membrane dystrophy	Epithelial (mnemonic : CML)	m/c type
meesman's epithelial dystrophy		Keratin gene mutation
Lisch epithelial dystrophy		X-linked dominant
Reis Buckler's dystrophy	Bowman layer (mnemonic : RST)	Fish net
Schnyder central crystalline dystrophy		Corneal lipid metabolism disorder
Thiel Behnke dystrophy		Honeycomb pattern opacities
Lattice corneal dystrophy		Amyloid deposit (Stained by Congo red)
Granular corneal dystrophy	Stromal	Hyaline deposits (Stained by masson trichrome)
macular corneal dystrophy		Least common , Autosomal Recessive (AR)
Gelatinous drop like dystrophy		AR, mulberry like appearance
Francois central cloudy dystrophy		Inborn error of Keratan sulphate metabolism
Fuch's endothelial dystrophy	Endothelial	Cornea guttata : Wart like excrescences on posterior cornea
Posterior polymorphous corneal dystrophy		AR inheritance, perinatal onset
Congenital hereditary endothelial dystrophy		



Granular corneal dystrophy

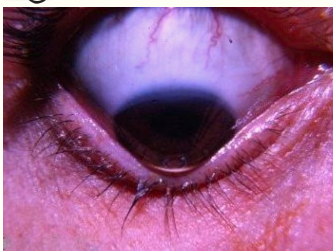
Keratoconus 👁

00:50:40

- Cone-shaped cornea.

- 1° ectasia/degeneration.

Signs :



1. munson's sign :
V-shaped lower lid



2. Rizzuti's sign :
Light thrown from one side refracts to form
a cone shape facing the opposite side.

3. Irregular myopic astigmatism.
4. **vogt's striae** : D/t stretching of cornea (Pathognomonic).
5. Fleischer's ring : D/t **iron** deposition.
6. Oil droplet reflex : On ophthalmoscopy.
7. Scissoring reflex : On retinoscopy.

Note :

Kayser Fleischer ring : D/t Cu
deposition in Wilson's disease

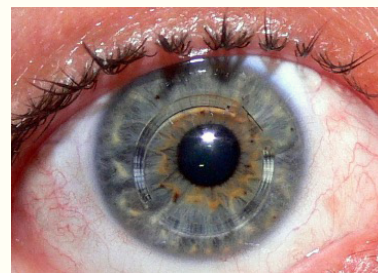
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Investigation :

Topography (Ioc).

Treatment :

- INTACS surgery : Intracorneal stromal ring segments.
- Rigid contact lens. Keratoplasty : Corneal transplantation.
- Collagen cross linking with riboflavin & UVA radiation (C3R) : Best initial treatment.
- Keratoplasty.



INTACS Sx :
Ring stretches the cornea

Note :

INTACS Sx also done in myopia.

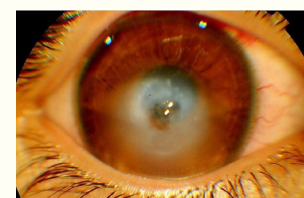
Keratoplasty

00:56:55

Types :

	Penetrating keratoplasty (Full thickness)	Lamellar keratoplasty (Partial thickness)	
		DSAEK	DALK
Removal of	All layers	• Descemet's membrane • Endothelium	• Epithelium • Bowman's membrane • most of stroma
Indications	-	• Bullous Keratopathy (Endothelial dysfunction) • Acute hydrops (Break in descemet's membrane)	• Stromal opacities • Acute hydrops in severe keratoconus

DSAEK : Descemet stripping automated endothelial keratoplasty.
DALK : Deep anterior lamellar keratoplasty.



Acute hydrops

Storage media for cornea preservation :

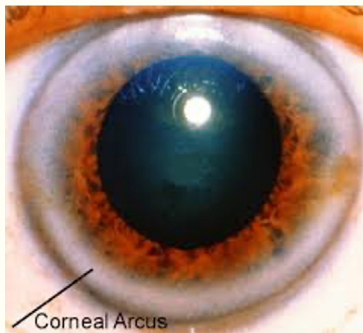
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Type	method
Short term	4°C moist chamber
Intermediate term (upto 2 days)	McCarey Kaufman medium
Long term	Cryopreservation

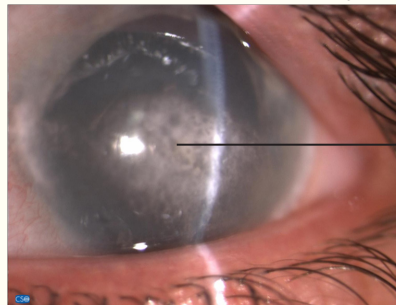
Miscellaneous

01:01:05

Arcus senilis : D/t lipid deposition near the limbus in old age



Band shaped keratopathy :



Band shaped deposition of Ca^{2+} in Bowman's layer

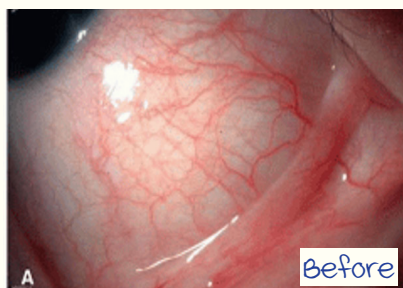
- Complication of anterior uveitis.
- Treatment : Chelation with EDTA.

Sclera

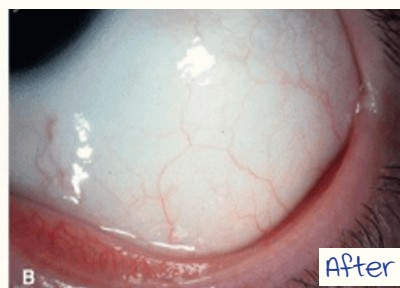
01:02:05

Episcleritis vs. Scleritis :

	Episcleritis	Scleritis
Inflammation of	Superficial vessels	Deep vessels
Hue	Reddish	Bluish/violaceous
Pain	mild	Deep seated, chronic
Response to phenylephrine (vasoconstrictor)	Redness disappears (Blanching)	Redness persists (No blanching)



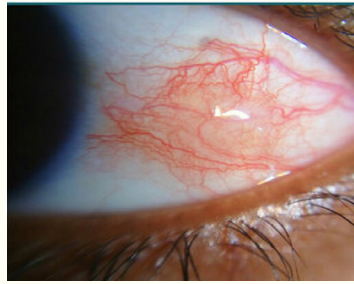
Before



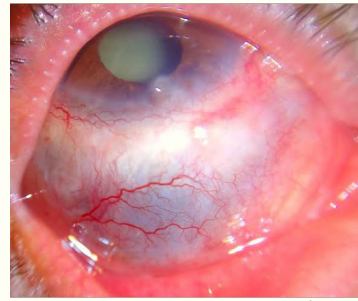
After

Episcleritis : Phenylephrine test

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Nodular episcleritis



Scleritis (Longstanding)
Bluish discoloration

Scleromalacia Perforans :

- Anterior necrotising scleritis.
- No redness/pain.
- Progression :
Chronic low grade inflammation → Scleral thinning → Perforation.
- m/c in women with rheumatoid arthritis.

NEURO-OPHTHALMOLOGY

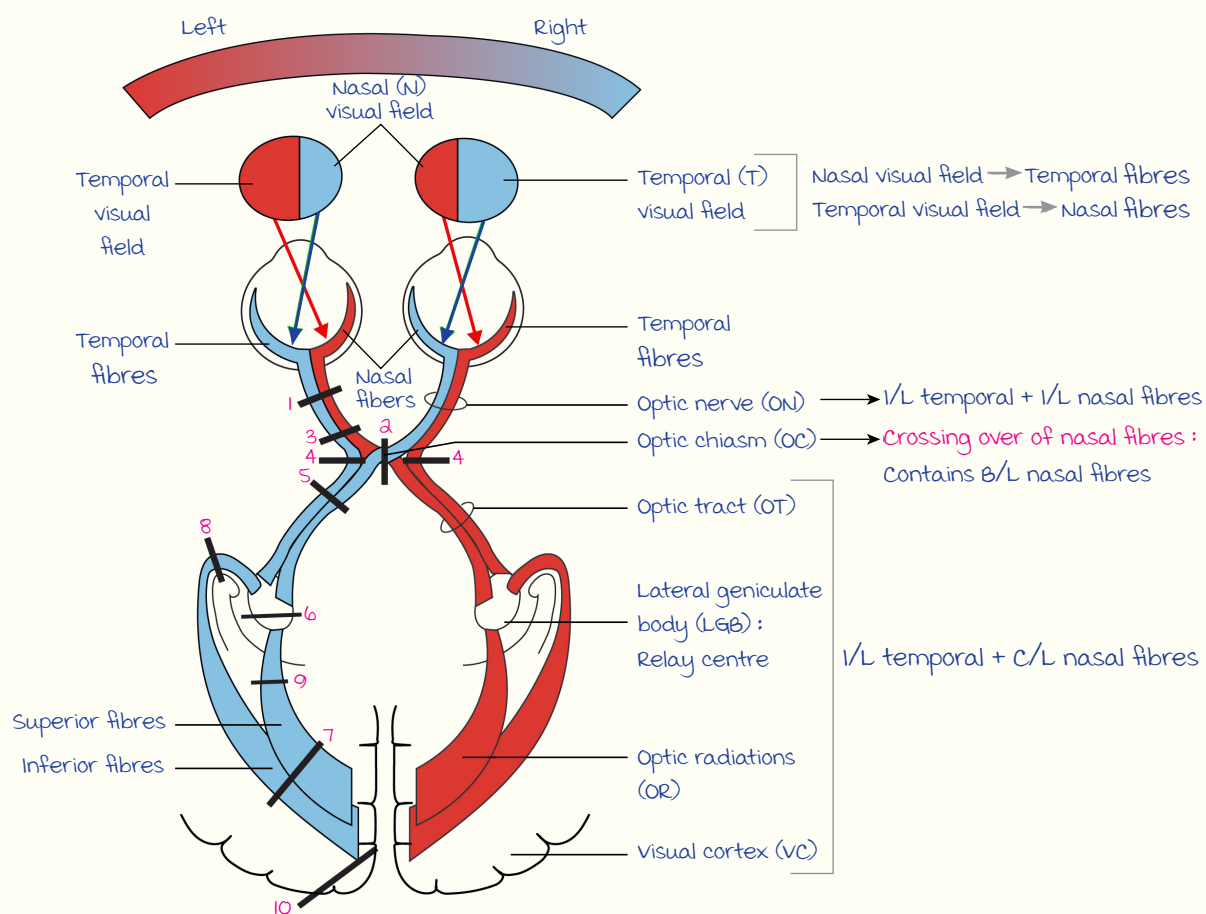
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Visual Pathway

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Retina :

Covers posterior part of eye till ora serrata.



Note :

FOF : Field loss is opposite to loss of fibres.

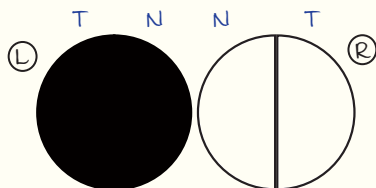
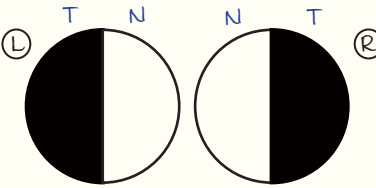
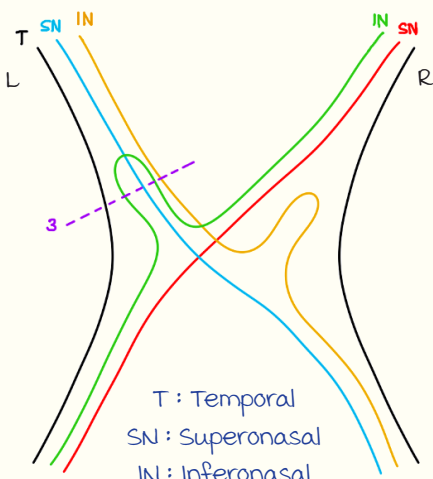
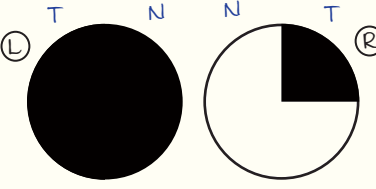
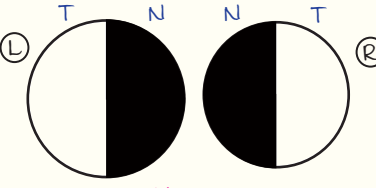
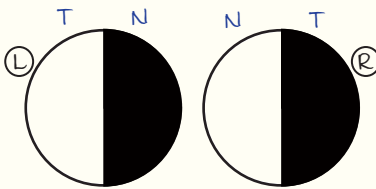
(Temporal fibres injured → Nasal field lost).

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Visual Field Defects

00:05:48

Left sided lesions are described below :

Fibres affected	Visual field defects (VFD)
1. Optic nerve : I/L temporal + I/L nasal	 I/L anopia (Blindness)
2. Optic chiasma  : B/L nasal. Causes : mnemonic → 3PC & AC • 3rd ventricular glioma • Pituitary adenoma • Craniopharyngioma (Occurs in Rathke's pouch) • Anterior communicating artery aneurysm • Cavernous sinus thrombosis	 Bitemporal hemianopia
3. Junction of ON & OC : Anterior knee of von Willebrand  T : Temporal SN : Superonasal IN : Inferonasal m/c cause : meningioma	 Junctional scotoma
4. Lateral optic chiasma : - B/L temporal (Always B/L) - very rare	 Binasal hemianopia
5. OT 6. LGB 7. OR 10. VC (Rare)  I/L temporal + C/L nasal	 C/L homonymous hemianopia

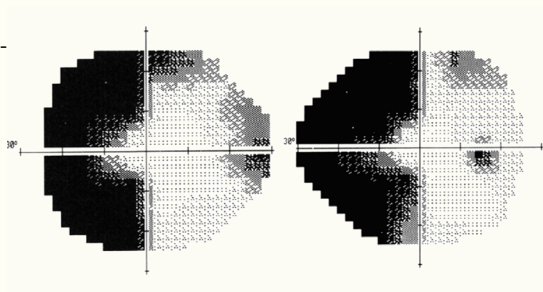
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Fibres affected	visual field defects
8. Inferior OR : Meyer's loop Inferior I/L temporal + C/L nasal	 C/L homonymous Superior quadrantanopia, AKA "Pie in the sky." mnemonic : TIPS (Temporal inferior pie in the sky)
9. Superior OR : Baum's loop Superior I/L temporal + C/L nasal F : Frontal lobe P : Parietal cortex T : Temporal lobe VC : Visual cortex	 C/L homonymous inferior quadrantanopia, AKA "Pie on the floor."
10. Occipital lobe lesion : 10 (a) Tip of cortex lesion : macular fibres of I/L temporal + C/L nasal Posterior cerebral artery 10 (b) visual cortex except tip : I/L temporal + C/L nasal except macular fibres	10 (a) C/L homonymous macular defect 10 (b) C/L homonymous hemianopia with macular sparing.

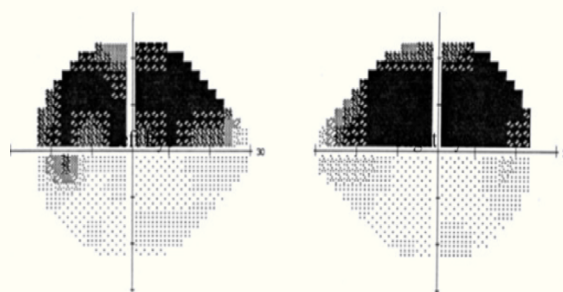
Difference b/w lesions at 5, 6 and 7 :

	OT lesion (5)	LGB lesion (6)	OR lesion (7)
Pupil	Abnormal : Wernicke's pupil	Normal	Normal
Optic atrophy	Present	Present	Absent (Optic radiations are post ganglionic : 4 th order)
Congruity	Incongruous	Incongruous	Congruous

----- Active space -----



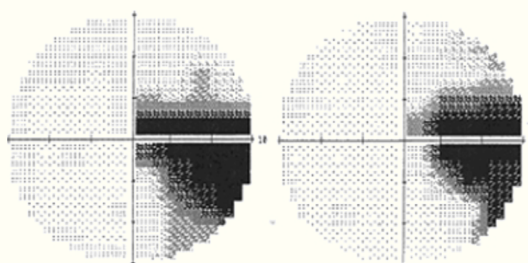
Left homonymous hemianopia with
macular sparing :
Right visual cortex except tip lesion.



Superior altitudinal visual field defect :

Causes : **GOA**

- **G**laucoma.
- **O**ptic disc drusen.
- **A**nterior ischemic optic neuropathy (AION).



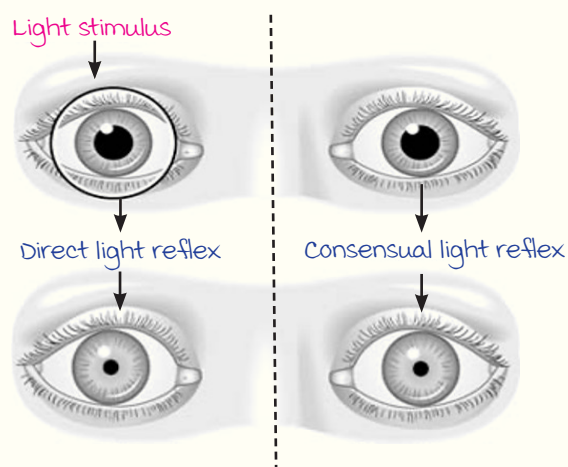
Right homonymous sector anopia/keyhole defect 🦋 :
Partial left LGB lesion.

Pupillary Reflex

00:25:49

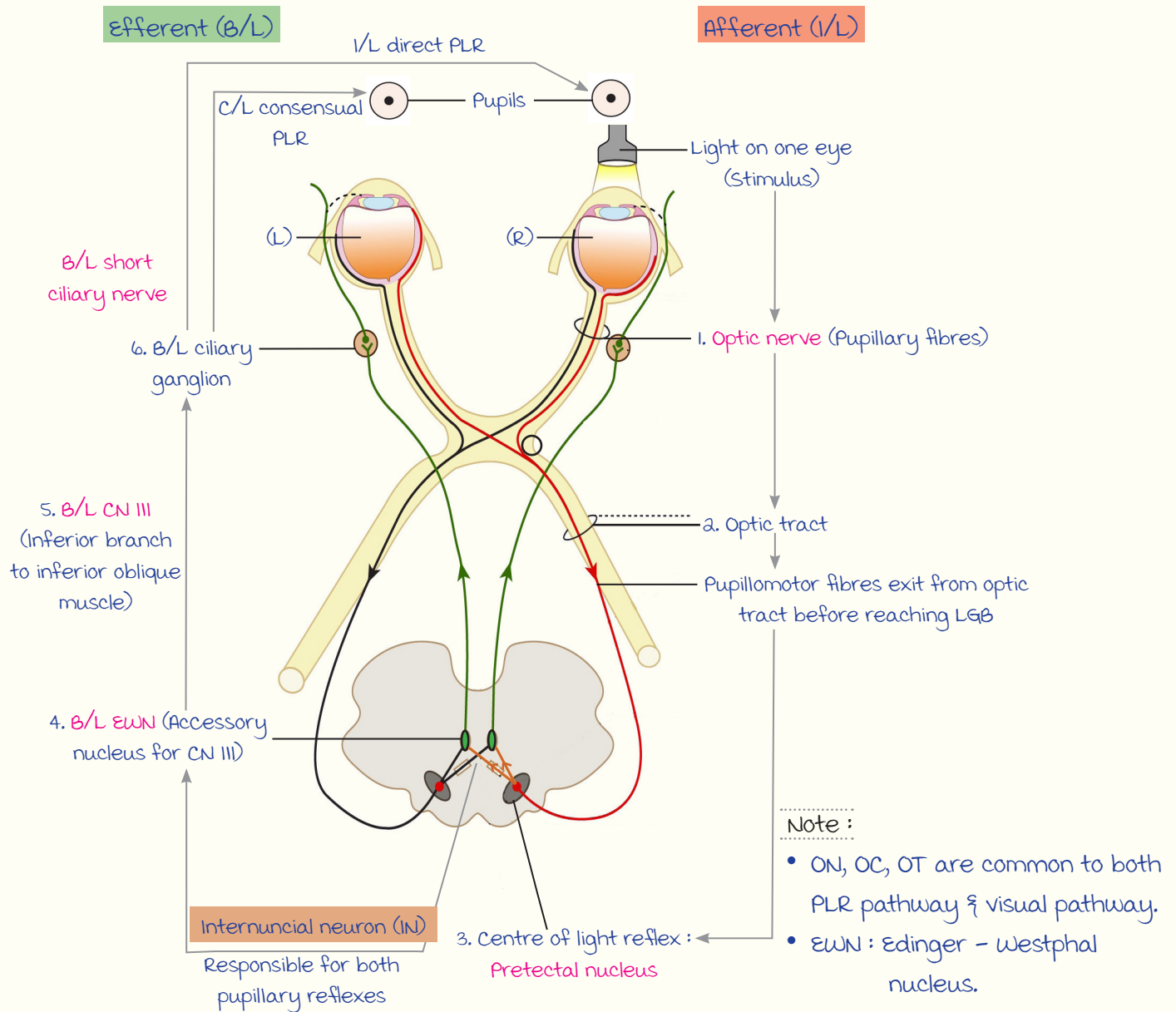
Light → Pupil → Constrict.

Types : Direct and consensual.



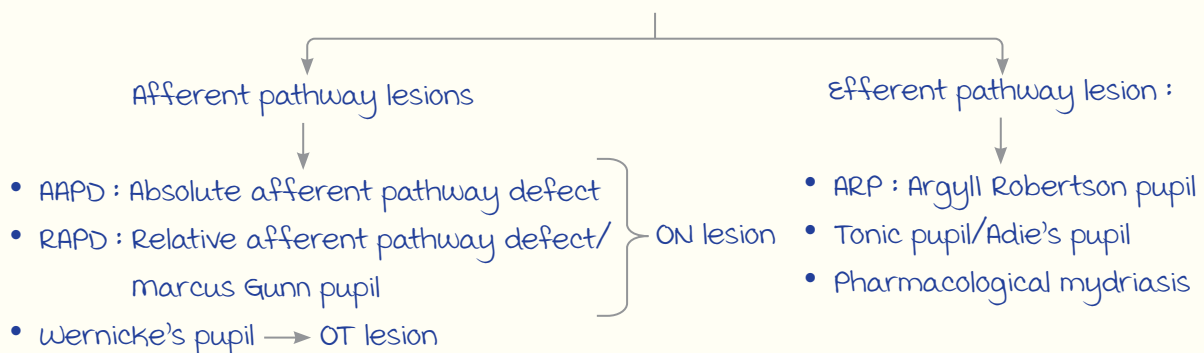
Pupillary Pathway

----- Active space -----



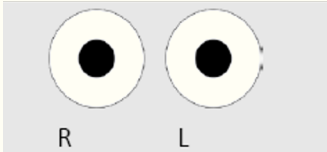
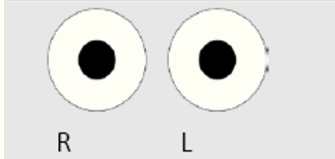
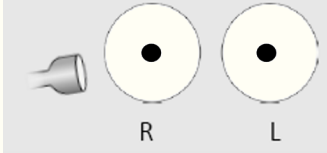
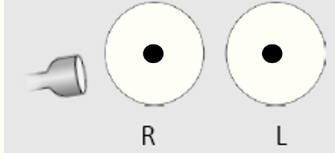
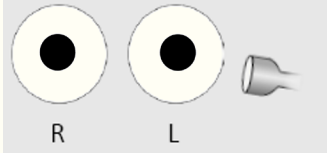
Lesions of Pupillary Pathway

00:31:13



----- Active space -----

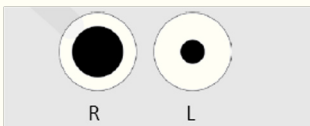
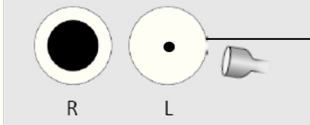
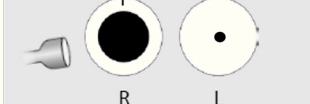
Afferent Pathway Lesions :

	AAPD	RAPD
ON lesion	Complete	Partial (E.g. : Optic neuritis)
Findings	 Normal	
	 Light stimulus on normal eye : B/L constriction	
	No B/L constriction  Light stimulus on affected eye :	 Swinging flashlight test ↓ Nerve fatigue ↓ B/L dilatation (Paradoxical dilatation/Pupillary escape phenomenon)
Pupil size	Isocoria	

Efferent Pathway Lesion :

ARP : mnemonic → ARP-PRA

- Accommodation Reflex Present Pupillary Reflex Absent.
- Cause : Neurosyphilis.
- Site of lesion : Posterior commissure.

	Tonic/Adie's Pupil	Pharmacological mydriasis
Cause	Lesion at ciliary ganglion/short ciliary nerve	Drugs : Atropine
Light reflex	 E.g. : Lesion/mydriatic application at right eye	
	 Light on normal side	1/L pupil does not constrict (AKA tonic pupil)
	 Light on lesion side	
Pupil size	Anisocoria	

	Tonic/Adie's Pupil	Pharmacological mydriasis
Accommodation reflex	Normal	Absent MOA : Cycloplegic action ↓ Relaxation of ciliary muscle ↓ Loss of accommodation.
Diluted Pilocarpine (0.1 %)	Constricts d/t denervation hypersensitivity MOA : Sphincter pupillae denervated ↓ Over-expression of receptors ↓ Hyper-reactive to stimuli.	Does not constrict (Even with 1% pilocarpine)

----- Active space -----

Note :

Pupil normally constricts with 1% pilocarpine.

Colour Blindness

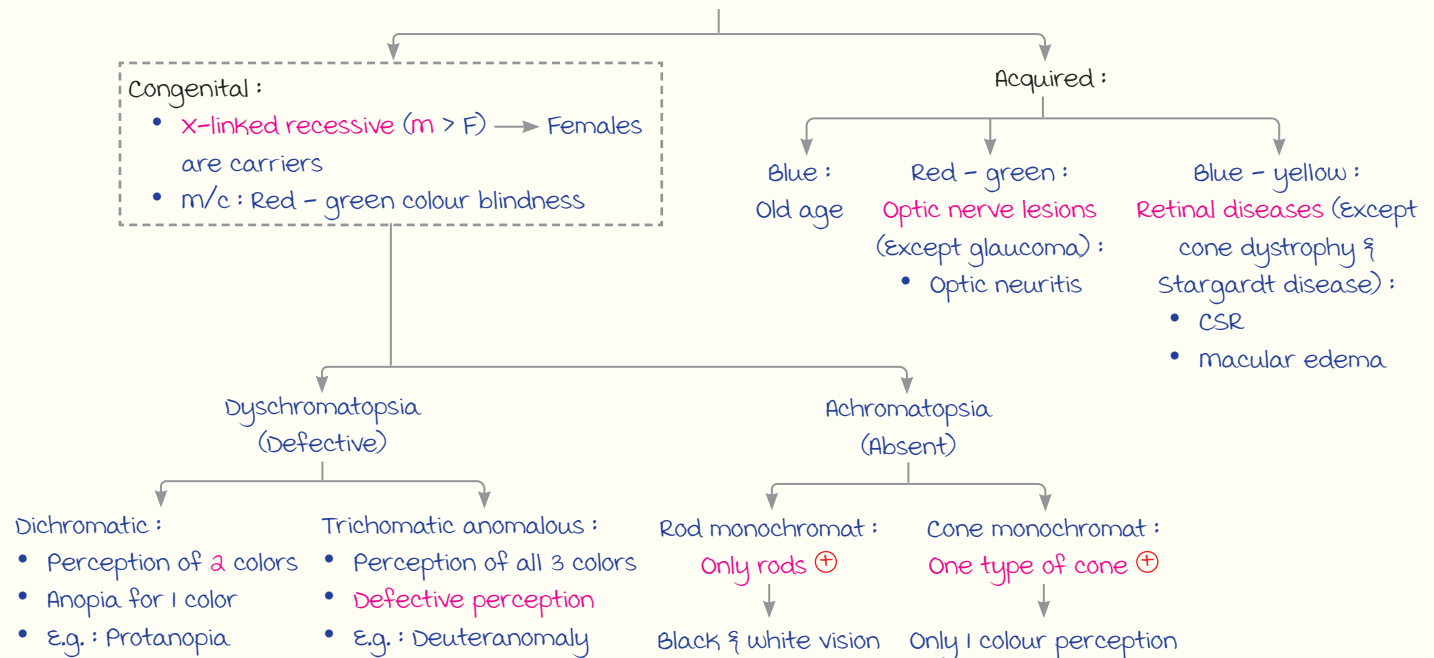
00:42:10

1° colors :

Red : Protan.

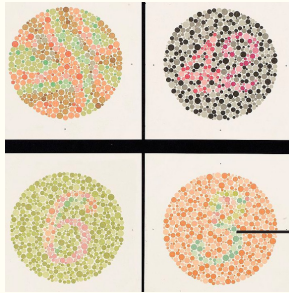
Green : Deutan.

Blue : Tritan.

Etiology :

----- Active space -----

Tests :

	Ishihara chart	Hardy-Rand-Rittler
Chart		
Features	- m/c used - Disadv. : Cannot detect tritan defects	used to test pediatric age group
Principle	Pseudo-isochromaticity	

	Farnsworth munsell 100 hue test	Farnsworth munsell 15 hue test
Chart		
Features	most sensitive	Differentiates between congenital & acquired

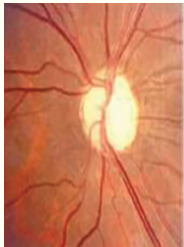
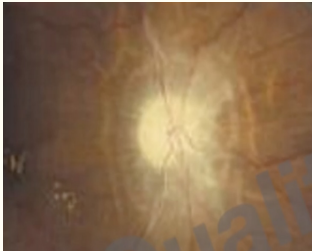
- Holmgren's wool.
- Lantern test.



Nagel's anomaloscope

Optic Atrophy

00:49:10

	Primary (1°) 	Secondary (2°)	Consecutive
Definition	Optic atrophy without antecedent optic disc swelling	Optic atrophy after optic disc swelling	Optic atrophy due to disease of inner retina and its blood supply
Disc appearance	Chalky white , well defined margins	 Dirty gray-white , ill-defined margins	waxy pallor of disc
Optic cup	 Chalky white	 Ill defined margins of OD	-

	Primary (1°)	Secondary (2°)	Consecutive
Causes	• Optic neuritis • Tumors • Trauma • Hereditary : LEBER's disease - mms : mitochondrial inheritance (mother to son) • Drugs : DICE - Digoxin - Isoniazid - Chloroquine - Ethambutol	Papilledema	-

----- Active space -----

Optic Neuritis, Papilledema & Miscellaneous Conditions

00:52:08

Optic Neuritis :

Inflammation of optic nerve.

Cause : multiple sclerosis (m/c).

Clinical features :

Mnemonic : 2 CUP mar.

- Central scotoma.
- Color blindness of red & green.
- u/L loss of vision.
- Uhthoff sign : Worsening of symptoms on ↑ body temperature.
- Pulfrich sign.
- Pain in eye on ocular movements.
- Marcus Gunn pupil (RAPD) → Earliest sign.

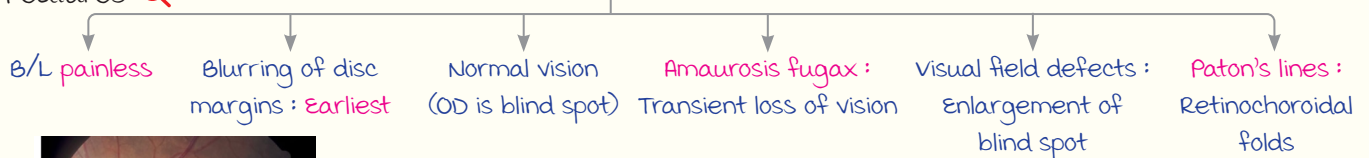
Treatment :

Steroids : IV f/b oral.

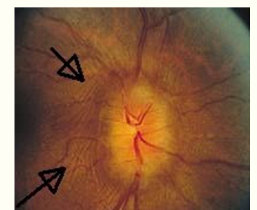
Papilledema :

Swelling of optic disc d/t ↑ ICP.

Features  :



Optic disc edema



Paton's lines

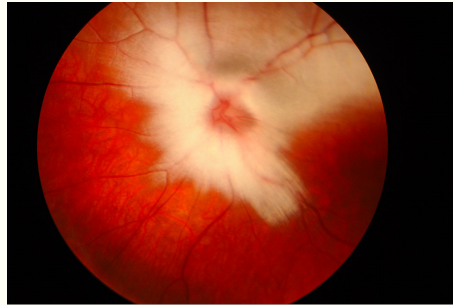
Treatment : Control ICP.

----- Active space -----

Miscellaneous :

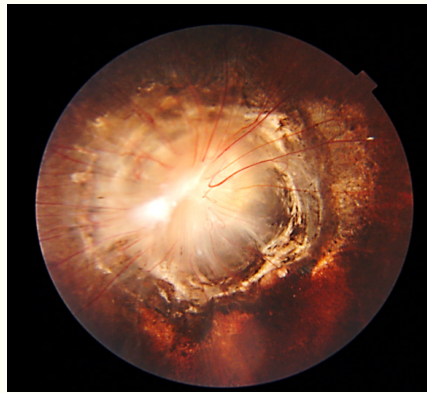
myelinated nerve fibres :

- myelination of OD & nerve fibres of retina.
- myelination : ↑ Speed of conduction.



myelinated nerve fibres

morning glory syndrome :



SQUINT

----- Active space -----

Extraocular Muscles (EOM)

00:00:57

6 EOMs act on eyeball & 1 EOM acts on eyelid (Levator palpebrae superioris).

muscles Acting on the Eyeball :

Action only in 1° gaze :

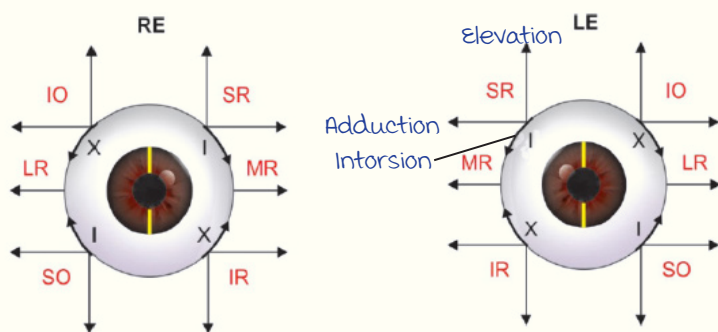
- medial rectus (MR) : Adduction.
- Lateral rectus (LR) : Abduction.

Action in 1°, 2° & 3° gaze : Cyclovertical action.

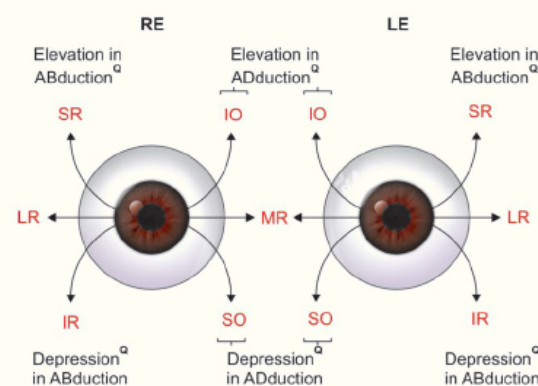
muscle	Actions		
	1° (Predominates in abduction)	2° (Predominates in adduction)	3°
Superior Rectus (SR)	Elevation	Intorsion (mnemonic : SIN)	Adduction (mnemonic : RAD)
Inferior Rectus (IR)	Depression	Extorsion	
Superior Oblique (SO)	Intorsion	Depression	Abduction (mnemonic : ABO)
Inferior Oblique (IO)	Extorsion	Elevation	

Nerve supply :

- LR6 : Lateral rectus supplied by Abducens N. (CN 6).
- SO4 : Superior oblique is supplied by Trochlear N. (CN 4).
- Others : CN 3.



Anatomical action of EOM



Clinical action of EOM

Note :

Superior oblique function can be confirmed by asking the pt. to look at the tip of their nose.

----- Active space -----

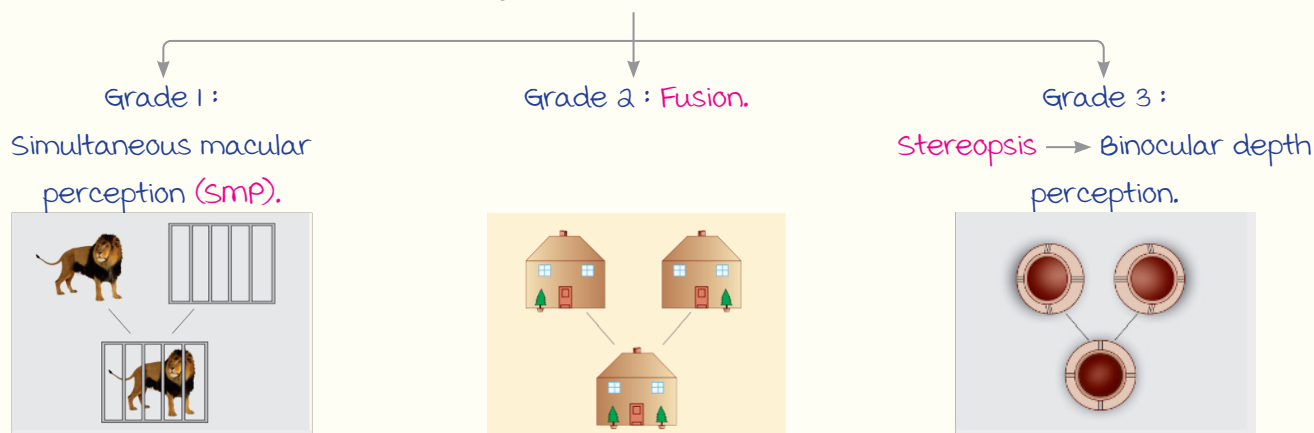
Binocular Vision :

Terminology	Action of the muscles 	Examples
Antagonist	Opposite action in same eye	Rt. SR & Rt. IR
Agonists/synergists	Same action in same eye	Rt. SR & Rt. IO
Yoke muscles/ Contralateral synergists	One muscle from each eye; helps to look in the same direction (version movement)	Rt. MR & Lt. LR

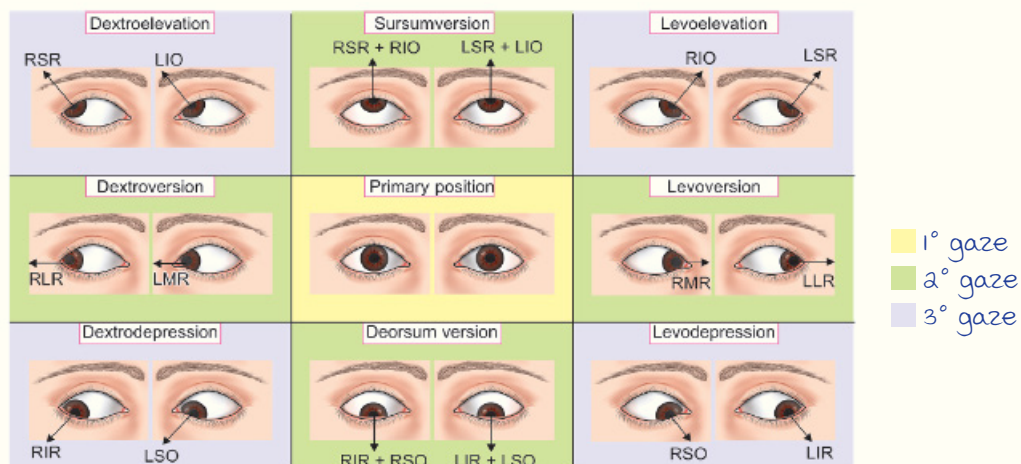
mechanisms :

- Hering's law : Equal innervation to yoke muscles.
- Sherrington's law : Reciprocal inhibition to antagonist (Relaxes).

Grades of binocular single vision (BSV) :



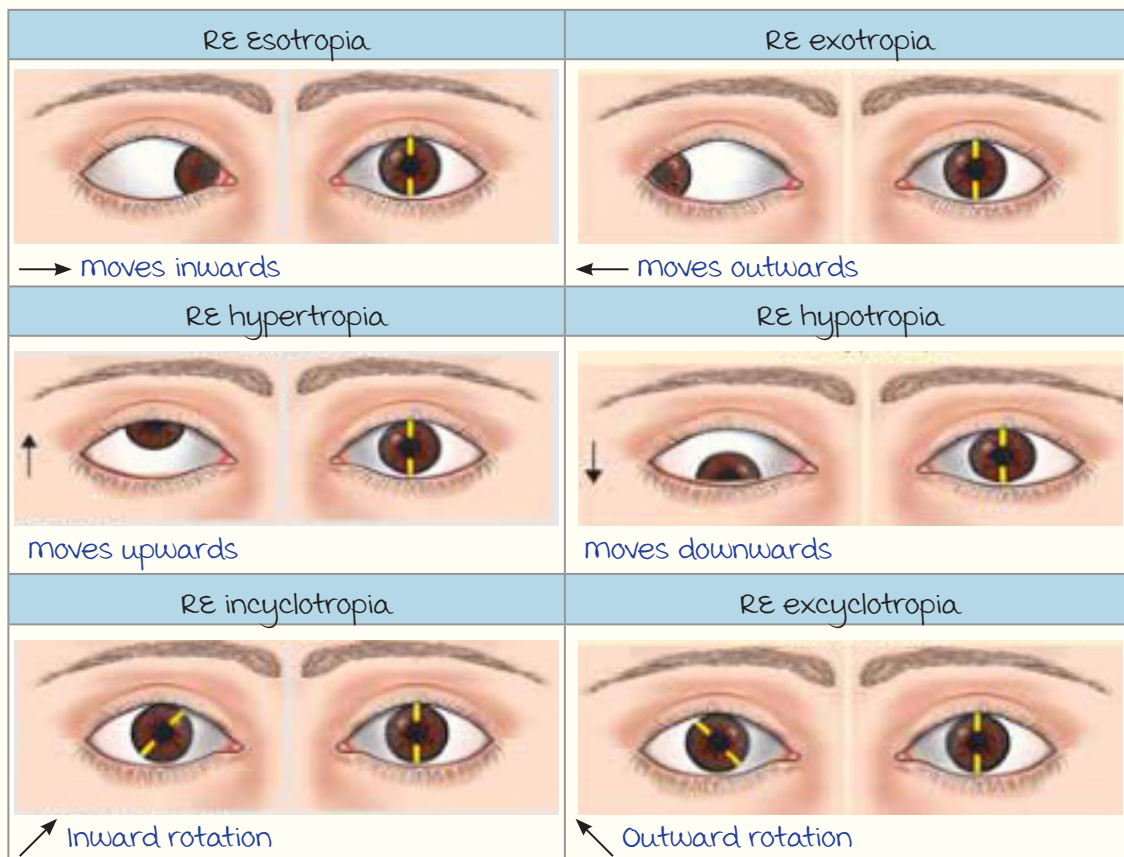
Positions of gaze & yolk muscles :



Directions of Deviation

00:17:52

----- Active space -----

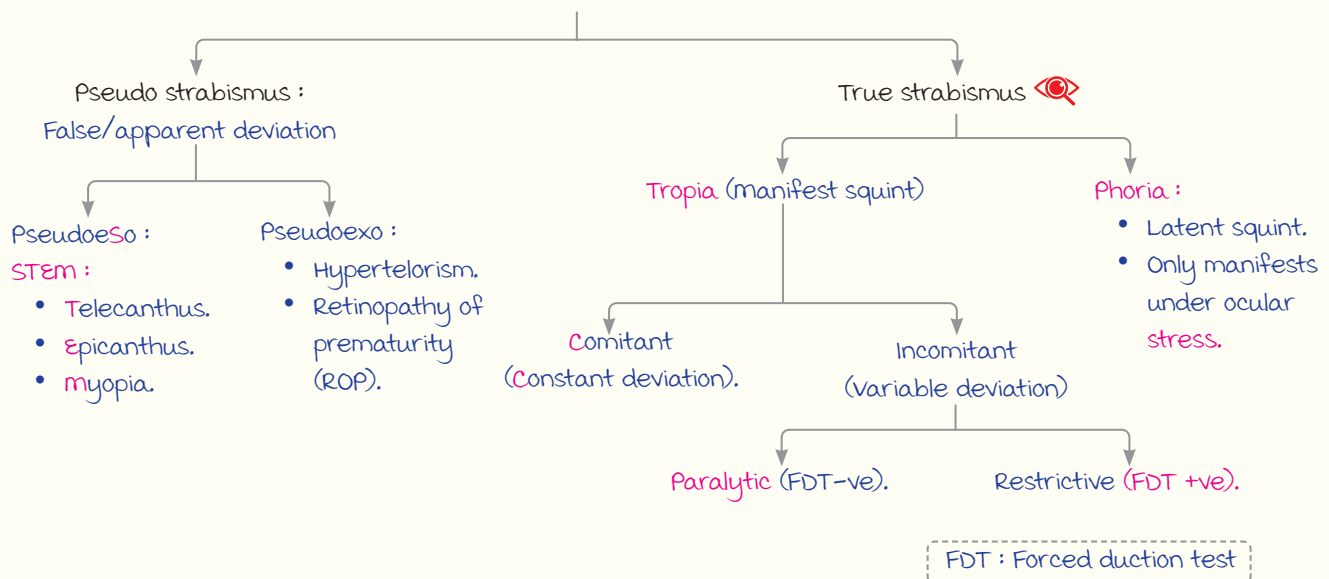


Note :

Cyclotropia is clinically not visible → Dx → using conjunctival marker.
→ Double Maddox rod test.

Types of Squint

00:20:47



----- Active space ----- Concomitant vs. incomitant squint  :

	Comitant	Incomitant
$1^\circ \text{ \& } 2^\circ$ deviation	$1^\circ = 2^\circ$	$1^\circ < 2^\circ$
Diplopia	⊖	⊕
Abnormal head posture		
Restricted ocular movements		

Investigations

00:26:00

- Ocular movement examination : In all directions of gaze.
- Calculate refractory error  : Done using retinoscopy to rule out hypermetropia (Esotropia seen in hypermetropia).
- measure fusional vergence :
 - measure strength reserve of EOM.
 - Dx : RAF ruler.
- measure ocular deviation :
 - Hirschberg's test.
 - Cover-Uncover test.
 - maddox rod.
 - PBCT (Prism bar cover test).
- measurement of diplopia :
 - Hess chart.
 - Lee's screen.
- Tests for sensory anomalies : Worth 4 dot test.



Measurement of Ocular Deviation :

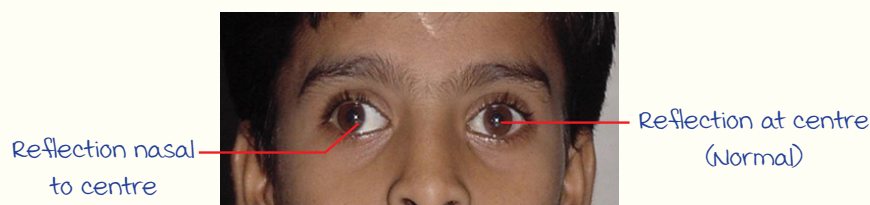
- Hirschberg's test  :

- Direction of deviation :

DOOR : Deviation Opposite Of Reflection

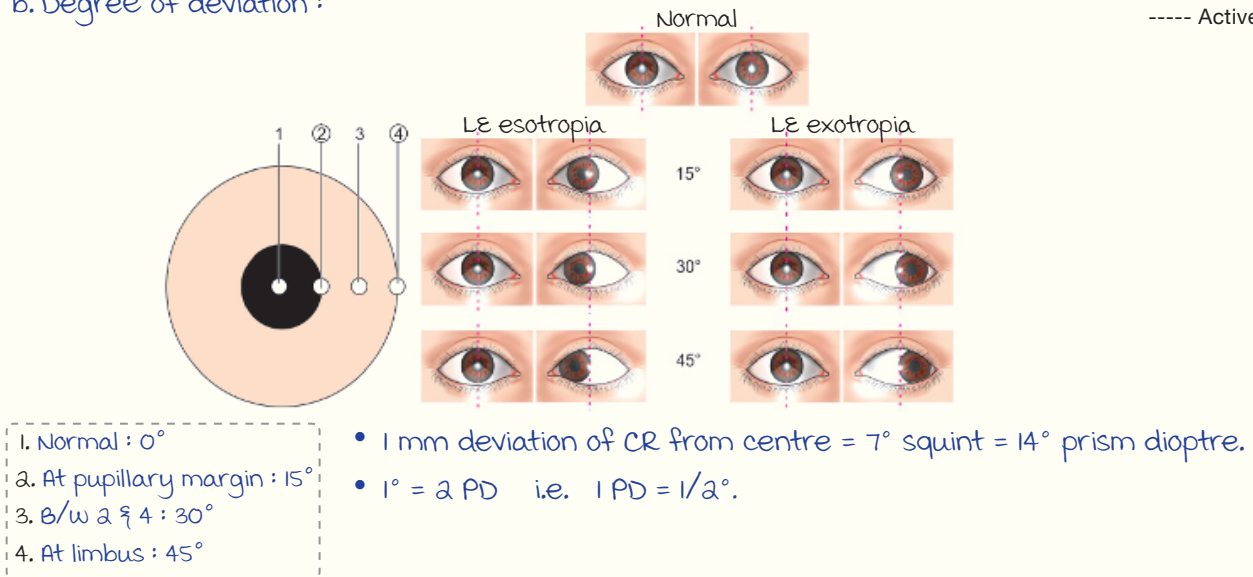


Temporal deviation (AKA : Exotropia).



b. Degree of deviation :

----- Active space -----



a. Cover & Uncover Test :

• Cover test $\rightarrow$ Tropia.• Uncover test $\rightarrow$ Phoria.

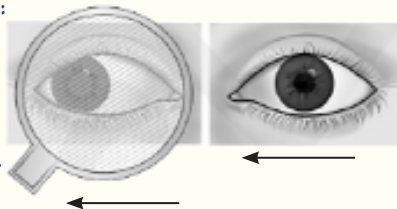
DOOM : Deviation Opposite Of movement.

a. Cover test :

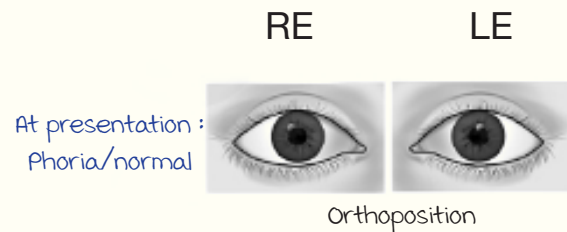
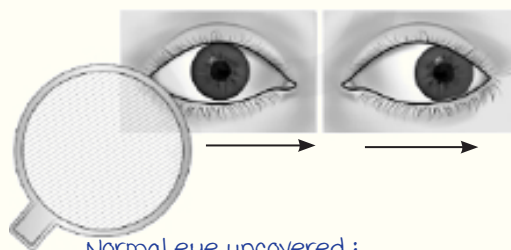
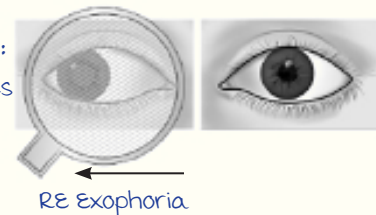


Normal eye covered :

- LE moves inwards to compensate
- No movement : Pseudostrabismus

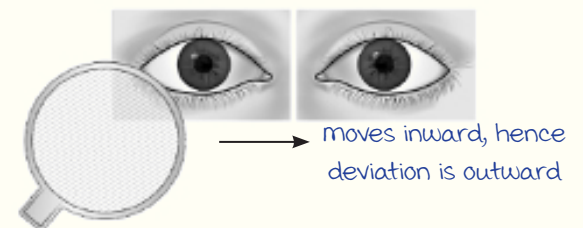


b. Uncover test :

On covering RE :
Normal eye does not move

Normal eye uncovered :

- LE returns to its original position
- LE exotropia confirmed



- RE returns to initial position on uncovering
- RE exophoria confirmed

----- Active space -----

3. Prism bar cover test (PBCT) :

- most accurate test to measure degree of tropia.
- DOOB : Deviation opposite of base of prism.



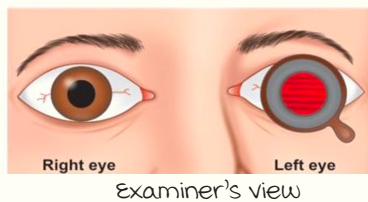
Prism bar

4. maddox rod test :

- Diagnose phoria at far fixation.
- macular function test.
- Cyclotropia measurement with double maddox rod test.



maddox rod



Patient's view

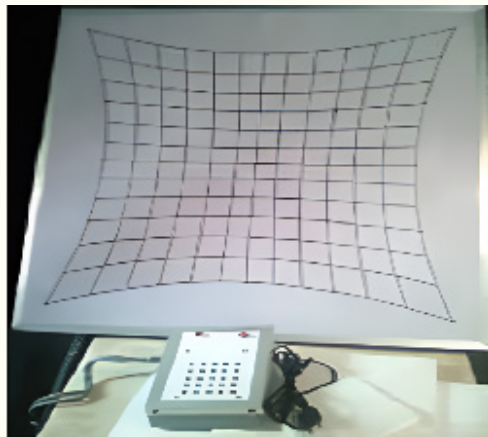
Results : (L) (R)

 → Normal (Orthophoria)

  → Crossed diplopia (Exophoria)

  → uncrossed diplopia (Esophoria)
measurement of Diplopia :

1. Hess chart :



2. Lee's screen :

Tests for Sensory Anomalies :

Differentiates b/w :

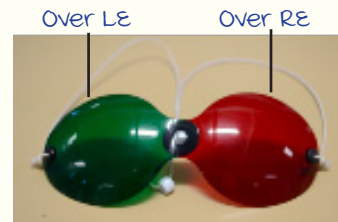
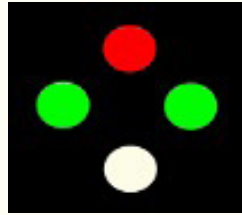
- BSV.
- Anomalous retinal correspondence (ARC) : Small angle squint.
- Suppression : Large angle squint.

Worth 4 dot test :

- Equipment :

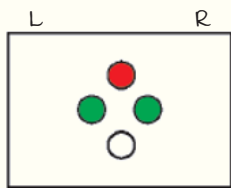
4 dots :

- 1 Red.
- 2 Green.
- 1 white.

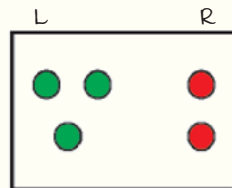


----- Active space -----

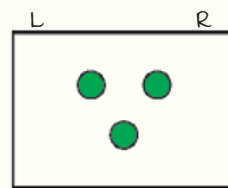
- Interpretation :



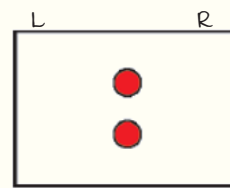
- Normal binocular single vision
- Harmonious ARC



Non-harmonious
ARC (Diplopia)



RE suppression



LE suppression

Tests for Stereopsis :

1. Titmus fly test :

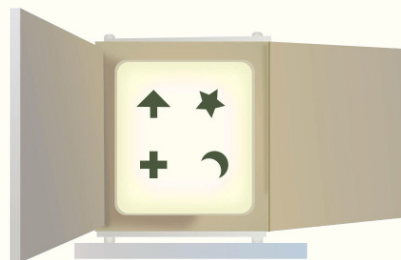


2. Random dot stereo tests :

Better → Eliminate mono-ocular cues of depth perception.



Frisby stereotest



Frisby-Davis distance stereotest

----- Active space -----

Paralytic Squint

00:49:02

- C/F : **Opposite** direction
 - Head tilt : **Same** direction
- } To action of paralysed EOM.

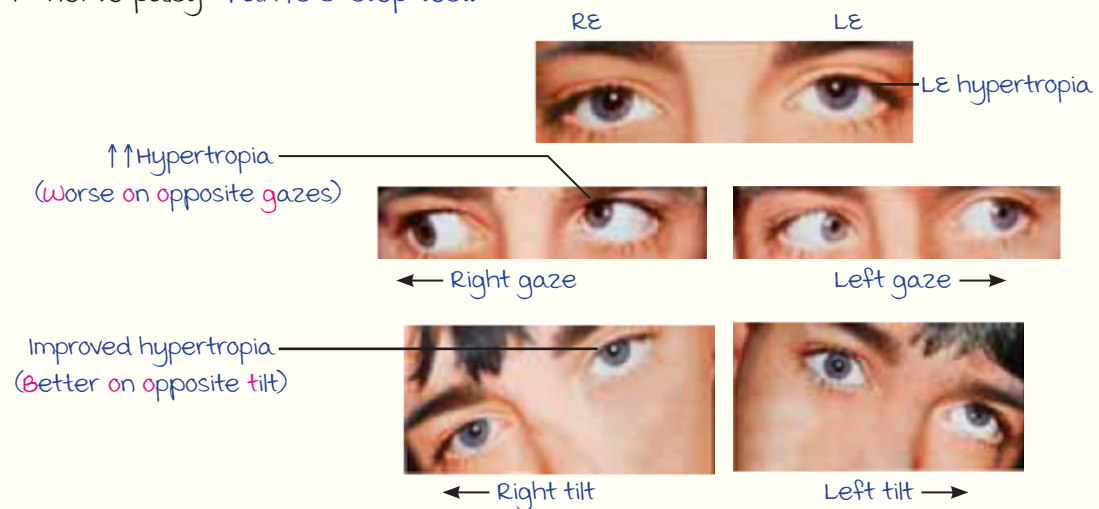
	muscles	Action	Palsy	Other C/F	mx
3 rd CN	LPS	Elevation of lid	Ptosis	Crossed diplopia	Wait and watch (CN III regenerates)
	SR, IO	Elevation	Hypotropia		
	MR	Adduction	Exotropia		
	Sphincter pupillae	miosis	mydriasis (Loss of light reflex)		
	Ciliary muscle	Accommodation	Loss of accommodation		
C/L 4 th CN	Superior oblique	Intorsion (1°)	Excyclotropia	• Chin depression • C/L Head tilt • Diplopia while coming down stairs	Confirmation of Dx : Park's 3 step test
		Depression (2°)	Hypertropia (C/L eye)		
		Depression in adduction	↑Hypertropia on adduction/opposite gaze		
6 th CN	Lateral rectus	Abduction	Esotropia	• uncrossed diplopia. • I/L face turn.	-

3rd nerve palsy :

Down & out pupil



Ptosis

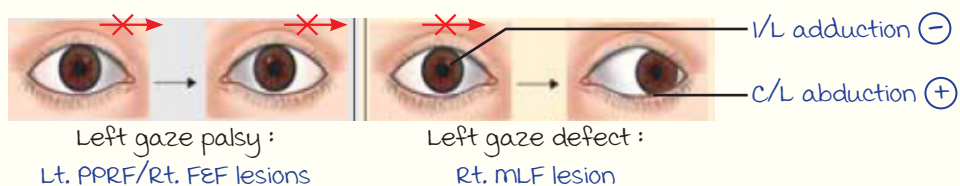
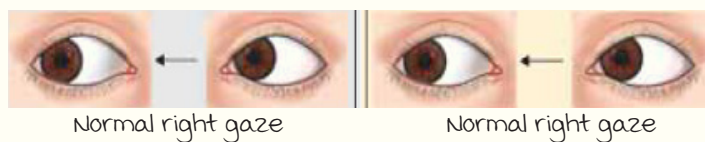
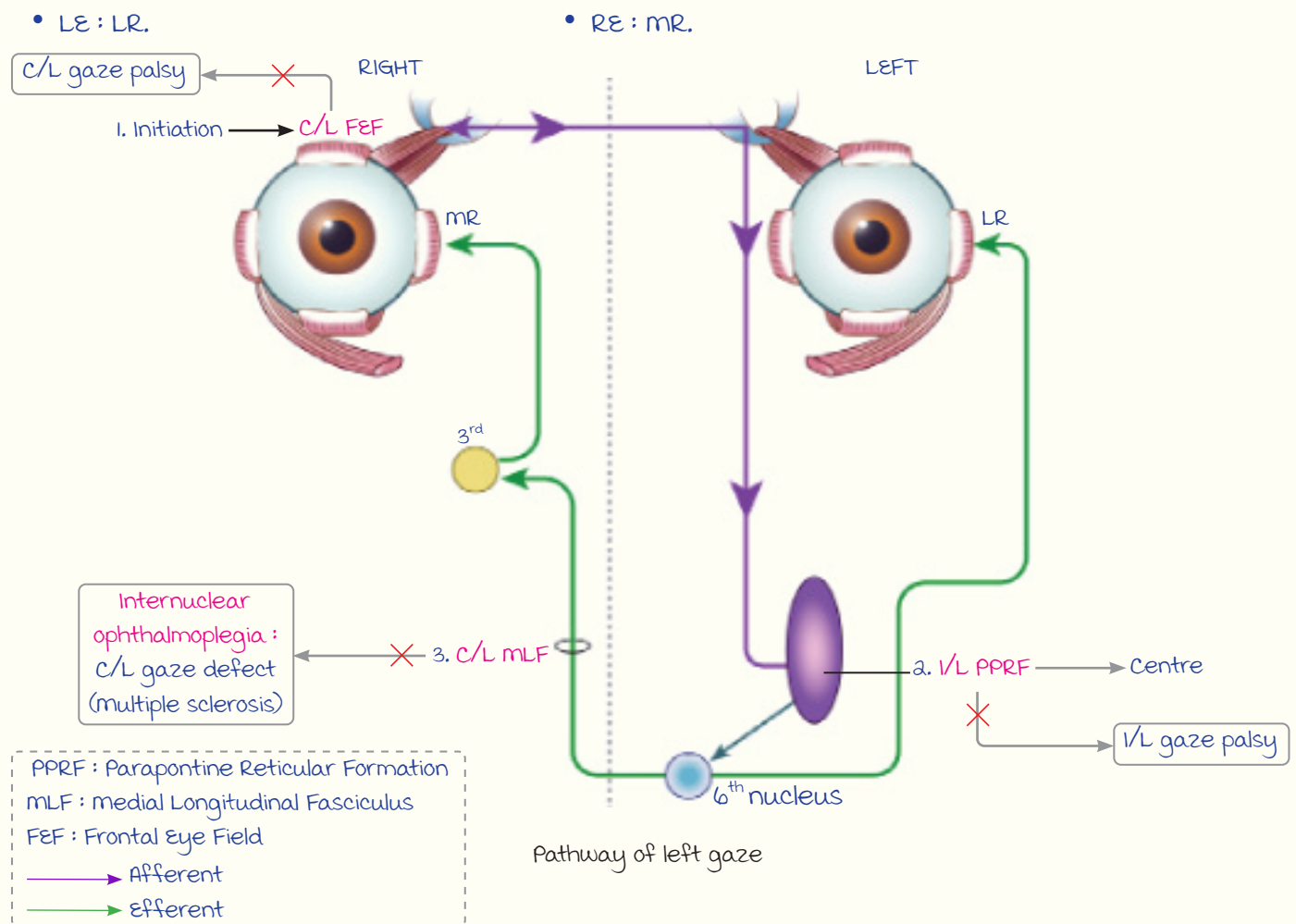
4th nerve palsy : Park's 3-step test.6th nerve palsy :

Horizontal Gaze

01:00:21

----- Active space -----

Left Horizontal Gaze :



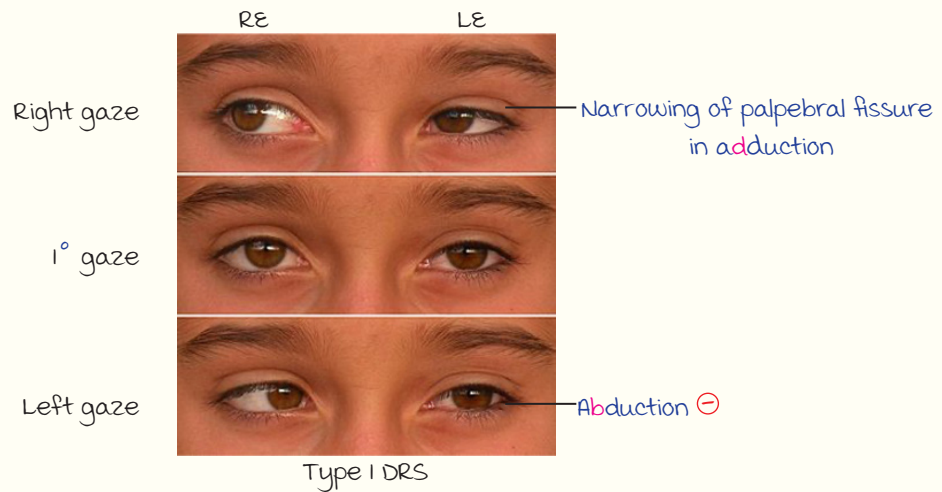
----- Active space -----

Restrictive Squint

01:06:16

Duane's retraction syndrome (DRS) :

	Limitation of movements	Palpebral fissure closure (+) Retraction of globe
Type 1	Abduction	Adduction
Type 2	Adduction	
Type 3	Abduction + adduction	

**Comitant Squint**

01:09:41

Esotropia :

m/c presentation of comitant squint.

1. Accommodative (Role of accommodation ⊕) .

	Refractive	Non-refractive
Cause	Hypermetropia (SO HYPER)	↑ accommodative convergence/accommodation ratio
Esotropia	Distant > Near	Near > Distant
Rx	Convex spectacles	Bifocal spectacles miotics : <i>ecothiophate</i>

2. Non-accommodative (Role of accommodation ⊖) .

- AKA *essential infantile esotropia*.
- 4 - 6 months of age.
- moA : Large angle squint (> 15°) → 2 eyes focus on different objects

↓
Suppression of deviated eye↓
Amblyopia (Lazy eye) of deviated eye :
Loss of vision without organic cause.

- No treatment.
- Prevention : Occlusion therapy (until 6 yrs of age)

↓

Patch normal eye for 'x' days

(x = Age of child)

↓ F/b

Patch deviated eye for 1 day.

----- Active space -----

Myasthenia Gravis

01:14:21

Clinical features 👁️ :

Ptosis :

- B/L.
- Worsens as day progresses.

Diplopia.

Cogan lid twitch sign.

Diagnosis :

Tensilon test : Definitive test.



Rt. Ptosis

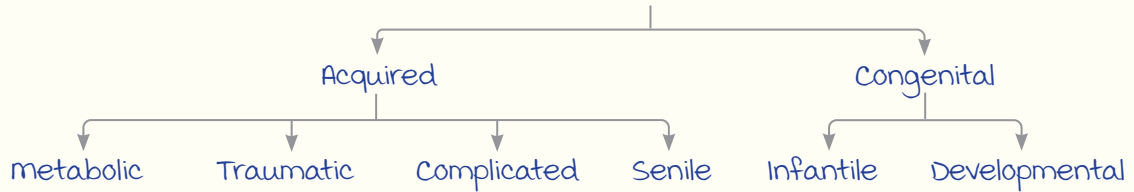
↓ Inj. Edrophonium



Improvement of ptosis

LENS AND BLUNT TRAUMA

Classification of cataract :



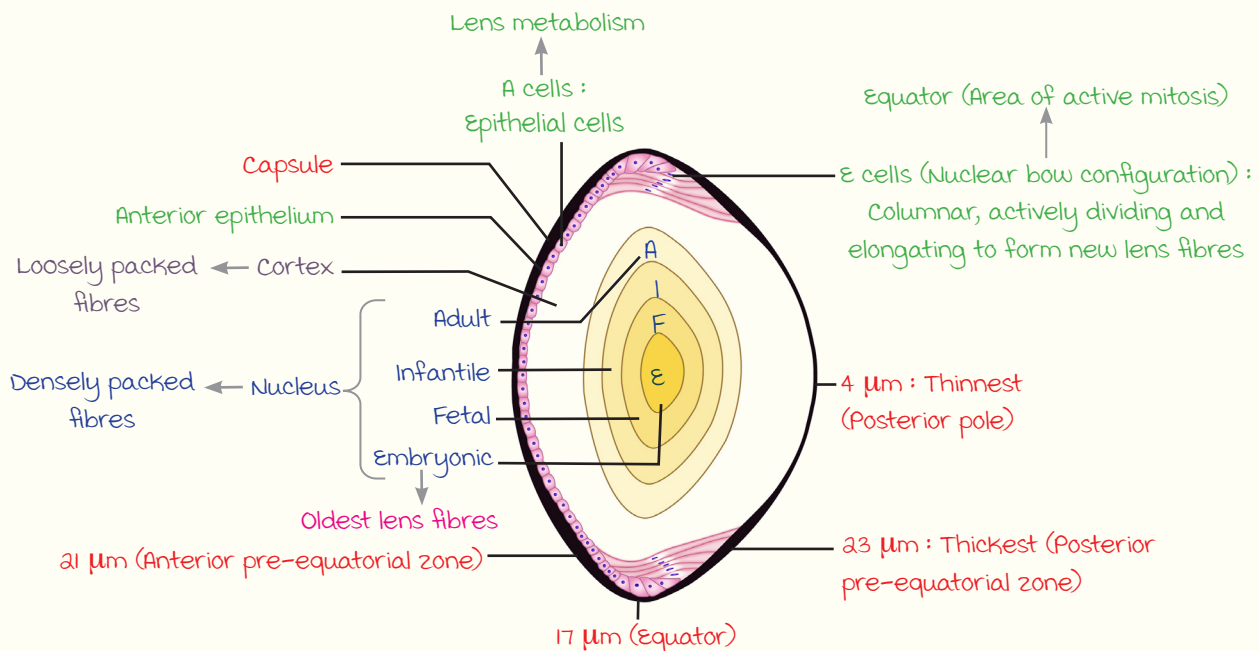
Anatomy & Metabolism of Lens

00:01:23

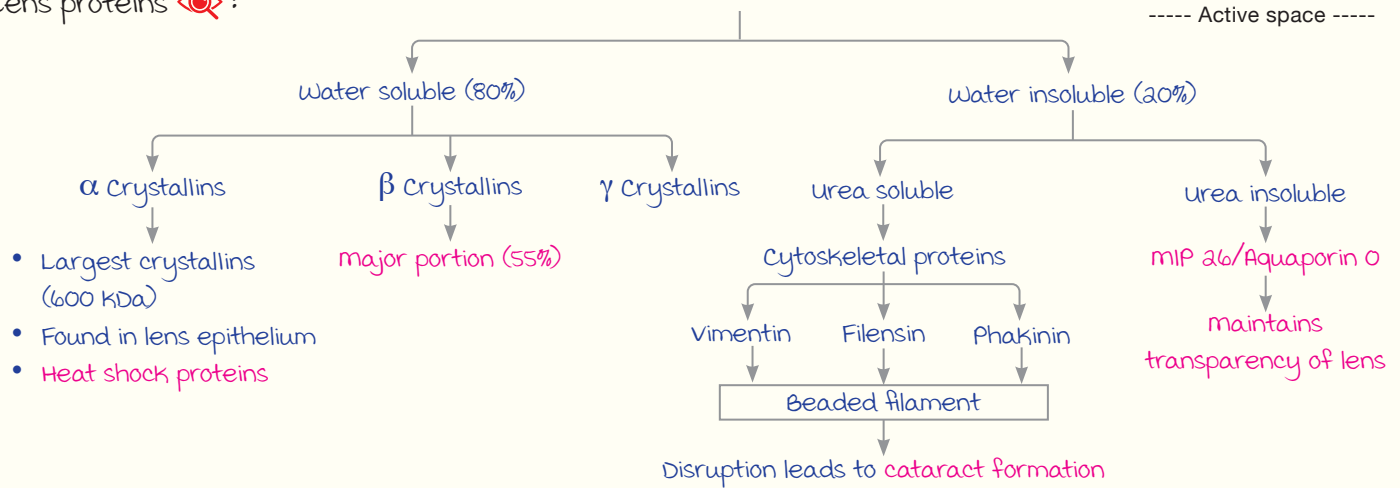
Anatomy :

- Colour : Transparent.
- Shape : Biconvex.
- Power : +16 D to +19 D
- Refractive index :
 - Center : 1.41 (max. in eye).
 - Periphery : 1.39.
- Embryological development : Surface ectoderm.
- Sutures (Fetal nucleus) :
 - Anterior (Y shaped).
 - Posterior (Inverted Y shaped).

Parts :



Lens proteins 👁️ :



Lens metabolism :

- Lens is avascular.
- Diffusion of glucose occurs via aqueous.

metabolic pathways 👁️ :

- Anaerobic glycolysis (> 80%).
 - Krebs's cycle & HMP shunt (~ 15%).
 - Sorbitol pathway (< 5%) : Glucose $\xrightarrow{\text{Aldose reductase}}$ Sorbitol $\xrightleftharpoons[\text{dehydrogenase}]{\text{Polyol}}$ Fructose.
 - Accumulate in lens
 - Cataract
- Normally inactive.
 - Activated by hyperglycemia (DM & galactosemia).

Metabolic Cataract

00:08:50

1. Parathyroid tetany :

Hypocalcemia → Children : Lamellar cataract.
 → Adults : Anterior & posterior subcapsular cataract.



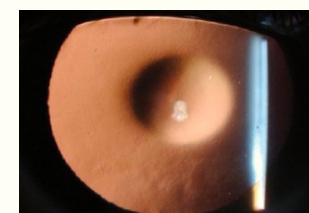
Snowflake cataract

2. Snowflake cataract :

DM → Carbamylation of crystallins → Cortical cataract.
 (Type 1 > Type 2) (Posterior > Anterior)

3. Oil droplet cataract 👁️ :

- Galactosemia → Accumulation of metabolite (Galactitol) & substrate (Galactose) → Posterior subcapsular cataract.
- Only reversible cataract.



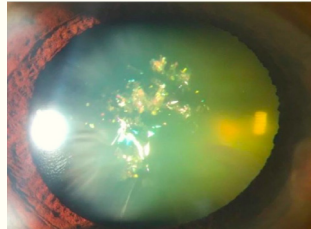
Oil droplet cataract

----- Active space -----

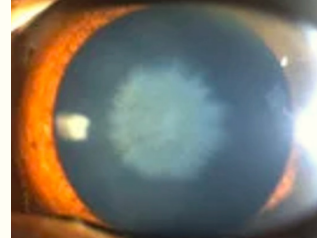
4. Sunflower cataract : D/t Wilson's disease.

5. Christmas tree cataract :

myotonic dystrophy → Posterior subcapsular cataract.



Christmas tree cataract



Sunflower cataract

Note :

Type 2 Dm → Pre-senile cataract 🦋.

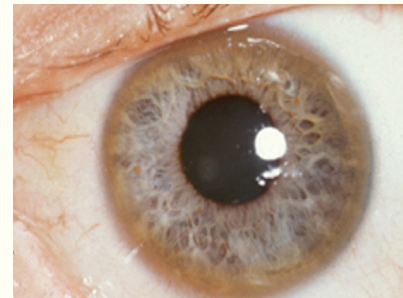
- Kayser Fleischer ring : A/w Wilson's disease.

- Golden - brown Cu^{2+} deposition in the Descemet's membrane near the limbus.

- Progression : Superior → Inferior → Lateral
↓
Circumferential.

- All cases with neurological manifestation and 50% cases with only hepatic manifestation.

- Fleischer's ring : A/w keratoconus. (D/t iron deposition).
- Pseudo - Fleischer's ring : A/w hypermetropia.



Kayser Fleischer ring

Traumatic Cataract

00:14:29

Noxious stimulus	effect	Cataract
IR rays	True exfoliation of lens capsule	Glassblower's cataract
Lightning/electric shock	Anterior capsular opacities	Unnamed

Blunt Trauma :

Contusion injury → Posterior subcapsular : stellate/rosette shaped cataract
(Tennis ball/Fist injury)

Signs :

1. Hyphema 🦋 (m/c) :

- Bleeding from major arterial circle of iris → Collection of blood in anterior chamber (AC).

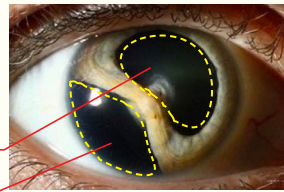
- Rx : Wait & watch (Head end elevation) $\xrightarrow{\text{Not resolving}}$ AC wash.



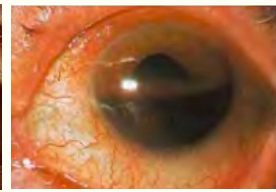
Stellate cataract

2. Iris signs :

- Iridodonesis : Trembling iris.
- Iridodialysis : Detachment of root of iris from ciliary body.
 - D shaped pupil.
 - Diplopia d/t 2nd pupil.



Iridodialysis

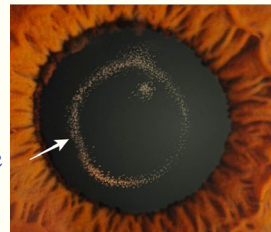


Hyphema

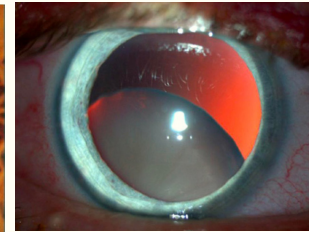
----- Active space -----

3. Lens signs :

- Phacodonesis : Trembling lens.
- Rosette shaped cataract.
- Vossius ring : Ring of pigment (d/t miotic pupil striking anterior surface of lens).
- Ectopia lentis : Subluxation of lens.



Vossius ring



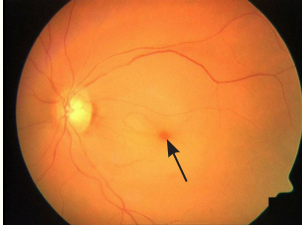
Ectopia lentis

4. Fundus/retinal signs :

- Cherry red spot at macula : d/t fluid accumulation in retina (Berlin's edema).
- Retinal detachment (RD) : m/c rhegmatogenous RD.
- Commotio retinae : Pale fundus.
- Traumatic optic neuropathy : Causes 1° optic atrophy.

5. 2° open angle glaucoma : Angle recession glaucoma.

Note :

Cherry red spot at macula	
mnemonic : Cherry Trees Never Grow Tall in Sand, mud & Grime	
• Central retinal artery occlusion (CRAO) • Trauma (Blunt) • Niemann - Pick disease • GM₁ gangliosidoses • Tay - Sachs disease • Sandhoff's disease • metachromatic leukodystrophy • multiple sulfatase deficiency • Gaucher's disease (Type 2 only) : Least common	
GM₁ gangliosidoses GM₂ gangliosidoses	

----- Active space -----

Complicated Cataract

00:24:25

Causes : mnemonic **UMAR**.

1. **U**veitis (m/c).
2. **M**yopia.
3. **A**ngle closure glaucoma.
4. **R**etinitis pigmentosa.

Appearance : **Breadcrumb + polychromatic lustre**.

Site : **Posterior subcapsular cataract**.

Senile Cataract

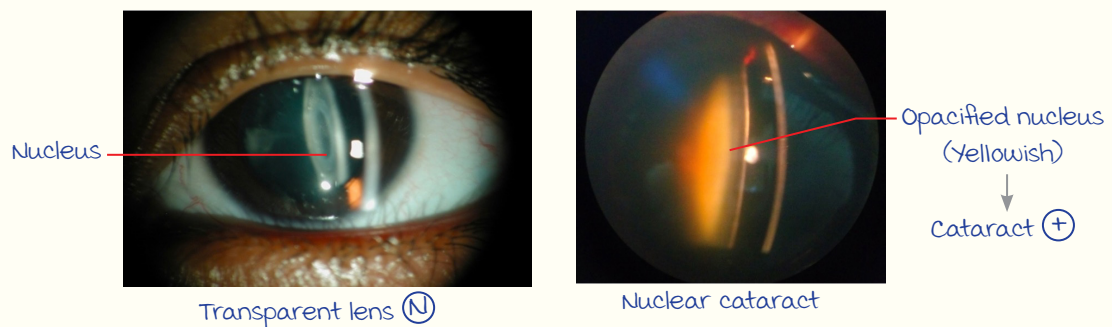
00:25:45

Age group : **50 – 70 years**. (mean : 60 yrs).

Types (Based on location) :

- a. **Nuclear**.
- b. **Cortical**.
- c. **Posterior subcapsular**.
- d. **Anterior subcapsular**.

Nuclear/Central Cataract :



Slit lamp oblique illumination

Pathogenesis : **Sclerosis of lens** → **↑ Refractive index**.

Clinical features :

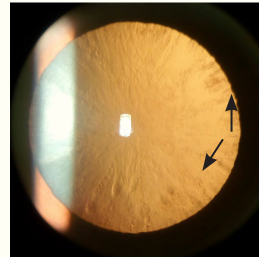
- **Index myopia** (Near - sightedness).
- **Second sight phenomenon** : Improved near vision.
- **LOV : Day > Night** (D/t central location)
 - Pupil dilates in low light → vision improves.
- **Xanthopsia** : Yellow colored vision d/t discolored lens.

Cortical Cataract :

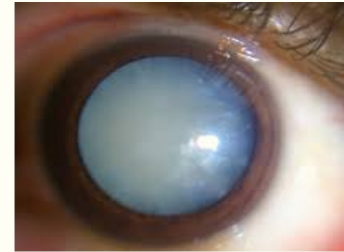
----- Active space -----

Involves periphery → Nyctalopia.

Pathogenesis 👁 :

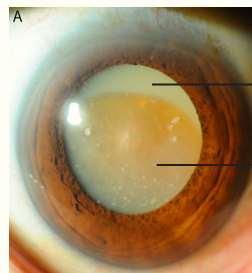


Cuneiform cataract
m/c inferonasally



Intumescent cataract :
Complications :

- Phacomorphic glaucoma (m/c)
- Intraoperatively →



Morgagnian/
Hyper-mature cataract
m/c complication :
Phacolytic glaucoma



Argentinian flag sign

Posterior Subcapsular/Cupuliform Cataract :

Causes : mnemonic → MNC Repay GST.

- Myotonic dystrophy.
- Neurofibromatosis type 2.
- Complicated cataract.
- X-Rays.
- Galactosemia.
- Steroids.
- Trauma (Blunt).

Clinical features 👁 :

- Glare → Difficulty in driving at night (m/c).
- Max. LOV (Near > Far ; Day > Night).

Anterior Subcapsular Cataract :

Rarest type.

Causes : AACG

- Acute Angle Closure Glaucoma.
- Amiodarone
- Atopic dermatitis → Shield cataract.
- Chlorpromazine
- Gold deposits.

----- Active space -----

Surgical Management of Cataract

00:36:48

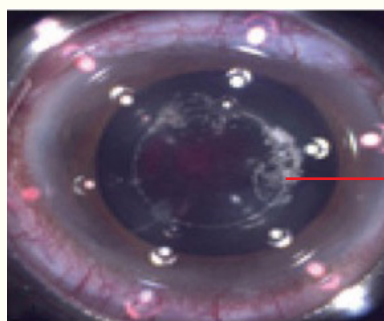
- ICCE (Intracapsular cataract extraction) : whole lens removed (No longer used).
- ECCE (Extracapsular cataract extraction) : Lens removed, capsule retained.

Techniques of ECCE :

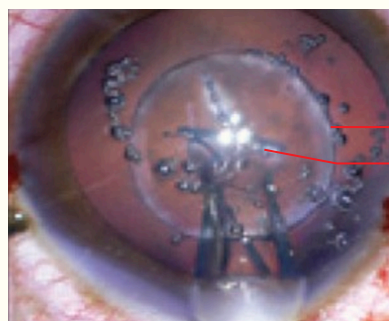
	Site of Incision	Size of Incision
1. Conventional ECCE	Limbus	8 - 10 mm
2. Small incision cataract surgery (SICS)	Sclera (Sclera - corneal tunnel)	5 - 7 mm
3. Phacoemulsification	Clear corneal	2.2 - 3.2 mm

4. FLACS (Femtosecond laser assisted cataract surgery) 👁 :

- Femtosecond : 10^{-15} sec.



CCC



Corneal incision

CCC

Nuclear

fragmentation

Steps
by laser

Nd : Glass laser

Principle → Photodisruption

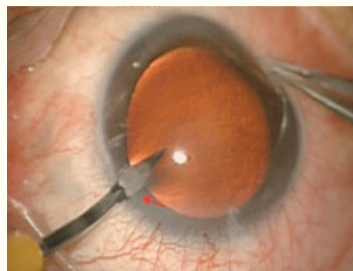
CCC : Continuous curvilinear capsulorhexis

Phacoemulsification :

Principle :

Piezoelectric crystal at tip of phacoprobe $\xrightarrow{\text{converts}}$ ultrasonic energy $\xrightarrow{\text{to}}$ mechanical energy
(To & fro motion).

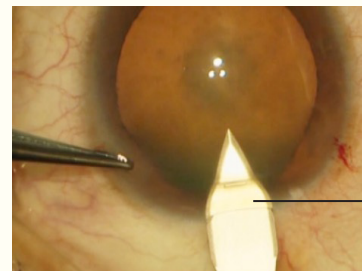
Steps :



1. Side port incision
using MVR blade

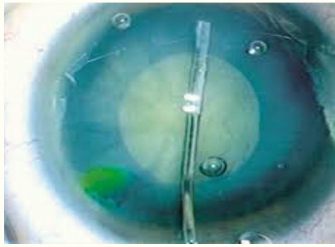


2. Staining the capsule + CCC 👁
Dye : Trypan blue

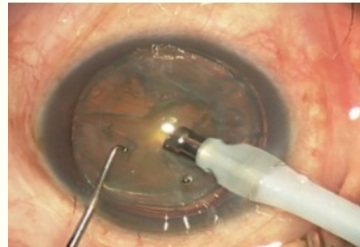


3. main incision

Keratome
(2.2/2.8 mm)



4. Hydrodissection
Separation of cortex & capsule



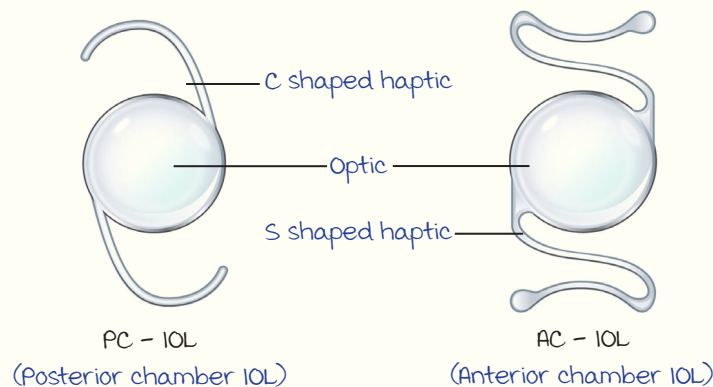
5. Nuclear fragmentation

6. Irrigation & aspiration
7. IOL implantation :
In empty capsular bag

----- Active space -----

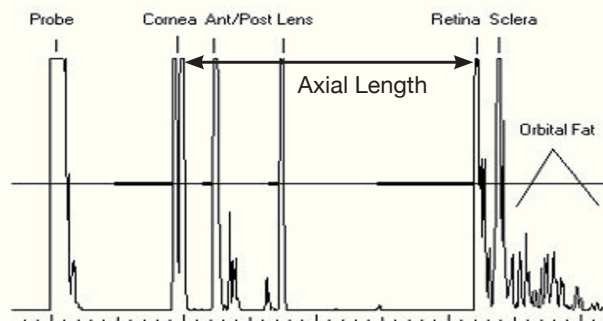
Intra - Ocular Lens (IOL) :

Types of IOL :



Biometry : Calculations of power of IOL to be implanted.

1. Keratometry : Corneal curvature.
2. A-scan USG : Axial length/AP diameter of eye (N AL ~24 mm).



3. IOL power calculation using SRK-T formula.

Complications of Cataract Surgery

00:45:36

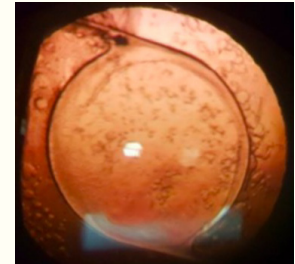
Operative :

- Posterior capsular rupture (PCR) → Lens falls into vitreous.
- uveitis - glaucoma - hyphema (UGH) syndrome → D/t ACIOL.
- vitreous loss (most serious).
- Expulsive choroidal hemorrhage : Bleeding from choroidal artery.
- Descemet membrane detachment : Reattached by injecting air bubble or gas in AC → Descemetopexy.
- Superior rectus damage (D/t bridle sutures).

----- Active space -----

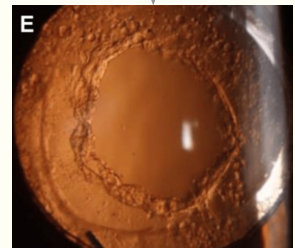
Postoperative :

- **Posterior capsular opacification (m/c)** 👁️ : AKA 2° cataract/After cataract.
 - Occurs 6 - 12 months postop.
 - Types
 - Elschning's pearls (90%).
 - Sommering's ring.
- Retinal detachment.
- **Irvine Gass syndrome :**
 - 6 - 10 weeks postop.
 - Cystoid macular edema + bullous keratopathy.
- Displacement of IOL :
 - Sunset syndrome : Inferior subluxation.
 - Sunrise syndrome : Superior subluxation.
 - Lost lens syndrome : Complete dislocation.
- Toxic anterior segment syndrome (TASS).
- **Flat/shallow AC :**
 - Wound leak.
 - Ciliary choroidal detachment.
 - Pupillary block glaucoma.
 - malignant glaucoma.



Elschnig's pearls

Rx

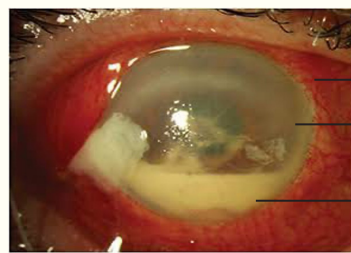


Nd YAG laser posterior capsulotomy

Endophthalmitis 👁️ :

- **Suppurative** inflammation of all structures **except sclera**.
- Causes :

Type	m/c pathogen
Early postop (< 7 days)	CONS (Staphylococcus epidermidis)
Late postop(> 6 weeks)	Propionibacterium acnes
Traumatic	B. cereus
Endogenous	Klebsiella, Candida



Congestion

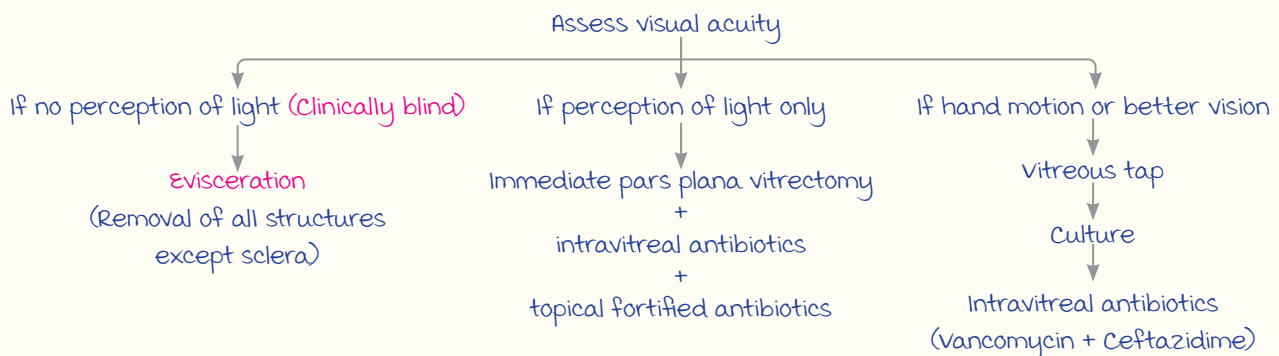
Hazy cornea

Hypopyon

Anterior segment involvement

Presentation of endophthalmitis

- C/f : Pain, redness, LOV, vitritis (Posterior segment involvement).
- Rx :

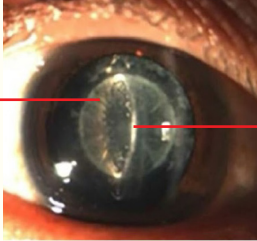
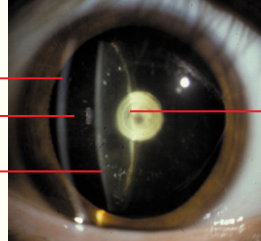


Congenital Cataract

00:54:23

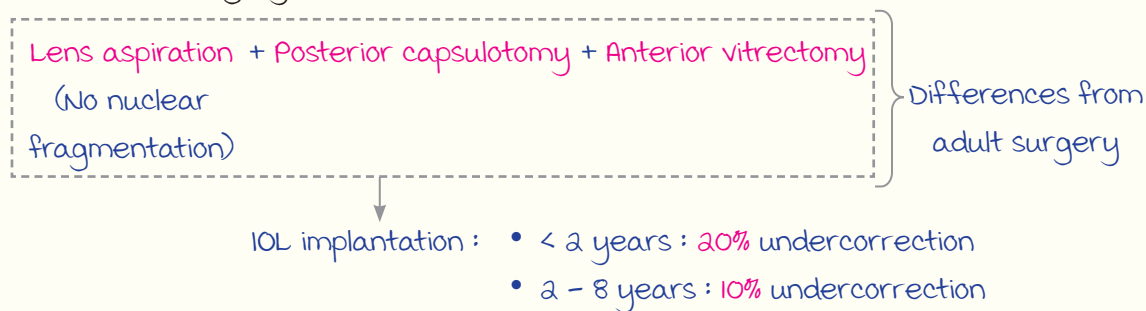
----- Active space -----

Types :

	Infantile cataract 	Developmental cataract
Age at presentation	< 1 yrs	> 1 yrs
LOV	Severe (D/t interference with foveal fixation)	minimal
Types	Lamellar/zonular cataract (m/c)  Riders (Spoke like opacities) Opacity in fetal nucleus	- Blue dot/Punctate cataract (m/c) : A/w Down's syndrome - Posterior polar cataract : Onion ring/whorled appearance  Cornea AC Anterior capsule Opacity at the centre of posterior capsule
Rx	Immediate surgery	-

Treatment :

ECCE based surgery :



Miscellaneous

00:58:41

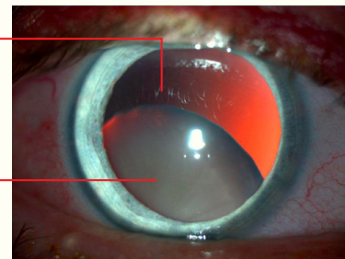
Ectopia Lentis :

Causes :

- Blunt trauma (m/c).
- marfan's syndrome (Superotemporal subluxation).
- Homocystinuria (Inferonasal subluxation).
- Weill-marchesani syndrome (Anterior subluxation).

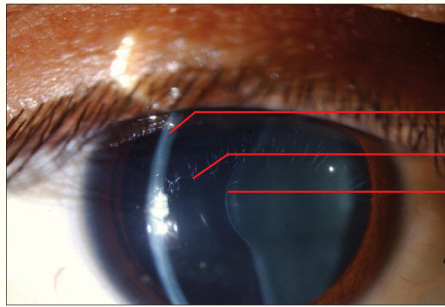
o/e : Golden red crescent in pupillary area

Inferiorly displaced lens



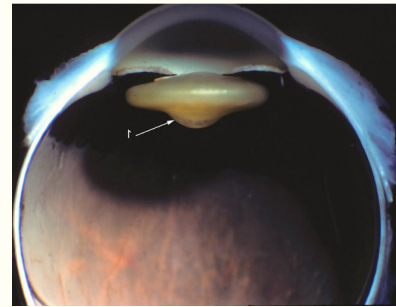
Clinical features : Diplopia d/t differential refractive power.

----- Active space -----

Lenticonus :

Anterior lenticonus

Causes : Spina bifida, Alport syndrome,
Waardenburg syndrome



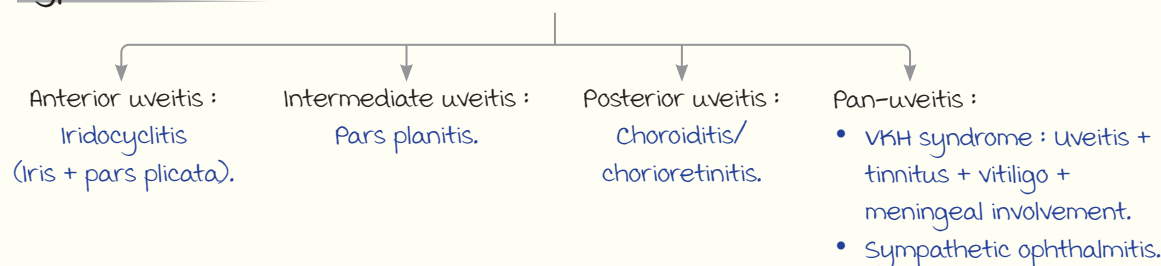
Posterior lenticonus

Cause : Lowe syndrome

UVEA

----- Active space -----

Types of uveitis :

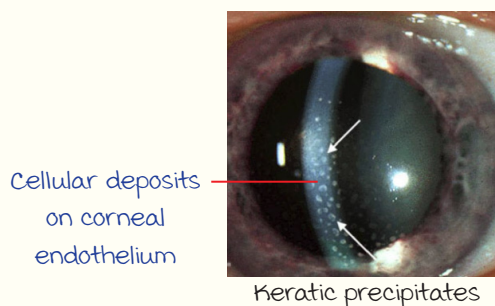


Anterior Uveitis

00:01:47

Clinical Features :

- Pain.
- Redness.
- LOV (Loss of vision).
- Lacrimation.
- Photophobia.



Signs 👁 :

1. Keratic precipitates (KP) :



2. Iris nodules :

- Koeppe nodules : At pupillary margin.
- Busacca nodules : Iris stroma.

3. Pupil 👁 :

- Acute : miosis.
- Chronic : Posterior synechiae → Festooned pupil (Irregularly dilated).

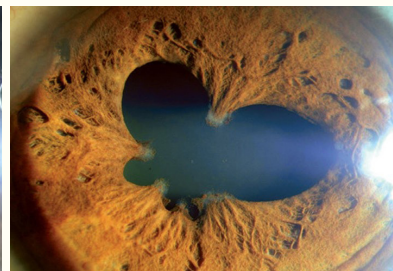
----- Active space -----



Koeppe nodules



Busacca nodules



Festooned pupil

4. Aqueous :

- Cells : marker of disease activity.
- Flare : Leakage of proteins → Turbidity (visible by Tyndall effect).

5. Occlusio pupillae :

Whitish membrane over pupil.

6. Cyclitic membrane :

membrane over ciliary processes → Hypotony.



Occlusio pupillae

Treatment :

Topical steroids + atropine (Relieves pain d/t cycloplegia).

Complications :

- Band shaped keratopathy.
- Complicated cataract.
- Glaucoma :
 - Acute : 2° OAG (Latanoprost c/i).
 - Chronic : 2° ACG.
- Phthisis bulbi.

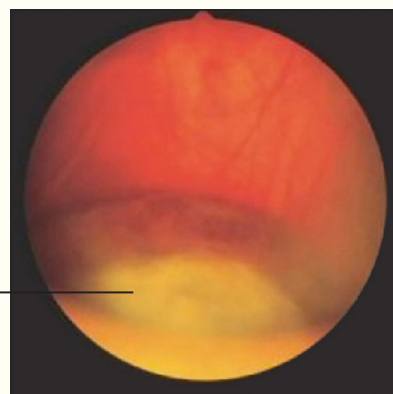
Intermediate Uveitis

00:09:15

m/c cause : Idiopathic.

Clinical Features :

- Snowbanking sign.
- Snowball opacities.
- Cystoid macular edema.
- Spillover anterior uveitis.

Coalescence of
snowball opacities

Snowbanking sign

Treatment :

----- Active space -----

Posterior sub-tenon steroids → Systemic steroids → Immunomodulator therapy → Pars plana vitrectomy.

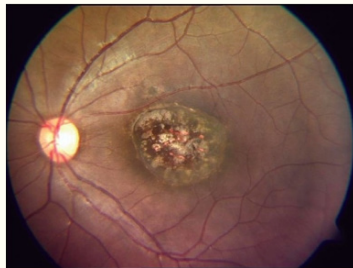
Posterior Uveitis

00:11:09

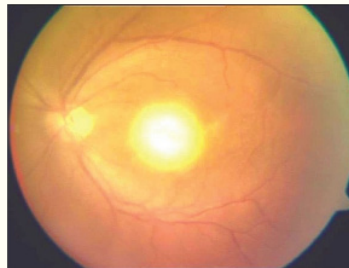
Toxoplasmosis :

m/c cause.

Signs :



Punched out scar



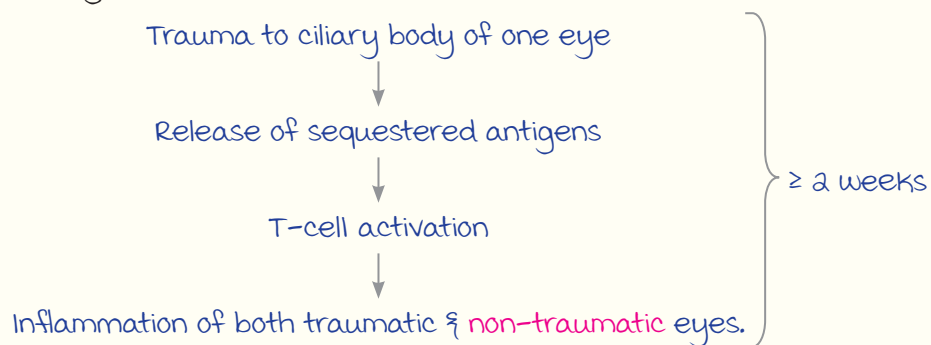
Headlight in fog appearance

Panuveitis

00:12:15

Sympathetic Ophthalmitis :Cause  : Penetrating trauma to ciliary body.

Pathogenesis :



Prevention : Enucleation of traumatic eye within 2 weeks.

Removal of whole eyeball
+ optic nerve

GLAUCOMA

----- Active space -----

Pathophysiology & Types

00:01:05

Pathophysiology :

- D/t $\uparrow$ production or $\downarrow$ clearance of aqueous.
- Optic neuropathy :
 - Optic disc changes : Progressive, **characteristic**.
 - Visual field defects : **Irreversible**.
 - IOP : $\uparrow$ / Normal (11 - 21 mmHg).

Flow of aqueous :

Trabecular outflow (90%) :

Trabecular meshwork

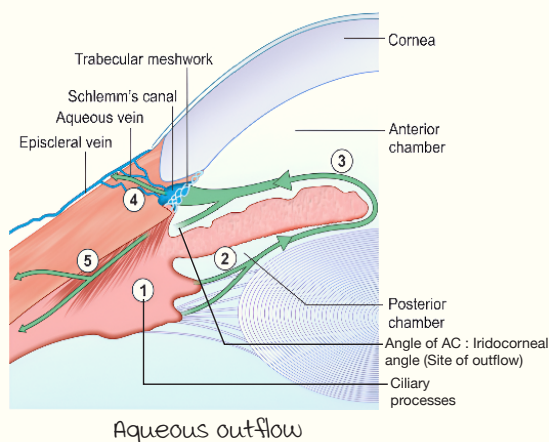
Schlemm's canal

Aqueous vein

Episcleral vein.

uveo scleral outflow

Drains into supra ciliary space.



Aqueous outflow

Types :

	Open Angle Glaucoma (OAG)	Angle Closure Glaucoma (ACG)
Angle b/w iris & cornea	Wide	Narrow
Pathology	Blockage of trabecular meshwork	Anterior displacement of iris
Anterior chamber	Deep	Shallow
Rx	• Trabeculoplasty • Trabeculectomy	• Iridectomy • Iridotomy

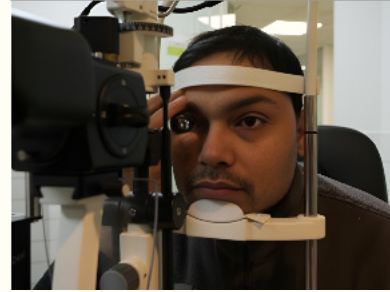
Investigations

00:07:03

----- Active space -----

Optic Disc Examination :

- Direct ophthalmoscopy.
- Slit lamp biomicroscopy : + 90 D convex lens used.
- Confocal scanning laser tomography (CSLO).
- Spectral domain OCT.
- Confocal scanning laser polarimetry (SLP).



Slit lamp biomicroscopy

Visual Field Examination :

Perimetry :

Kinetic :

- Goldmann perimeter
- Bjerrum tangent screen

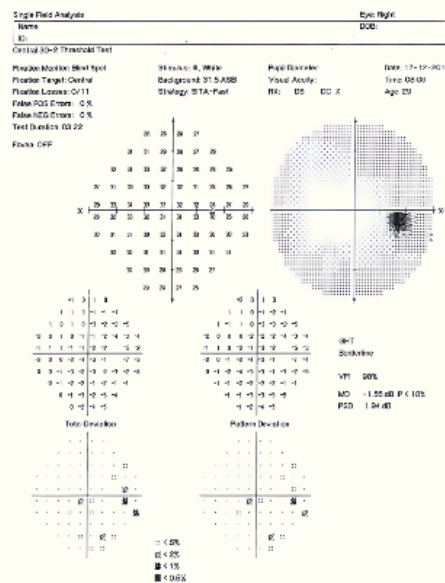


HFA machine

Static (Preferred) :

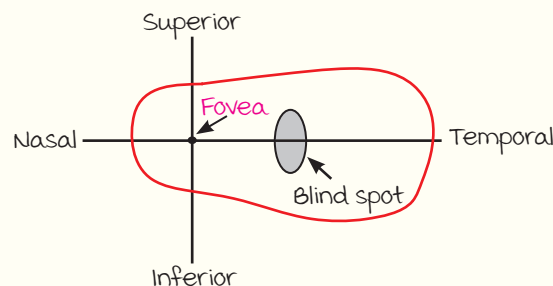
- Humphrey field analyzer (HFA)
- Octopus static perimeter
- Dicon

Output →



Normal visual field :

- Horizontally oval with an infero-nasal notch.
- Superior : minimum.
- Temporal : maximum.



----- Active space -----

IOP measurement/Tonometry :

1. Indentation tonometry :



Schiøtz tonometer
Not preferred d/t variability



Rebound tonometer
Choice for self-monitoring

2. Non-contact tonometry : Choice for screening in camps.

3. Transpalpebral tonometry (Diaton, proview) : Visualized through eyelid
(Not commonly used).

4. Applanation tonometry (AT) 👁 : Best

- Principle : Imbert Fick law $\rightarrow \text{Pressure} = \frac{\text{Force}}{\text{Area}}$
- Types :

Fixed area :

1. Goldmann's :

- Gold standard
- Cobalt blue light + fluorescein dye used
- Area of applanation : 3.06 mm^2

2. Perkin's : Handheld $\rightarrow$ Preferred in children / bedridden patients

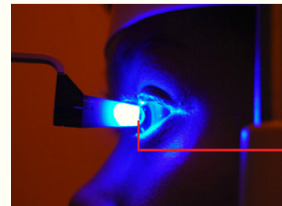
3. Mackay-Marg : Irregular, edematous cornea

4. Pascal : Dynamic contour testing post-LASIK

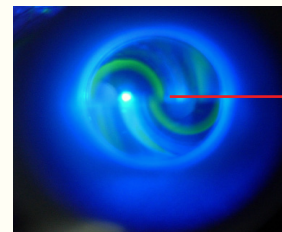
5. Pneumatic

6. Draeger

Fixed force (Not used)



Touches the cornea



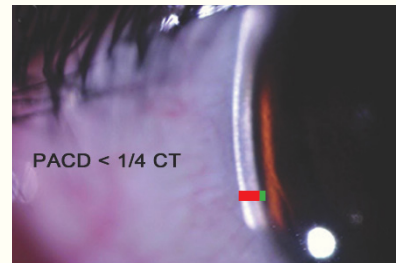
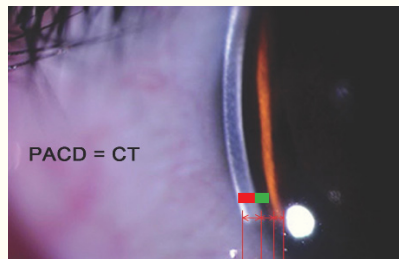
Inner margins of both mires should touch

AT rings/mires

Examination of Angle of Anterior Chamber :

1. Van Herick's method 👁 :

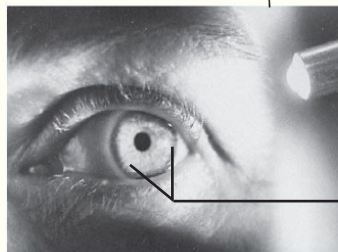
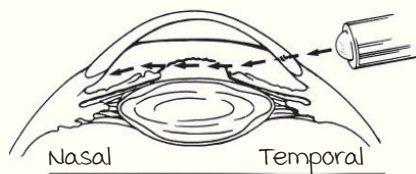
- Comparison of peripheral anterior chamber depth (PACD) & corneal thickness (CT).
- Performed in slit lamp.
- $\text{PACD} = \text{CT}$: Deep AC $\rightarrow$ OAG.
- $\text{PACD} < 1/4 \text{ CT}$: Shallow AC $\rightarrow$ ACG.



■ : CT
■ : PACD

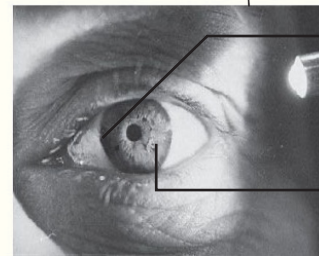
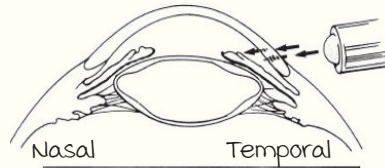
----- Active space -----

2. Oblique Flashlight Test :



Open angle

Temporal
& nasal iris
illumination



Shadow
over
nasal iris

Temporal iris
illumination

Shallow AC → Angle closure glaucoma

3. Gonioscopy 🧐 : Best.

- Principle : Overcomes total internal reflection.
- c/i : Dilated pupil.
- Structures seen :

mnemonic : Can't see this stuff.

Posterior Ciliary body

Scleral spur

Trabecular meshwork

Anterior Schwalbe's line (Peripheral termination of descemet's membrane).



4 mirror Gonioscope

Other methods :

- Ultrasonic biomicroscopy (UBM).
- Anterior segment ASOCT.

Primary Open Angle Glaucoma (POAG)

00:23:53

Features 🧐 :

- Chronic disease.
- Family history ⊕
- Age : > 50 yrs.
- Genes : Optineurin, WDR 36, MYOC.

----- Active space -----

Symptoms 👁 :

- Headache.
- Delayed dark adaptation.
- Frequent change in **presbyopic spectacles**.

Double angulation of blood vessel



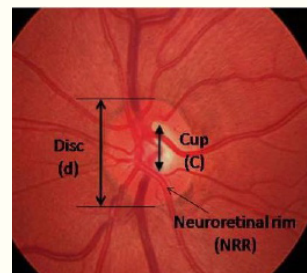
Bayonetting sign

Signs :

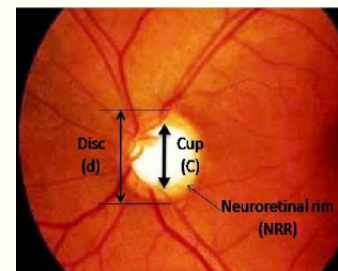
IOP : > 21 mmHg OR > 5 mmHg difference b/w both eyes.

OD changes:

1. Loss of neuroretinal rim.
2. Vertical cupping (c/d ratio > 0.5).
3. Bayonetting sign.
4. Laminar dot sign.
5. Peripapillary atrophy (PPA)



Normal disc
(c/d ratio = 0.3)



Glaucomatous changes

α zone : **Specific**.

β zone : Also seen in myopia

(As temporal myopic crescents).

visual field changes 👁 :

Earliest defect

→ **Isopter contraction**
(Constriction of peripheral VF)

↓
Baring of blind-spot

↓
Paracentral scotoma

(Earliest clinically visible manifestation)

↓ Connects with blind spot

Seidel's scotoma :

Comma shaped

↓ Arches to opposite side

Bjerrum/Arcuate scotoma

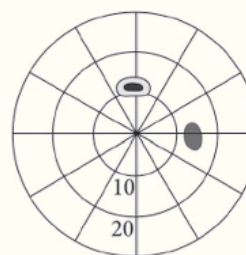
↓
Ring/double arcuate scotoma

↓
Roenne's central/peripheral nasal step
(D/t unequal contraction of upper & lower hemispheres)

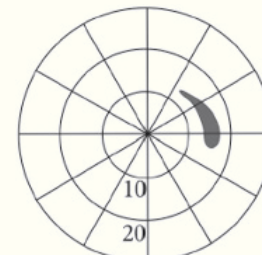
↓
Loss of central vision

↓
Loss of **temporal island vision**.
(Last to be lost)

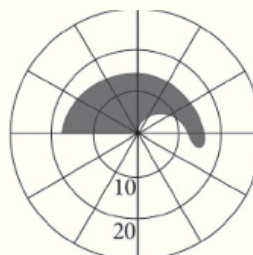
Occur in **Bjerrum's area** affecting arcuate nerve fibres.



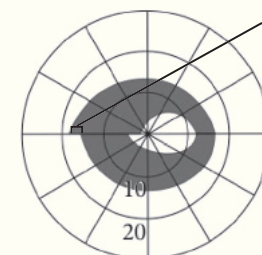
Paracentral scotoma



Seidel's scotoma



Bjerrum/Arcuate scotoma



Ring/Double arcuate scotoma

→ Roenne's nasal step

Treatment :

----- Active space -----

medical mx : Topical drugs.

Drug		MOA	S/E
Prostaglandin $F_{2\alpha}$ ($PGF_{2\alpha}$) Latanoprost (DOC) : OD dosage (Around 9pm)		$\uparrow$ uveoscleral outflow	- Hypertrichosis ($\uparrow$ Growth of eyelashes) - Hyperpigmentation of iris - Cystoid macular edema - Uveitis
β blockers : Timolol (Non-selective β blockers)		$\downarrow$ Aqueous production	- Corneal anesthesia - Nasolacrimal duct (NLD) blockage - Blepharoconjunctivitis
α -agonists	Apraclonidine	$\downarrow$ Aqueous production + $\uparrow$ Trabecular outflow	- Eyelid retraction - Mydriasis - Follicular conjunctivitis
	Brimonidine	$\downarrow$ Aqueous production	CNS suppression d/t crossing of BBB (C/I in children) : - Drowsiness - Apnea - Depression
Carbonic anhydrase inhibitors : Acetazolamide		Only oral	-
Rho kinase inhibitors : Netarsudil		$\downarrow$ Aqueous production + $\uparrow$ Trabecular outflow	-

Surgical mx :

1. Laser trabeculoplasty :

- Photocoagulation with argon laser > Frequency doubled Nd : YAG laser.
- TOC for prophylaxis in other eye.

2. Trabeculectomy.

3. Non-penetrating Sx :

- Visco canalostomy.
- Deep sclerectomy.

4. Seton Sx/drainage devices :

In resistant cases :

- Express mini shunt.
- Ahmed glaucoma valve.

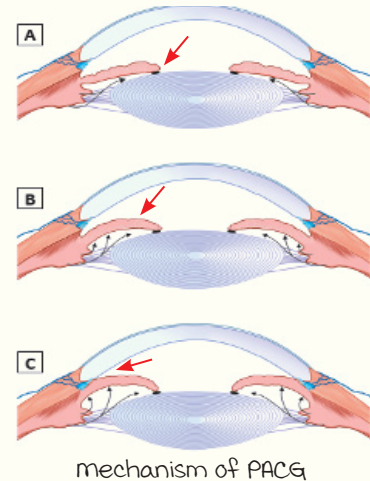
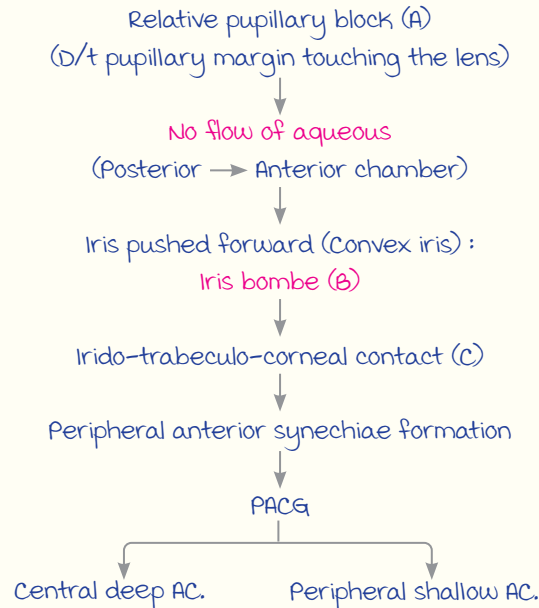
5. MIGS (micro invasive glaucoma Sx).

Primary Angle Closure Glaucoma (PACG)

00:40:00

Acute disease

Pathogenesis :

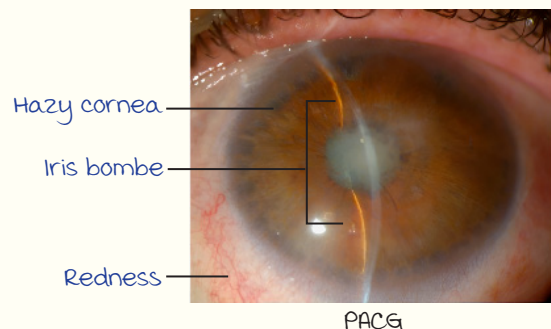


Symptoms :

- Headache.
- Severe eye pain : IOP > 40 mmHg (Rock hard eye in digital tonometry).
- Distorted vision : D/t hazy cornea (High IOP disrupts corneal endothelium).
- Redness of eye.
- Vomiting.
- Halos around lights : Prismatic dispersion d/t corneal edema.

Signs :

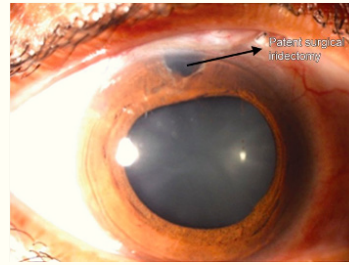
- **Eclipse sign** : Shadow over nasal iris. (Oblique flashlight test)
- **Vogt's triad** (Slit lamp examination) :
 - a. Iris atrophy.
 - b. Posterior synechiae (Adhesion b/w pupillary margin & lens).
 - c. Glaucomflecken.
- Pupil : **vertically oval mid-dilated** pupil, which is fixed.



Treatment :

Acute emergency Rx :

- IV mannitol (DOC)
 - Oral acetazolamide
- } ↓ IOP
- Pilocarpine eyedrops (miotic) : Pulls iris to center and open angle (S/E : Headache).



Laser iridotomy

----- Active space -----

Definitive Rx :

Laser iridotomy (TOC)
using Nd-YAG laser

Surgical iridectomy

Congenital Glaucoma

00:50:14

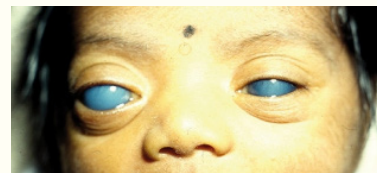
D/t Trabeculodysgenesis.

Signs :

1. Buphthalmos :

Enlarged eyeballs :

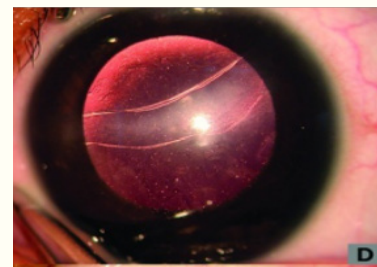
- Deep AC.
- ↑ Axial length → myopia.
- Corneal diameter > 13 mm.
- Blue sclera.



Blue sclera

2. Haab's striae : Horizontal breaks in descemet's membrane.

3. Hazy cornea : Earliest sign.



Haab's striae

Symptoms :

- Blepharospasm.
- Photophobia.
- Lacrimation.

Treatment 🏹 :

- Trabeculectomy + trabeculotomy : TOC.
- Goniotomy : If cornea is clear.

Secondary Glaucoma

00:53:05

Lens Induced :

- Phacomorphic : Intumescent cataract, 2° ACG.
- Phacolytic : morgagnian cataract, 2° OAG.

Pigmentary :

- Presents as 2° open angle glaucoma.
- Krukenberg spindles : On corneal endothelium.

Pseudoexfoliation :

m/c cause of 2° open angle glaucoma.

Target sign : Deposit of exfoliative material on lens.

Fnocks : Deposition of exfoliative material on pupillary margin.

Neovascular glaucoma :

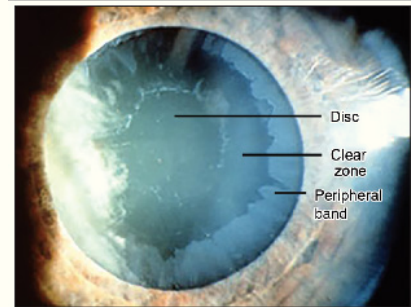
D/t neovascularization of iris/rubeosis iridis.

Causes :

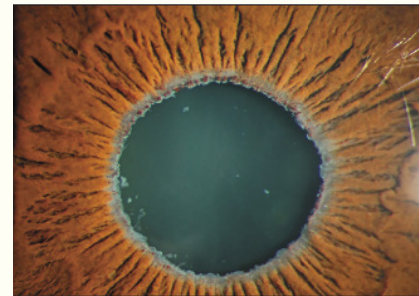
- Proliferative diabetic retinopathy (PDR).
- Ischemic CRVO.

Progression :

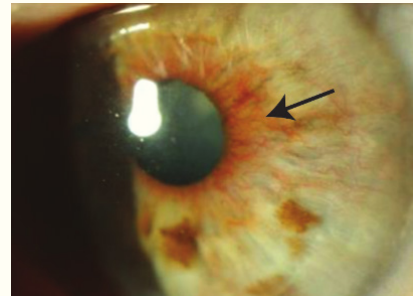
Hyphema $\rightarrow$ 2° OAG (AKA neovascular glaucoma).



Target sign



Fnocks



Neovascularization of iris

OPTICS, REFRACTION & ACCOMMODATION

----- Active space -----

Tests for Vision

00:00:55

Tests for Far Vision :

1. Snellen's chart :

- Distance of examination : 6 metres/20 feet.
- Normal : 6/5, 6/6.
- Least recordable vision : 1/60.
- 1/6 → Finger counting (FC) → Hand movement (Hm)
- Perception of light (PL) → Blind.

Principle :

Each letter subtends an angle of 5 minutes of arc when viewed from its respective distance.



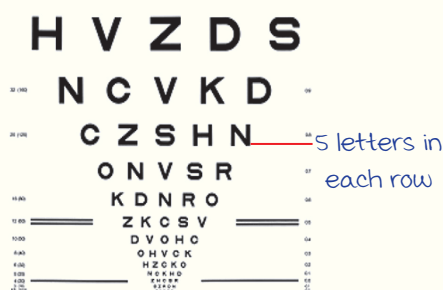
Snellen's chart

Note :

minimum angle of resolution (MAR) : 30 - 60 sec of arc.

2. Bailey-lovie/Log/ETDRS chart :

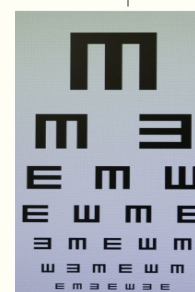
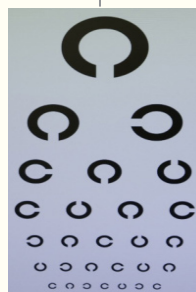
Not used in India.



3. Landolt C-ring.

4. Snellen's E-chart.

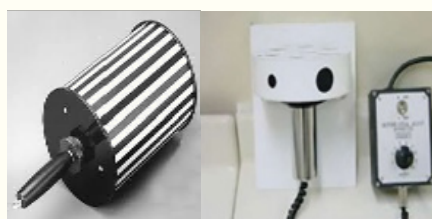
If patient is illiterate



Tests used in infants/pre-verbal children :

1. Optokinetic nystagmus :

The child normally focuses on alternating light & dark bands.



OKN drum

Catford drum

2. Preferential looking test :

The child prefers to look at the patterned areas.



Teller acuity cards



Keeler gratings

----- Active space -----

3. Cardiff acuity test :



4. Blink reflex :

- Blinks on exposing eye to light.
- Present since birth.

5. menace reflex :

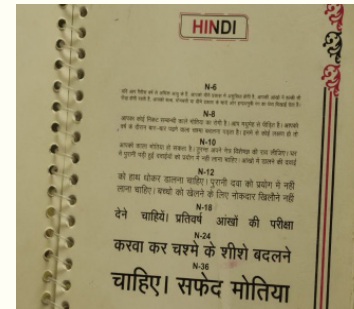
- Any object brought close to the child's eye → Child closes eyes.
- Develops at 4 to 6 months of age.

Tests for Near Vision :

Distance of examination : 30 - 35 cm.

1. Snellen's near chart :

- Best near vision : N_6 .
- Poor near vision : N_{36} .
- m/c used.

2. Jaeger's chart : J_1 to J_7 vision.

Snellen's near chart

Tests For Contrast Sensitivity :

mnemonic : CRAP BF.

- Cambridge test.
- Regan chart.
- Arden grating chart.
- Pelli-Robson chart.
- Bailey-Lovie contrast chart.
- FACT chart.

Pelli Robson chart :

used in case of cataract,
where contrast vision is lost early.



Normal Optics Of Eye

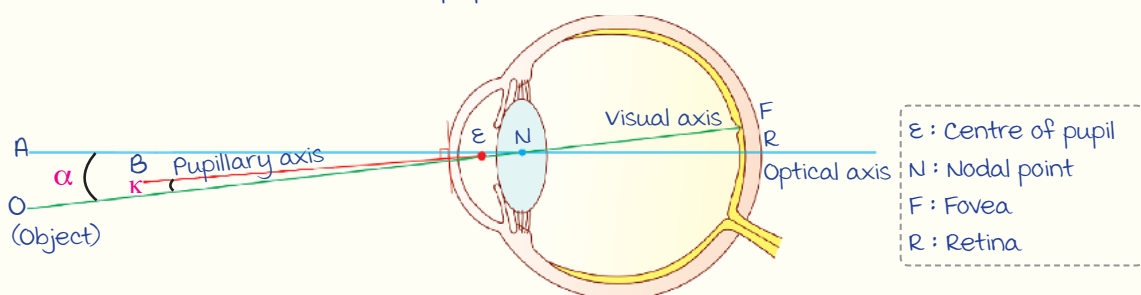
00:12:07

Axes :

1. Anatomical/optical axis : $\overline{ANR}$.
2. Visual axis : $\overline{OENF}$.
3. Pupillary axis : $\overline{BE}$ (Axis to the centre of the pupil).

Angles :

1. Kappa (κ) :
B/w pupillary and visual axes.
2. Alpha (α) :
B/w anatomical and visual axes.



Gullstrand's Schematic Eye :

----- Active space -----

Purkinje images  : Reflections of light based on 4 refractive surfaces in the eye.

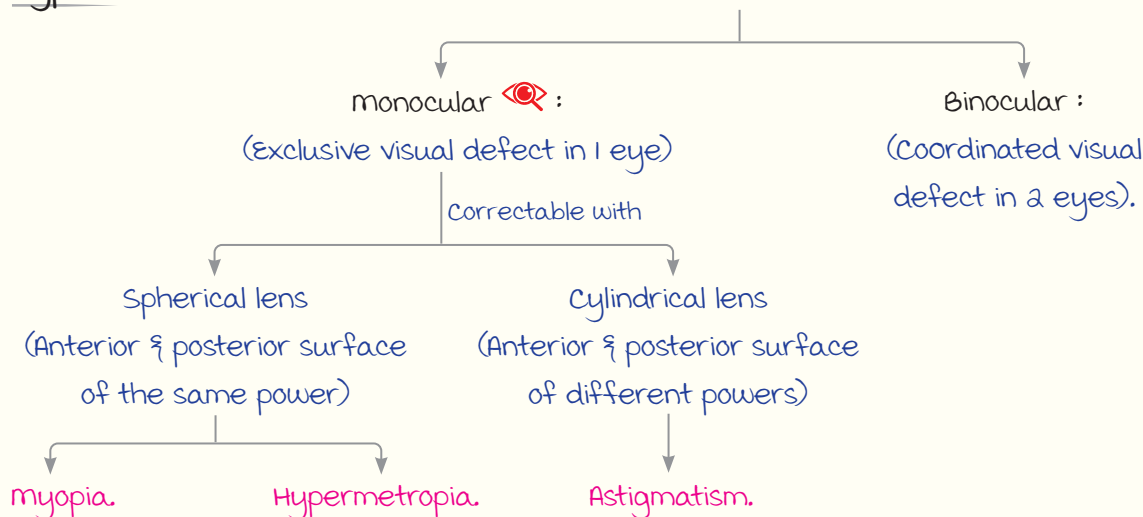
Purkinje images	I	II	III	IV
Surface	Cornea		Lens	
	Anterior	Posterior	Anterior	Posterior
Power	+48.8 D (max)	-5.88 D	+19 D	

Net power : +43D to +44D

- **Aphakia** : 2 Purkinje images (I & II).

Refractory Errors

00:15:06

Types :

	myopia	Hypermetropia
Power	more (Image in front of retina)	Less (Image behind the retina)
Corrective lens	(-)/concave (Diverging)	(+)/convex (Converging)
Axial length (Axial myopia)	↑ (Buphthalmos)	↓
Curvature	↑	↓
Refractory index (Index myopia)	↑ (Nuclear cataract)	↓
Position of lens (Positional myopia)	Anterior (Weil marchesani syndrome)	Posterior

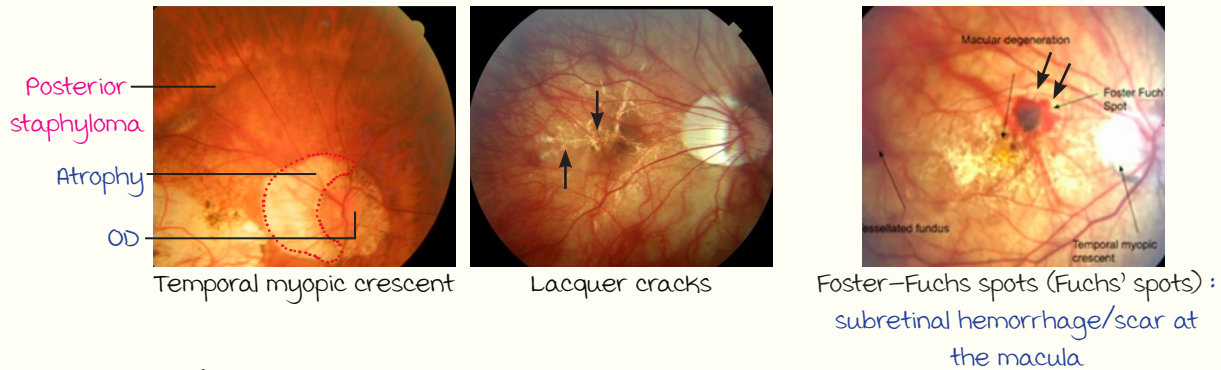
----- Active space -----

Myopia

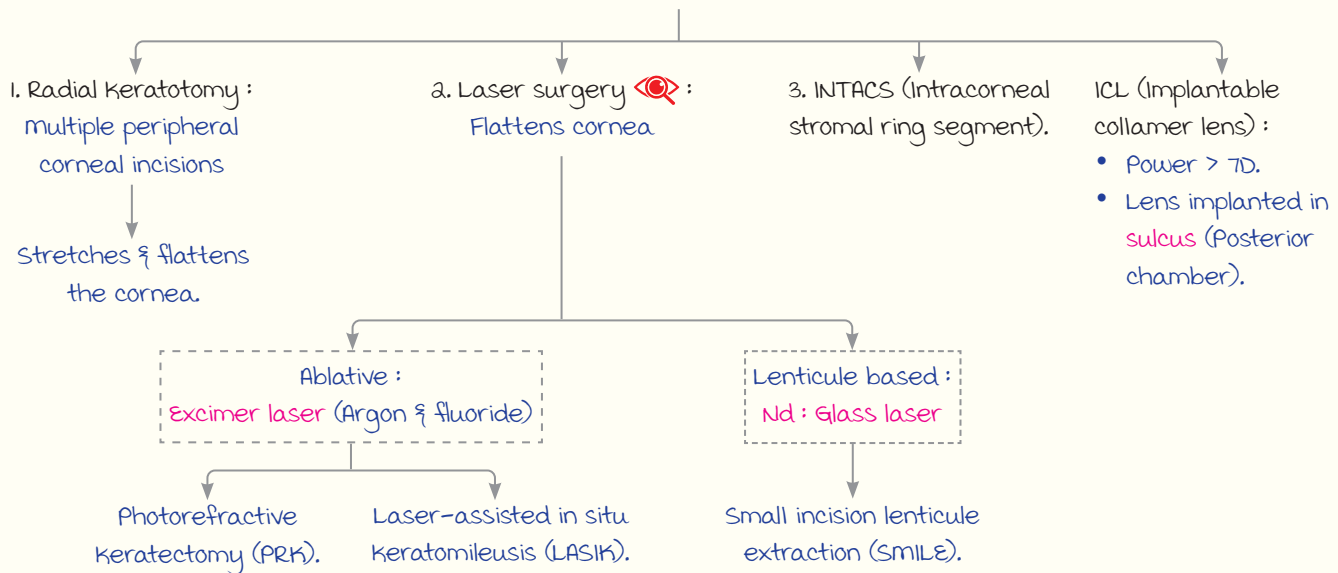
00:19:54

Fundus Signs :

Only seen in **pathological myopia/degenerative myopia** ($AL > 26 \text{ mm}$ or $\text{power} > 6D$).



Surgical Correction :

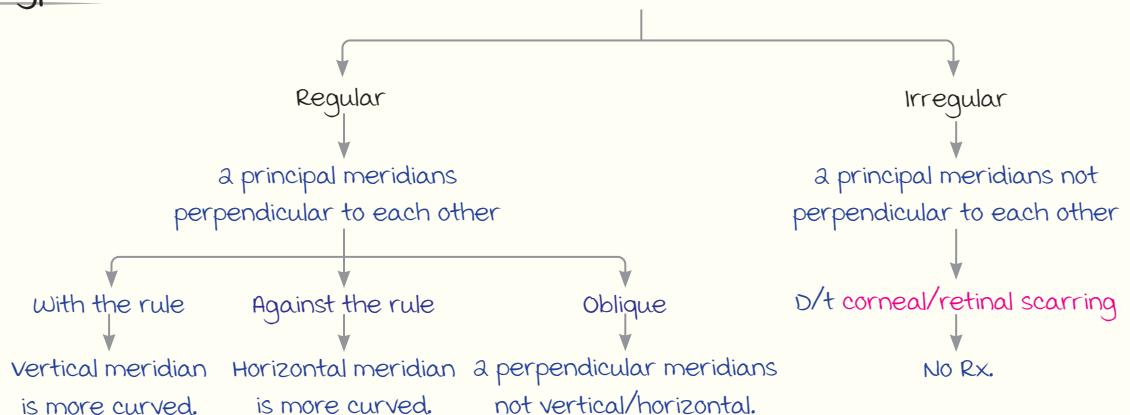


Astigmatism

00:25:56

Horizontal & vertical axes : unequal powers.

Types :



Treatment :

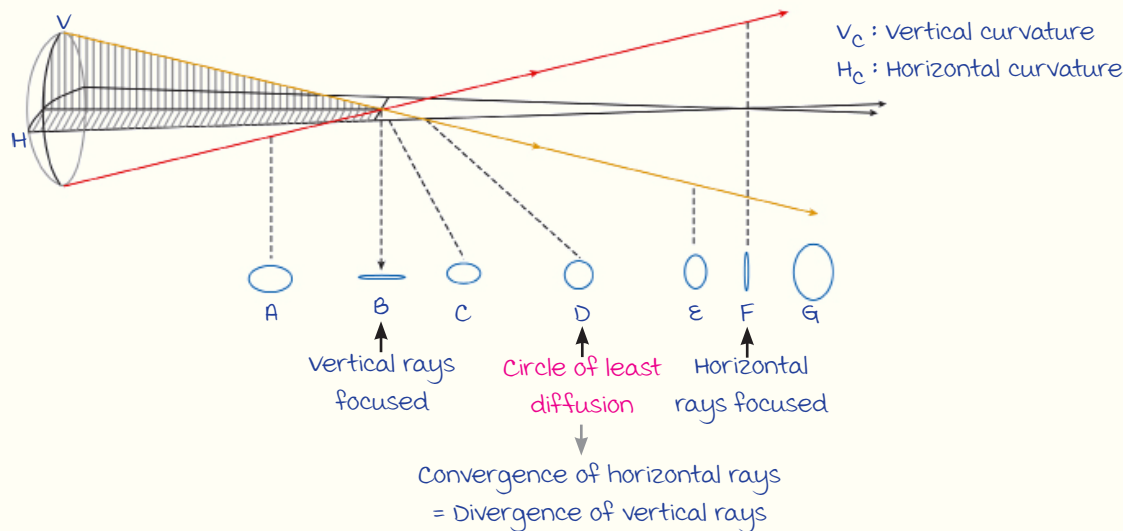
----- Active space -----

- Cylindrical glasses/Toric contact lens.
- Sx : Astigmatic LASIK, limbal relaxing incision.

Axis determination :

mnemonic : With CV, against VC.

	180°	90°
With the rule	Concave (-)	Convex (+)
Against the rule	Convex (+)	Concave (-)

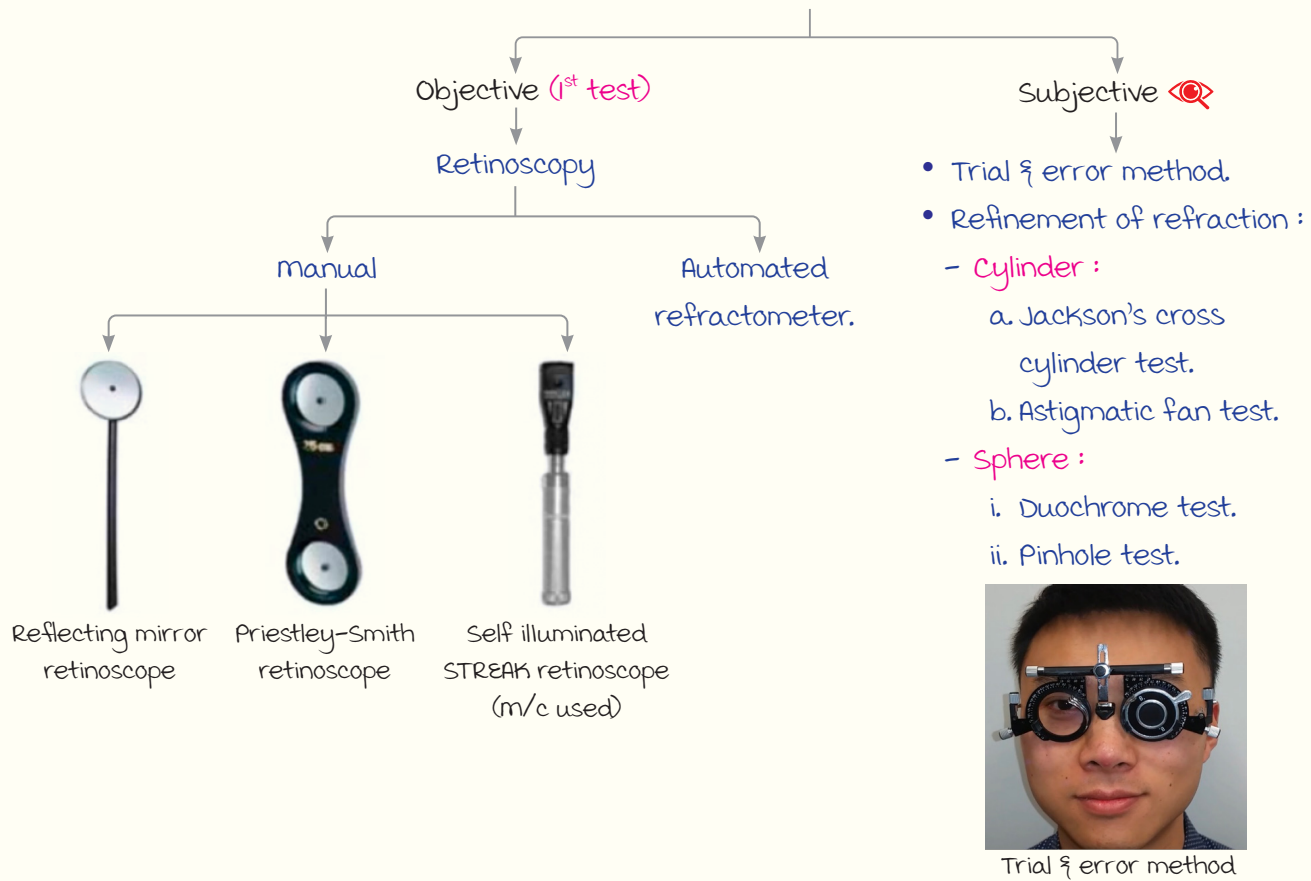
Sturm's Conoid :Refraction of light rays when refracted through a toric surface (Different V_c & H_c).Correction of Refractory Errors

00:30:25

Prescription of Spectacles 👁 :

Lens	use
(-) Spherical	myopia
(+) Spherical	Hypermetropia
(-) Cylinder	Simple myopic astigmatism
(+) Cylinder	Simple hypermetropic astigmatism
(-) Spherical/(-) Cylinder	Compound myopic astigmatism
(+) Spherical/(+) Cylinder	Compound hypermetropic astigmatism
(-) Spherical/(+) Cylinder OR (+) Spherical/(-) Cylinder	mixed astigmatism

----- Active space -----

Determination of Refractory Errors :Retinoscopy :

- Distance of examination : 1 mt.
- method : Comparing movement of red streak to movement of retinoscope.

	Against	No movement	With
Power	> 1 D myopia	1 D myopia	< 1 D myopia
Retinoscopy value	variable	Zero	variable

← Can also be either emmetropia/hypermetropia.

Retinoscopy values :

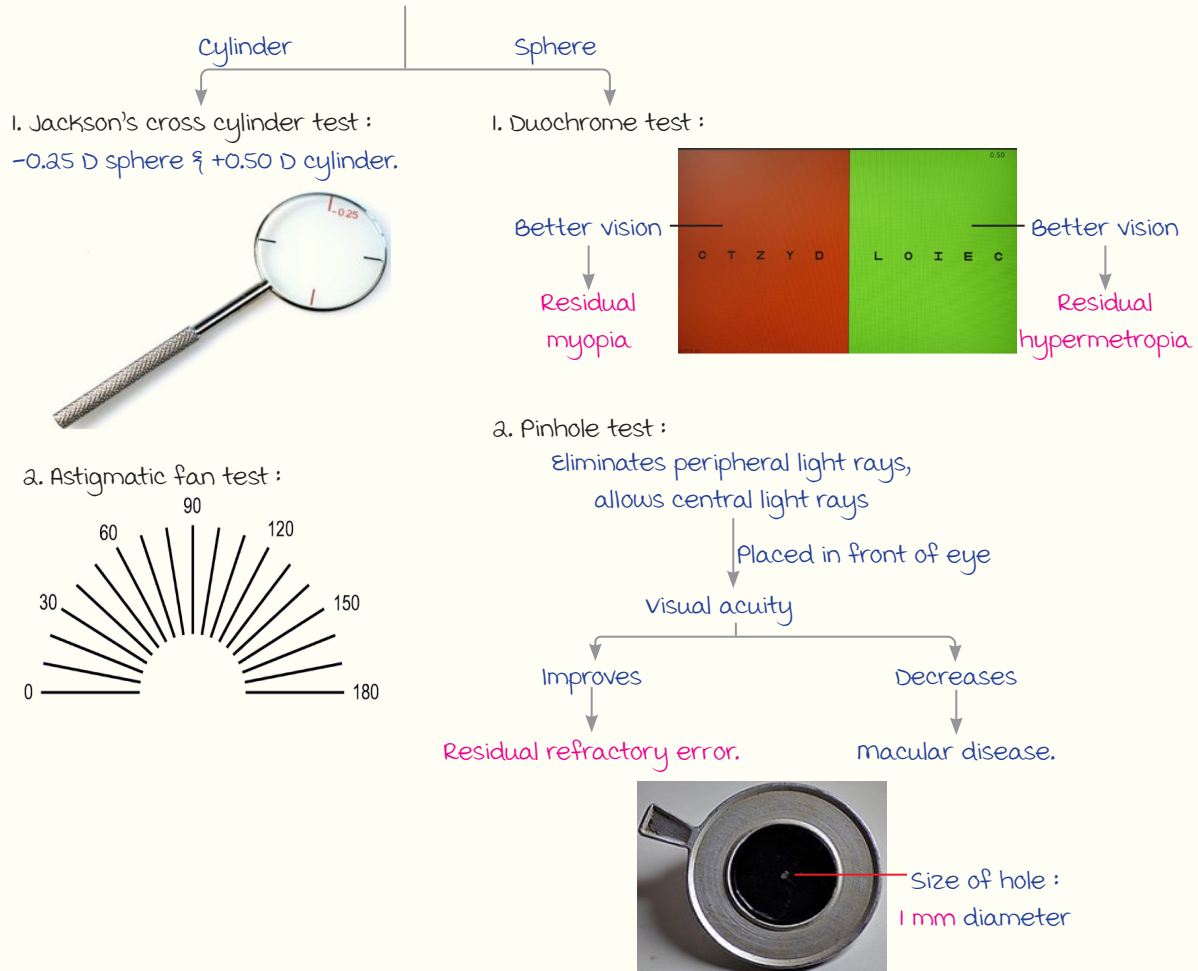
- Correction factors (C) :
 - Distance (d) of examination : $C_1 = 1/d$.
 - Tonus allowance of cycloplegic drug : C_a .
- Corrected retinoscopic findings (RF) :

$$RF = RV - C_1 - C_a$$

Cycloplegics	C_a
Atropine	1
Cyclopentolate	0.75
Homatropine	0.50

Refinement of Refraction :

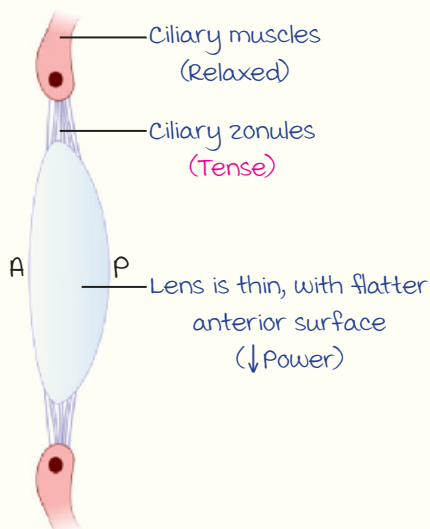
----- Active space -----

Accommodation

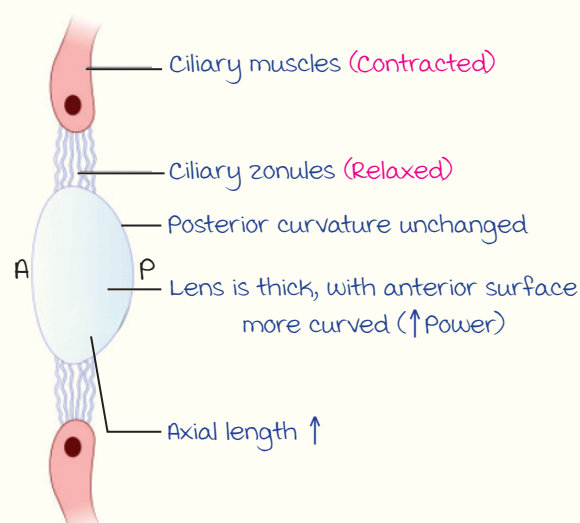
00:43:25

Helps near vision (Power needs to ↑).

Far vision :



Near vision :



----- Active space -----

Presbyopia :

- Physiological insufficiency of accommodation (Difficulty in near vision).
- Age > 40 yrs.

Treatment : Convex spectacle (+).

Spasm of Accommodation :

- AKA pseudomyopia.
- Issue with distant vision (D/t spasm of ciliary muscles).

Treatment : Cycloplegics → Atropine (Relaxes ciliary muscle).

Newborn :

At birth :

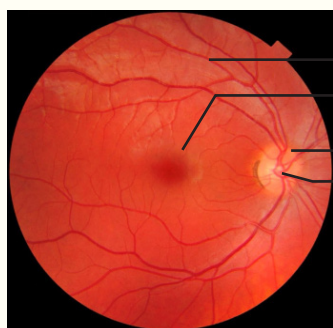
- Axial length (AL) : ~17 mm (Smaller than normal).
- Refractory error : 2 - 3 D hypermetropia.
- Becomes emmetropic by 6 - 7 yrs (Adult AL : 24 mm).

RETINA

----- Active space -----

Anatomy

00:01:00



- Arc-like vessels
 - macula (Appears avascular & temporal to optic discs)
 - Neuroretinal rim
 - Optic cup
- } Optic disc (C/d ratio : 0.3)

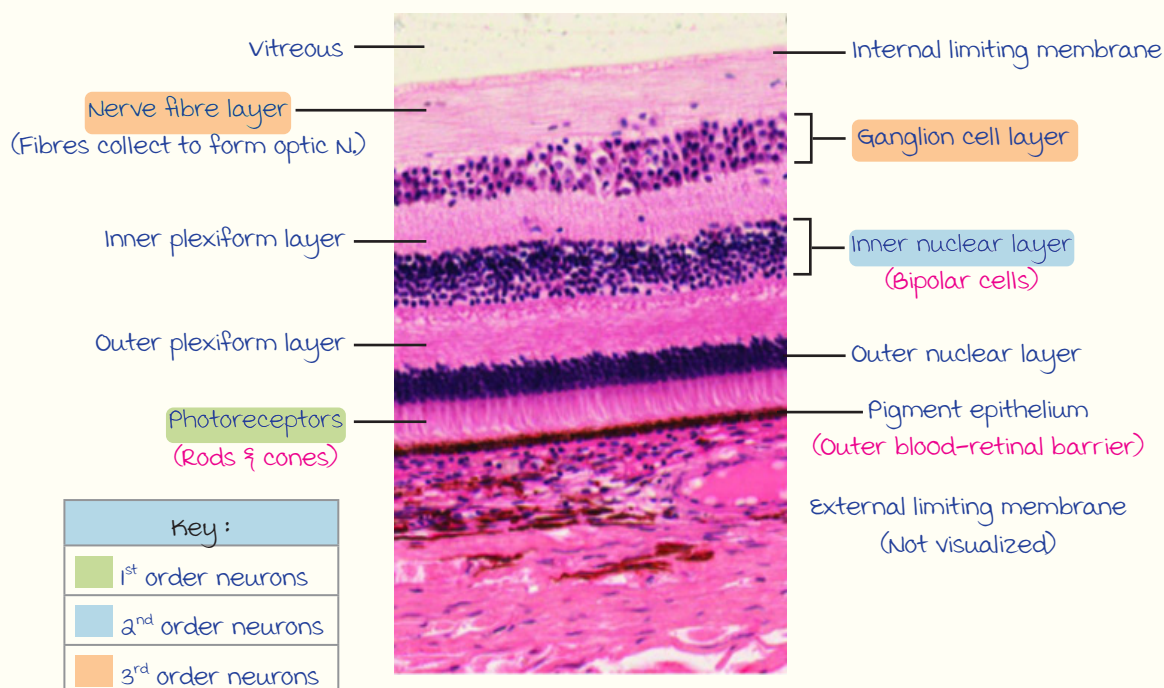
Fundus :

- Vitreous
 - Retina
 - Choroid : Red.
- } Transparent.

Ora serrata :

- Peripheral termination of retina.
- Site of intravitreal injection.

Layers of retina (Inside to outside) 👁 : mnemonic **ING (IOP)_a**



Subretinal space : Space b/w inner 9 layers (Neurosensory retina) & pigment epithelium.

----- Active space -----

Investigations

00:07:46

Indocyanine Green Angiography (ICGA) :

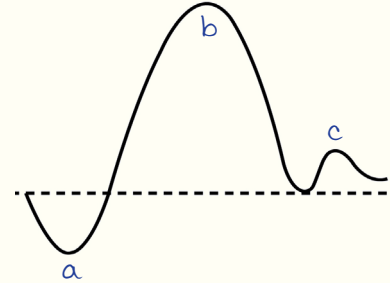
To visualise choroidal vasculature.

Electroretinography (ERG) :

- Records retinal **action potential**.
- Detects photoreceptor pathology.

Interpretation :

- a wave : Photoreceptors activity.
- b wave : Bipolar cells activity.
- c wave : metabolic activity of pigment epithelium.

**Electrooculography (EOG) :**

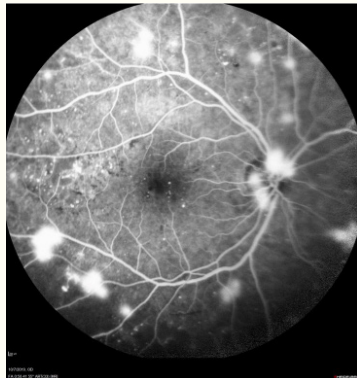
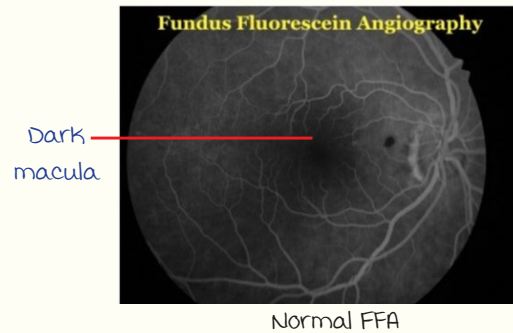
- Records retinal **standing potential** from outer retina. (Photoreceptors + Pigment epithelium)
- measurement : **Arden's ratio** = $\frac{\text{Light peak}}{\text{Dark trough}}$ (Normal > 185% / 1.85).
- Detects photoreceptor pathology.

Ophthalmoscopy :To visualise fundus (1st investigation).

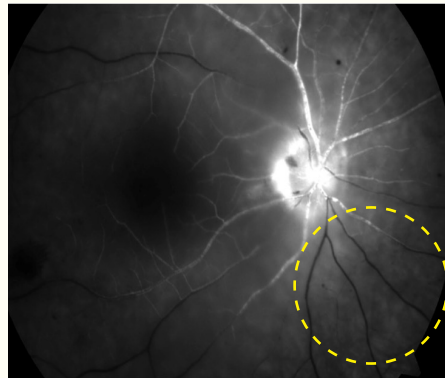
	Direct ophthalmoscope	Indirect ophthalmoscope
Appearance		
Operation	Hand-held	Worn around the head
External lens	No	Yes (Power : +20 D)
Distance of examination	As close to patient's eye as possible	Arm's length
Image	Virtual & erect	Real & inverted
magnification	15x	• +20 D lens : 3x • +14D lens : 5x
Area visualised	Small : • Central fundus • Optic disc & macula	Large : • Peripheral fundus • Ora serrata
Field of view	10°	45° with +20 D lens
Area viewed	2 DD	8 DD

Fundus Fluorescein Angiography (FFA) :

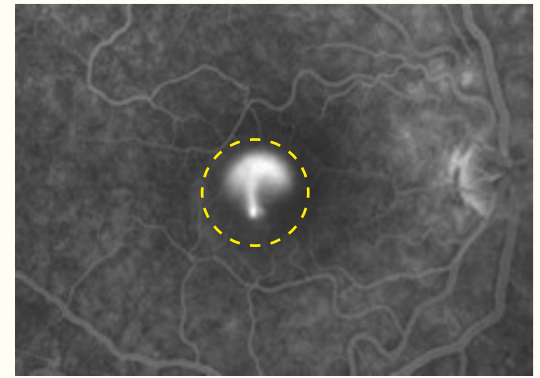
- Visualises fluorescent blood vessels.
 - Diagnoses :
 - Vascular disorders.
 - macular disorders :
- Normal macula appears dark.



Leakage from blood vessels



Blocked blood vessels

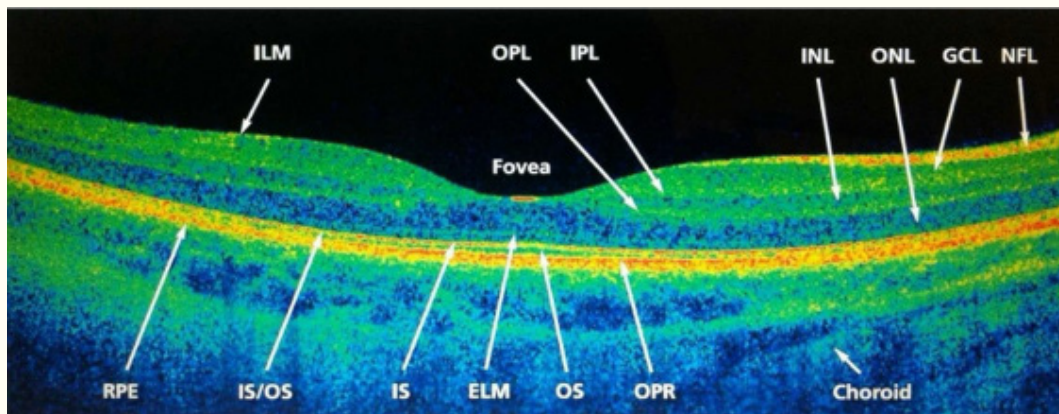


macular disease

Optical Coherence Tomography (OCT) 👁 :

To visualise retinal layers.

- RPE : Always red.
- Fovea : Depressed contour.

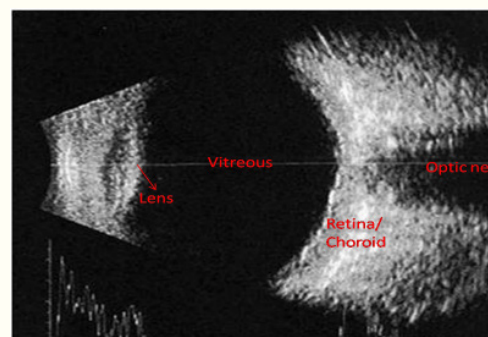


B (Brightness) Scan USG :

Indications : Dense cataract/corneal opacity.

Structures observed :

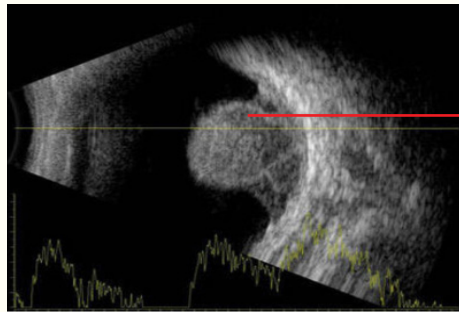
- vitreous (Black/hypoechoic).
- Retina } white/hyperechoic.
- Choroid }



----- Active space -----

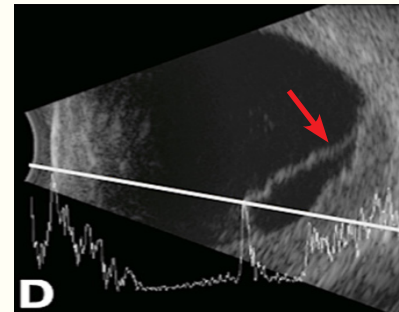
Pathologies on B-scan :

- Retinoblastoma/Choroidal melanoma : Hyperechoic mass from white area.



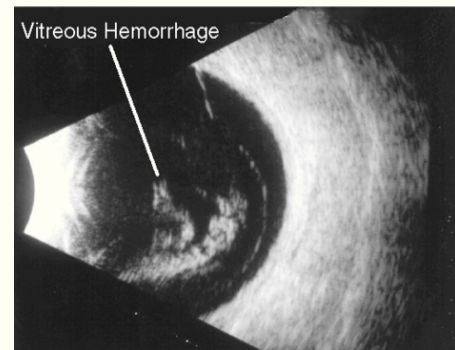
Choroidal melanoma

Collar stud/
mushroom
appearance
(Specific)



Retinal detachment

- Vitreous hemorrhage (VH) :
 - Source : Retinal blood vessels.
 - m/c cause of VH :
 - i. Diabetic retinopathy.
 - ii. Blunt trauma (Young adults).
 - iii. Exales disease (Recurrent)
 - c/f : LOV.



Retinoblastoma

00:20:53

Types :

- Non-hereditary (60%) : Always U/L.
- Hereditary : Non-familial (33%) and familial (7%).
- Sporadic (m/c) : Non-hereditary + Non-familial (93%).

Genetics :

- m/c inheritance : Autosomal dominant (AD).
- Gene : RB on Chr. 13q14.

Knudsons 2 hit hypothesis : Tumor presents only on loss of heterozygosity.

malignancy :

- m/c malignancy : PNET/pinealoblastoma.
- m/c 2° malignancy : Osteosarcoma.

Clinical Features :

- Leukocoria (m/c) : White pupillary reflex
- Squint (2nd m/c) : Esotropia > Exotropia.
- 2° glaucoma + Buphthalmos (3rd m/c)

Spread  : Direct through optic nerve.

Note :



Investigations :

CT scan : **IOC.**

MRI : To assess metastasis.



CT : Retinoblastoma

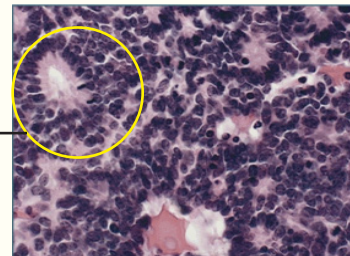
Calcified tumor (m/c cause of **intraocular calcification**)

----- Active space -----

Biopsy (Gold standard) :

Findings :

1. Homer wright rosettes (Pseudo-rosettes).
2. Fleurettes.
3. **Flexner wintersteiner rosette** (Pathognomonic) :
 - Empty central lumen.
 - Surrounding tumor cells.

Treatment :

Enucleation (TOC).

Chemotherapy :

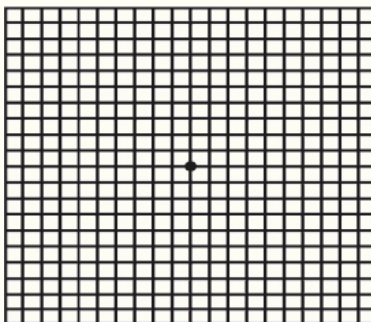
- **Carboplatin.**
- **Etoposide.**
- **Vincristine.**

Macular Disorders

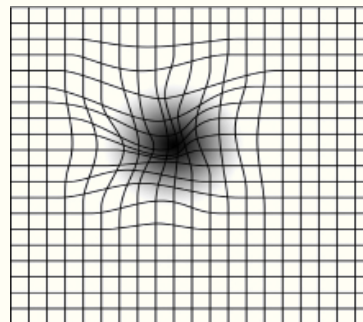
00:27:23

Tests :

1. Amsler's grid : Patient asked to look at the central dot.



Normal test : **Straight lines**



macular disorder : **Wavy lines**

2. Photostress test :

- **↑ Photostress recovery time** in macular disorder.
- To differentiate macular with optic nerve disorders.

----- Active space -----

Central Serous Retinopathy (CSR) :

- Damage to retinal pigment epithelium (RPE).

- Fluid in subretinal space.
- Central LOV.

Risk factors 🦋 :

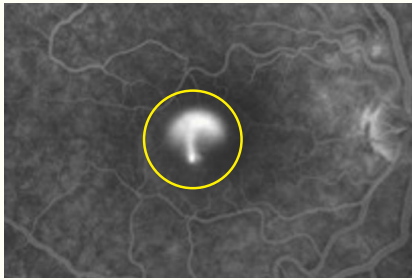
- Steroids.
- Cushing's syndrome.

- H. pylori infection.
- Type A personality.

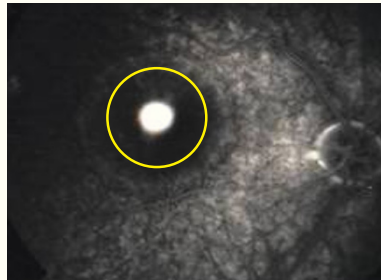
Treatment :

- Self resolving, treat risk factor.
 - Photodynamic therapy with verteporfin
 - Laser photocoagulation
 - Steroids c/i.
- } If persistent > 4 months.

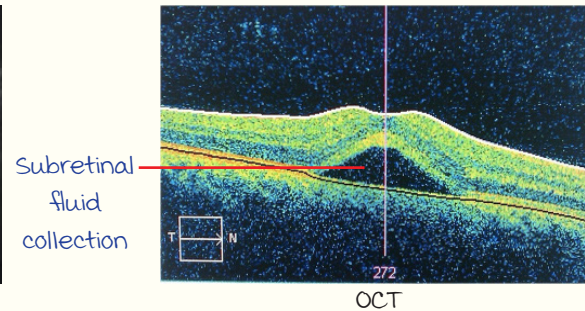
Investigations :



FFA : umbrella appearance



FFA : Ink blot appearance

Cystoid macular edema (CME) 🦋 :

- Fluid in outer plexiform layer.

- Central LOV.

Risk factors : mnemonic → RUN PRIDE.

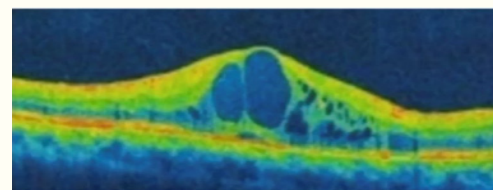
- Retinitis pigmentosa.
- Uveitis.
- Niacin/nicotine.

- Prostaglandin ($\text{PGF}_{\alpha\alpha}$) analogues.
- Retinal vein occlusion.
- Irvine-Gass syndrome (Post-op cataract complication).
- Diabetic retinopathy.
- Epinephrine in aphakia.

Investigations :



FFA : Flower petal appearance



OCT : Cystic spaces

Treatment : Intravitreal steroids.

Age Related macular Degeneration (ARMD) 👁️ :

----- Active space -----

Causes :

- Old age.
- Smoking.

a. Dry/non-neovascular/
non-exudative ARMD (90%) :

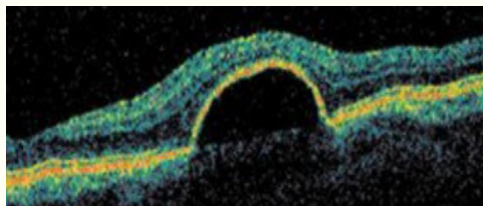
- Symptoms : No / Gradual LOV.
- Stages :
 - Early } Drusen ⊕ (Eosinophilic deposits)
 - Intermediate }
 - Late → Geographic atrophy of RPE ⊕.
- Rx : Antioxidants.

b. Wet/neovascular/exudative/late ARMD (10%) :

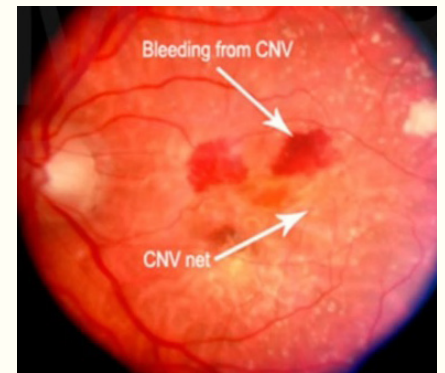
- Symptoms : Sudden LOV.
- Signs :
 - RPE detachment.
 - Choroidal neovascularisation
- ↓
Subretinal hemorrhage.
- Rx : Intravitreal anti-VEGF drugs.



Drusen



RPE detachment



Choroidal neovascularisation

Fundus Dystrophies

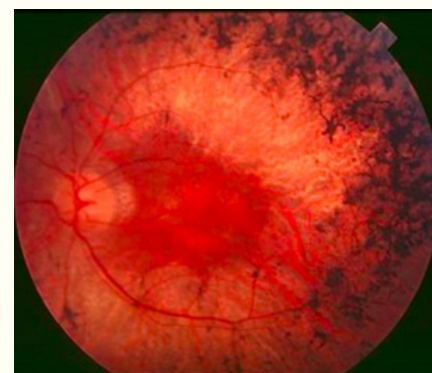
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Retinitis Pigmentosa 👁️ :

- Rod-cone dystrophy (Rods > cones).
- m/c inheritance : Autosomal recessive.
(m/c hereditary fundus dystrophy)

Clinical features :

- Nyctalopia, ring/donut scotoma/
tunnel vision (Rods affected).
- Bony spicule pigmentation.
- Waxy, pale optic disc.
- Systemic association :
Lawrence-moon-Biedl syndrome (RP + Obesity + Polydactyly).
- Arteriolar attenuation.



Bony spicule pigmentation

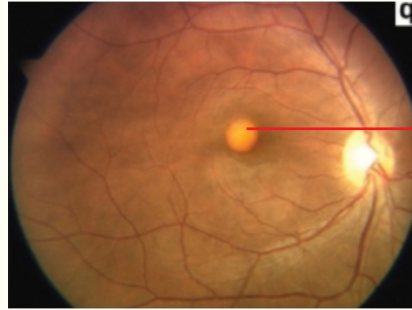
IOC : ERG.

Treatment : Genetic counselling (No Rx).

----- Active space -----

Best macular Dystrophy :

- Cone dystrophy → Day blindness, central scotoma.
- Autosomal dominant.



Egg yolk/vitelliform lesion

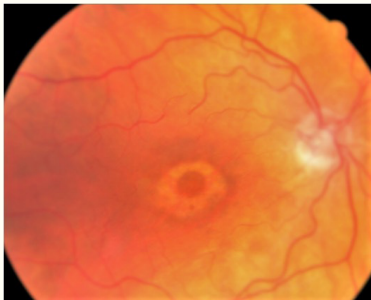
↓ Breaks

Pseudohypopyon/
vitelliruptive lesion

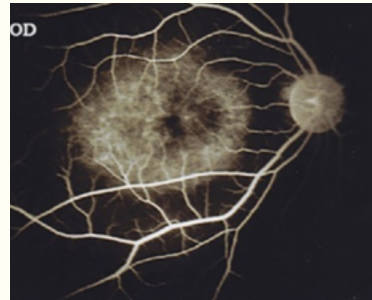
IOC : EOG (ERG normal, Arden's ratio < 150%)

Stargardt Disease :

- Cone dystrophy.
- Seen in juveniles.



Bull's eye maculopathy



FFA : Dark/silent choroid sign

Note :

Bull's eye maculopathy also seen in chloroquine toxicity.

Vascular Disorders

00:44:51

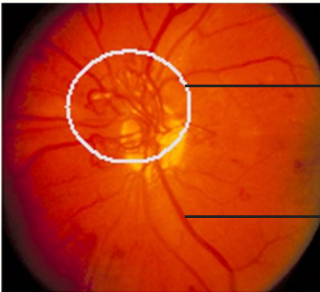
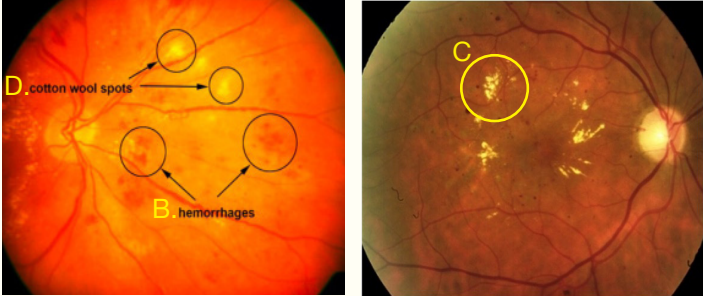
Diabetic Retinopathy :

Risk factor : Duration of Dm.

Screening : → T1DM : 5 years after Dx.
 → T2DM : Immediately on Dx.

Early treatment diabetic retinopathy study (ETDRS) classification :

----- Active space -----

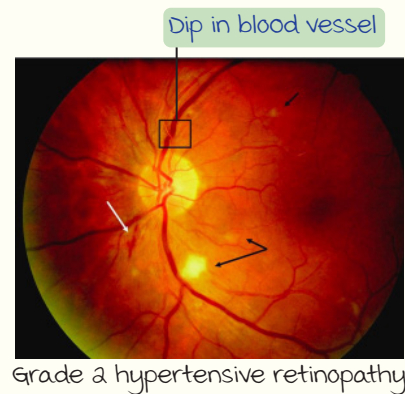
	Proliferative DR (PDR)	Non-proliferative DR (NPDR)
Signs	Neovascularisation : - NVD : At disc - NVE : Elsewhere  NVD - Requires treatment d/t leaky newly formed vessels.	a. micro-aneurysms (Earliest) b. Hemorrhages (Flame shaped, dot & blot) c. Exudates (Hard) : Lipid deposits d. Cotton wool spots (Soft exudates) : Neuronal debris 
Rx :	- Panretinal photocoagulation - Anti-VEGF drugs	Glycemic control

Hypertensive Retinopathy :

D/t vasoconstriction of retinal vessels.

Keith Wagner classification :

Grade	Retinal findings
1	Arteriolar attenuation → A : V ratio = 1 : 3 (Normal = 2 : 3)
2	Salu's sign
3	- Bonnet sign/Gunn's sign - Flame shaped hemorrhage (white arrow) - Cotton wool spots (black arrow) - Copper wiring of arterioles
4	- Papilledema - Silver wiring



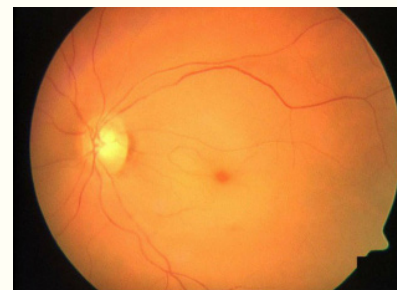
CRAO :

Causes : Atherosclerosis (m/c).

Symptoms : Sudden severe LOV (Emergency).

Fundus signs :

- Pale (Retinal edema).
- Cattle tracking
(Segmentation of vessels D/t thrombi).

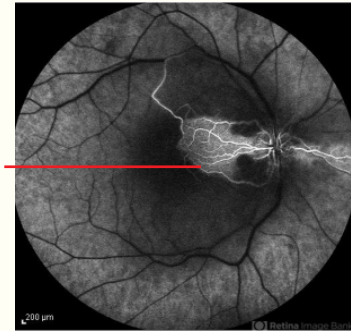


Cherry red spot at macula :
D/t fluid accumulation in
ganglion cell layer

----- Active space -----

Treatment  :

- Ocular massage (Dislodges emboli.)
 - Sublingual isosorbide nitrate (Vasodilation.)
 - IV mannitol
 - Aspiration of aqueous
- } ↓ IOP.

Cilioretinal vessels
(Supply macula)

FFA : Blocked blood vessels

CRVO :

	Non-ischemic	Ischemic
Pathogenesis	Stasis → Vasodilation ↓ ↑ Vascular permeability	-
c/f	macular edema	Severe flame-shaped hemorrhages (Tomato ketchup appearance) Rubeosis iridis (Iris NV) → 100 day glaucoma (Neovascular glaucoma)
Rx	Intravitreal steroids	Panretinal photocoagulation (Argon laser)

BRVO :

- Branch retinal vein occlusion.
- m/c affects superotemporal quadrant.



BRVO : Sectoral involvement



Tomato ketchup appearance

Retinopathy Of Prematurity (ROP)  :

Risk factors :

- Preterm infants < 34 weeks.
- Birth weight < 1750 grams.
- Preterm infants 34 - 36 WOG + risk factors.

Screening of infants :

- Born ≥ 28 weeks and ≥ 1200 g : Screen after 4 weeks of birth.
- Born < 28 weeks or < 1200 g : Screen within 2 - 3 weeks of birth.

Treatment :

- Laser photocoagulation using diode red laser.
- Lens sparing pars plana vitrectomy (For tractional retinal detachment).

Retinal Detachment

01:00:22

----- Active space -----

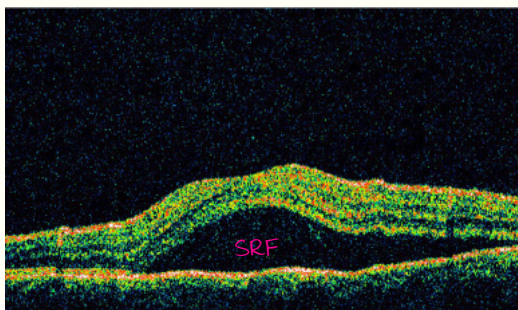
Separation of neurosensory retina from retinal pigment epithelium d/t fluid collection in subretinal space.

Types :

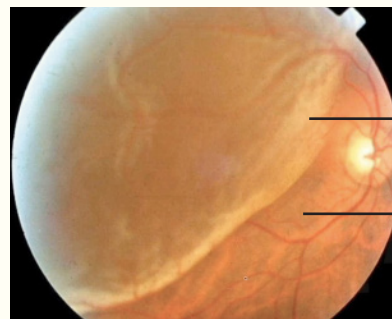
	Rhegmatogenous	Tractional	Exudative
m/c cause	Pathological myopia	Diabetic retinopathy	Choroidal melanoma
Retinal break	⊕	⊖	⊖
Syneresis (Vitreous liquefaction)	⊕	⊕	⊖
Vitreo-retinal traction	⊕	⊕	⊖
Photopsia	⊕	⊖	⊖
Floaters	⊕	⊕	⊖
Curtain falling before eye/LOV	⊕, Sudden	⊕, Gradual	⊕, Sudden
Detached surface	Convex	Concave	-
Retinal mobility	mobile	Immobile, taut	-
Signs of chronicity	- Retinal atrophy - High watermarks - Intraretinal cysts	Demarcation lines	-
Other signs	Shaffer's sign (Tobacco dust appearance)	-	Shifting fluid
RAPD/marcus Gunn pupil	⊕	⊕	⊕

Investigations :

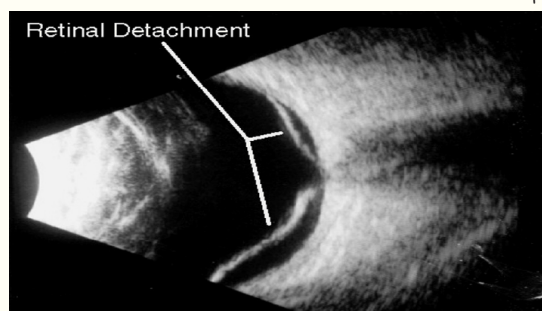
IOC : Indirect ophthalmoscopy.



OCT : Subretinal fluid collection



Fundoscopy

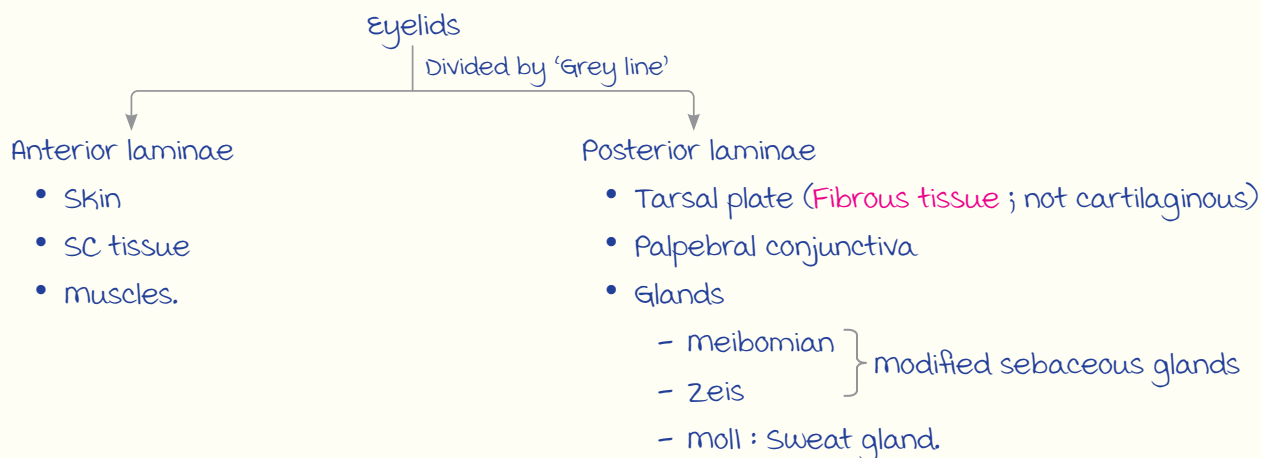


B-scan

EYELIDS AND ORBIT

Anatomy of Eyelids

00:00:20



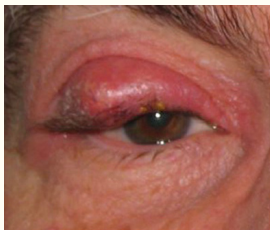
muscles of eyelid :

muscle	Nerve supply	Action	effect of injury
Levator palpebrae superioris	CN III	Elevation of upper eyelid	Ptosis
muller's muscle	Sympathetic fibres		
Orbicularis oculi	CN VII	Closure of eyelids	Lagophthalmos

Eyelid Pathologies

00:02:25

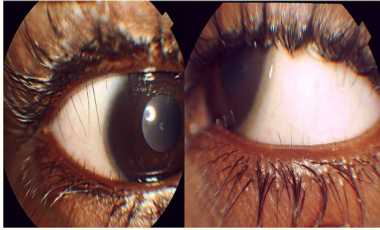
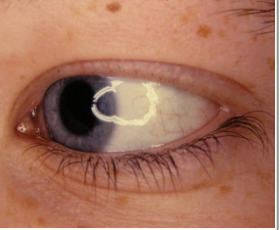
Chalazion vs. Sty/External Hordeolum :

	Chalazion 	Stye/external hordeolum
Pathology	Lipogranulomatous inflammation of meibomian gland	Suppurative (<i>S. aureus</i>) inflammation of Zeis gland
c/f	Painless circumscribed (Nodular) swelling not involving lid margins 	Painful generalized swelling 
Rx	Incision & drainage/ intralesional triamcinolone	Hot compressions

Eyelash Pathologies

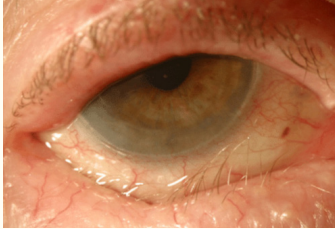
00:05:12

----- Active space -----

	Trichiasis 	Distichiasis	madarosis
Pathology	Eyelash turned inwards → Corneal injury → Corneal opacity → LOV	Additional posterior row of eyelashes	Absent eyelashes
Appearance			
Rx	Epilation/electrolysis of lash follicles	-	-

Lid Margin Pathologies

00:07:35

	Ectropion	Entropion
Pathology	Eversion of lid margin	Inversion of lid margin
Appearance		

Tylosis : Thickening of lid margin.

Ankyloblepharon : Fusion of upper & lower lid margins (Congenital/acquired : Burns).

Note :

Symblepharon (Conjunctival disorder) : Fusion of bulbar & palpebral conjunctiva.

Ptosis

00:09:37

Congenital Ptosis :



Left eye congenital ptosis

Sign :
Absent upper
eyelid crease

----- Active space -----

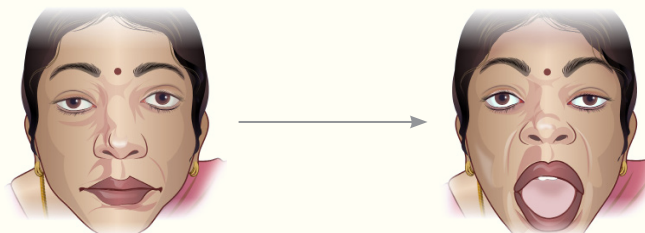
Treatment :

Severity	Surgery
mild ptosis & good levator function/Horner's syndrome	Fasanella servat operation
moderate ptosis and fair levator function	Levator resection surgery
Severe ptosis and poor levator function	Frontalis sling surgery

Surgery is done around 5 years of age.

Marcus Gunn Jaw Winking Syndrome :

Synkinetic phenomenon b/w trigeminal & oculomotor nerve.



Ptosis on closing mouth

Ptosis resolves on opening mouth

Orbital Pathologies

00:12:18

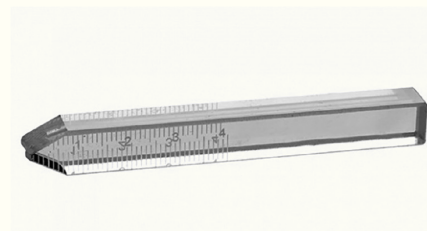
Proptosis/Exophthalmos :

Eyeball protrusion > 21 mm from lateral orbital rim.

Assessment :



Hertel's exophthalmometer : Better option



Leude's exophthalmometer : Easier to use (used in children)

Cause of proptosis	Feature
Orbital varices	↑ On bending forward/valsalva manoeuvre - u/L phleobiths (Bag of worms)
Orbital lymphangioma	↑ D/t URTI
Capillary hemangioma (Child)	↑ On crying
Encephalocele (infant)	
Carotid cavernous fistula	- B/L, pulsatile 75% cases : Traumatic - CN VI : Earliest affected nerve - IOC : DSCA (Digital Subtraction Angiography)

m/c causes  :

----- Active space -----

	Adults	Children
u/L	Thyroid associated ophthalmopathy (Grave's disease)	Orbital cellulitis
B/L		Leukemia/AML (Chloroma) > Neuroblastoma

Cavernous Sinus Thrombosis vs Orbital Apex Syndrome vs Orbital Cellulitis :

	Cavernous sinus thrombosis 	Orbital apex syndrome	Orbital cellulitis
Laterality	u/L → B/L	u/L	
CN affected	III, IV, VI (earliest), V _i , V _a	II, III, IV, V _i , VI	II, III, IV, VI
a. Ophthalmoplegia	Complete (Sequential)	Complete (Concurrent)	Complete
LOV	No	Yes (D/t CN II involvement)	
Corneal sensations	Lost (D/t V _i involvement)		Intact
b. Chemosis, proptosis	+++	+	+++
Fever, headache	Present	Absent	
mastoid edema	Present	Absent	
Chronology of symptoms	a = b	a → b	b → a
m/c cause	-	-	Ethmoidal sinusitis

V_i : Ophthalmic n. V_a : maxillary n.

m/c Associations with Orbit :

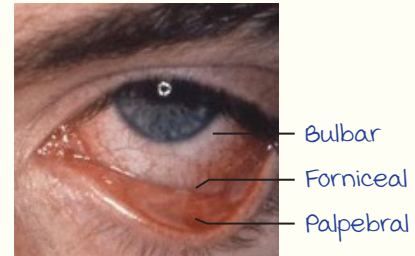
- m/c cystic orbital lesions : Ductal cysts of lacrimal gland.
- m/c cystic orbital tumor : Epidermoid and dermoid.
- m/c orbital neoplasm (Paediatric age) : Epidermoid and dermoid.
- m/c orbital malignant tumor (Paediatric age) : Rhabdomyosarcoma.
- m/c orbital and periorbital tumor in children : Capillary hemangioma.
- m/c benign orbital tumor (Adults) : Cavernous hemangioma.
- m/c malignant orbital tumor (Adults) : B cell NHL (Lymphoma).
- m/c intrinsic lacrimal gland lesion/lacrimal gland epithelial neoplasm : Pleomorphic adenoma.
- m/c orbital metastasis (Paediatric age) : Neuroblastoma.
- m/c peripheral neural tumor of the orbit : Plexiform neurofibroma.
- m/c primary source of orbital metastases :
 - Breast : 42%.
 - Lung : 11% (most deaths).
 - Neuroblastoma (Paediatric age).

CONJUNCTIVA

Anatomy :

Types :

- Bulbar : Lines sclera.
- Palpebral : Lines inner eyelids.
- Forniceal : Between bulbar & palpebral conjunctiva.



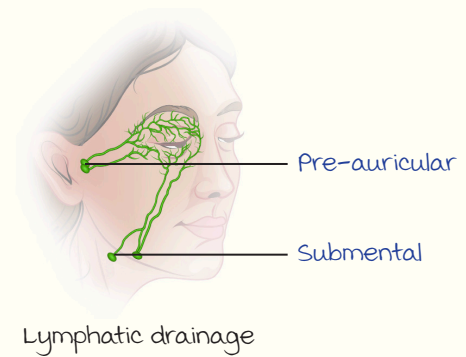
Lymphatic drainage :

- medially → Submandibular lymph nodes.
- Laterally → Preauricular lymph nodes.

Histology :

Goblet cells :

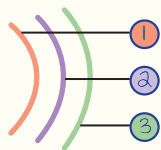
- Present in conjunctival epithelium.
- Secretes mucin → Forms innermost layer of tear film.



Tear Film

00:01:58

Anatomy :



	Layer	Produces	Source	Disease
①	Innermost	mucin	Goblet cells	Conjunctival disease
②	middle	Aqua	Lacrimal gland	Aqueous deficient dry eye (Sjogren's syndrome)
③	Outer	Lipid	meibomian gland	Evaporative dry eye

Investigations :

1. Tear break up time.
2. Schirmer test.
3. Phenol red thread test.

Conjunctivitis/Eye Flu

00:03:42

Clinical Features :

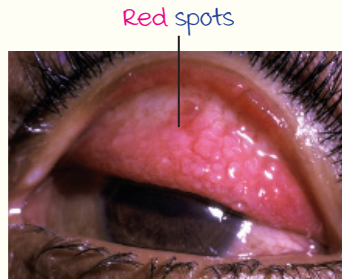
- Redness.
- Irritation.
- mild pain.

Signs :

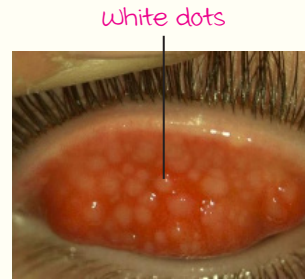
----- Active space -----



1. Discharge



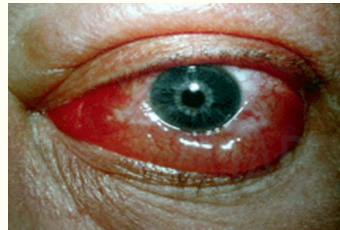
2. Papillae



3. Follicles



4. Subconjunctival hemorrhage



5. Chemosis :

Swelling of bulbar conjunctiva

Symptom Groups and Etiologies :

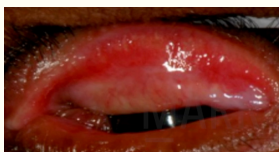
	Papillae	Follicles
mucopurulent discharge	Bacterial	Chlamydia
watery discharge	Allergic	Viral

Bacterial & Viral Conjunctivitis

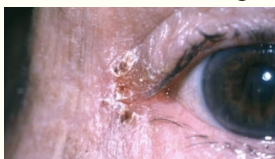
00:05:24



Acute pseudomembranous type



Acute membranous type



Angular conjunctivitis



Acute hemorrhagic conjunctivitis

Type 	Characteristics
Acute/Hyperacute purulent/ Blennorrhoea	- Neisseria gonorrhoeae - Copious discharge + intense pain + swelling of upper eyelids - Rx : Ceftriaxone
Acute membranous	Bleeds on removal
Acute pseudomembranous	Does not bleed on removal
Angular/Diplobacillary	- Moraxella axenfeld - Affects lateral canthi : Excoriation - Rx : Zinc boric acid drops
Acute hemorrhagic/ Adenoviral conjunctivitis	Etiology : - Picornavirus - Adenovirus TII - Coxsackie A24 - Enterovirus type 70

----- Active space -----

Chlamydial Conjunctivitis

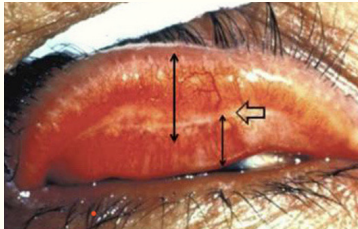
00:08:53

Trachoma/Egyptian Ophthalmia 👁 :

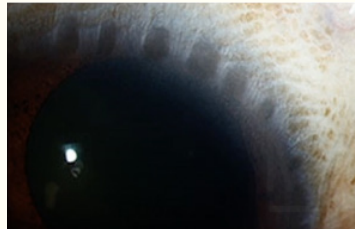
Spread : Finger, fly, fomites.

Signs :

1. Sago grain follicles.
2. Arit's line : D/t concurrent inflammation & healing.
3. Herbert's pits at limbus.
4. Pannus : Vascularization of cornea.



Arit's line



Herbert's pits



Pannus

Treatment : SAFE strategy.

- Surgery : Trichiasis
- Antibiotics : Azithromycin 1 gm single dose.
- Facial cleanliness.
- Environment changes.

Complication : Blindness.

Infections → Inflammation → Scarring of eyelids → Trichiasis : Irritation of cornea by eyelashes → Corneal opacity.

Ophthalmia Neonatorum :

Onset (Hours after birth)	Cause
Earliest	Chemical (Silver nitrate) conjunctivitis
24 - 48 hours (most severe)	Neisseria gonorrhoeae
> 1 week (m/c)	Chlamydia trachomatis



Prevention :

Erythromycin eye ointment : First 1 hr.

Allergic Conjunctivitis

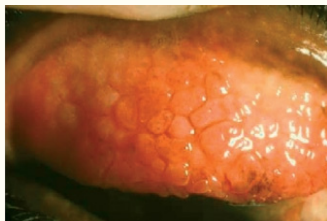
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----- Active space -----

VKC/Spring Catarrh :

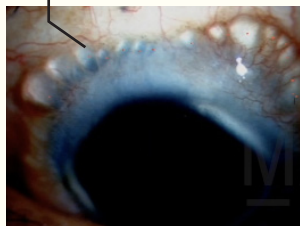
- m/c in **boys** of 5 – 15 yrs.
- Hypersensitivity : **Type I.**
- Ropy discharge + itching.
- Summer/spring.

Signs :

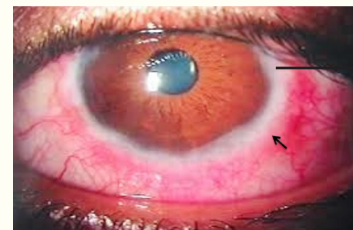


Cobblestone papilla

Eosinophilic deposits at limbus



Horner - Trantas dots



White ring around the limbus

Pseudogerontoxon

Rx :

Olopatadine (DOC) : Antihistamine + mast cell stabilizer.
+

Topical steroids : For immediate relief.

Giant Papillary Conjunctivitis :

- Hypersensitivity : **Type 4.**
 - Mechanically induced.
- Etiology :
- Soft < Hard contact lens.
 - Ocular prosthesis.
 - Protruding suture ends.

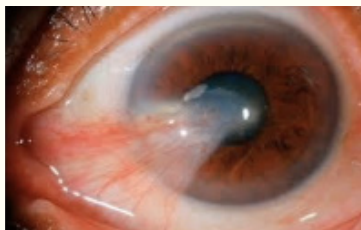
Pterygium

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Causes :

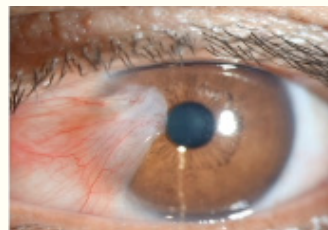
- Limbal stem cell deficiency.
- UV rays, dust.

Progression :



Over pupil

Obstruction of visual axis



Not over pupil

Visual defect d/t astigmatism

Treatment : **Excision** + autografting.

